

SGA 7

Exposé I

Summary of the First Lectures by A. Grothendieck

written up by P. Deligne

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0. Preliminaries

0.0. Terminology. Recall the following definitions:

(0.0.1) trait: the spectrum of a discrete valuation ring. The generic and closed points of a trait are often denoted by η and s .

(0.0.2) strictly local: for a ring, local henselian with separably closed residue field; for a scheme, the spectrum of such a ring.

(0.0.3) geometric point of S (in this exposé): a morphism $\bar{x} : \text{Spec}(\bar{k}) \rightarrow S$ with image $x \in S$, where $k(\bar{x}) \stackrel{\text{dfn}}{=} \bar{k}$ is a separable closure of $k(x)$. For a henselian trait $S = \text{Spec}(V)$, a generic geometric point of S (that is, one with image η) is often denoted by $\bar{\eta}$. The residue field of the valuation ring $\bar{V}$, the integral closure of V in $k(\bar{\eta})$, is a separably closed algebraic extension of $k(s)$ and defines a geometric point $\bar{s}$ of S centered at s .

(0.0.4) strict localization of X at the geometric point $\bar{x}$: SGA 4 VIII 4.4.

(0.0.5) essentially of finite type over S : for a local (respectively, henselian local; respectively, strictly local) S -scheme, the spectrum of a local ring (respectively, the henselization of a local ring; respectively, a strict localization) of a scheme of finite type over S .

(0.0.6) $F \mapsto F(n)$: Tate twist.

(0.0.7) An endomorphism T of a finite-dimensional vector space is quasi-unipotent if its eigenvalues are roots of unity, that is, if there exist $N \geq 1$ and $M \geq 1$ such that $(T^N - 1)^M = 0$.

0.1. Review of ℓ -adic cohomology

Let X be a scheme of finite type over an algebraically closed field k of characteristic exponent p , and let ℓ be a prime different from p .

The ℓ -adic cohomology groups of X were defined in SGA 4 and SGA 5.

a) $H^i(X, \mathbb{Z}/\ell^n)$ is the i -th cohomology group of the étale site $X_{\text{ét}}$ of X with coefficients in $\mathbb{Z}/\ell^n$;

b) by definition,

$$H^i(X, \mathbb{Z}_\ell) = \varprojlim_n H^i(X, \mathbb{Z}/\ell^n);$$

c) by definition,

$$H^i(X, \mathbb{Q}_\ell) = H^i(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

The ℓ -adic cohomology groups of X with proper supports are defined in the same way from the groups $H_c^i(X, \mathbb{Z}/\ell^n)$ of SGA 4 XVII.

In each of the following cases, one knows how to prove that these groups $H^i(X)$ have reasonable finiteness properties:

(0.1.1) for cohomology with proper supports;

(0.1.2) when X is proper over k ;

(0.1.3) when X is smooth over k (by (0.1.1) and Poincaré duality, SGA 4 XVIII);

(0.1.4) when Hironaka's resolution of singularities is available for schemes of finite type over k of dimension at most $\dim(X)$;

(0.1.5) when $i = 0, 1$.

In all these cases, the groups $H^i(X, \mathbb{Z}/\ell^n)$ are finite, the groups $H^i(X, \mathbb{Z}_\ell)$ are finitely generated, the groups $H^i(X, \mathbb{Q}_\ell)$ are finite-dimensional, and (except perhaps in case (0.1.5)) there are split short exact sequences

$$0 \longrightarrow H^i(X, \mathbb{Z}_\ell) \otimes \mathbb{Z}/\ell^n \longrightarrow H^i(X, \mathbb{Z}/\ell^n) \longrightarrow \mathrm{Tor}_1(H^{i+1}(X, \mathbb{Z}_\ell), \mathbb{Z}/\ell^n) \longrightarrow 0.$$

Moreover, in each case the proof of the finiteness theorem also proves a generic constructibility theorem: if k is the field of fractions of an integral noetherian scheme S , and X is the generic fibre of a morphism $f_1 : X_1 \longrightarrow S$ of finite type, then there exists a nonempty open subset $U \subset S$ such that over U the sheaves $R^i f_* \mathbb{Z}/\ell^n$ are locally constant and their formation commutes with every base change.

0.2. We no longer suppose that k is algebraically closed. Let $\bar{k}$ be an algebraic closure of k (or a separable closure, which amounts to the same thing here). We consider the “geometric” cohomology groups $H^i(X_{\bar{k}})$, where $X_{\bar{k}}$ is obtained from X by extension of scalars from k to $\bar{k}$. By transport of structure, the Galois group $\mathrm{Gal}(\bar{k}/k)$ acts on these groups. In $\mathbb{Q}_\ell$ -cohomology, in any of the cases (0.1.1)–(0.1.5), this gives continuous representations of $\mathrm{Gal}(\bar{k}/k)$ in a linear group $\mathrm{GL}(n, \mathbb{Q}_\ell)$.

0.3. Review of traits (0.0.1)

For the proofs, we refer to [4]. Let S be a henselian trait, and let η , s , V , $\bar{V}$, $\bar{\eta}$, and $\bar{s}$ be as in (0.0.1) and (0.0.3). Let p denote the characteristic exponent of $k(s)$, and let $\mathrm{Gal}(\bar{\eta}/\eta)$ (respectively, $\mathrm{Gal}(\bar{s}/s)$) denote the Galois group $\mathrm{Gal}(k(\bar{\eta})/k(\eta))$ (respectively, the corresponding group for $\bar{s}/s$). By transport of structure, $\mathrm{Gal}(\bar{\eta}/\eta)$ acts on $k(\bar{s})$; hence there is a homomorphism from $\mathrm{Gal}(\bar{\eta}/\eta)$ onto $\mathrm{Gal}(\bar{s}/s)$ whose kernel is the inertia group I . The subfield of $k(\bar{\eta})$ fixed by I is a maximal unramified extension of $k(\eta)$.

The profinite group I has a largest pro- p subgroup P . The quotient I/P is the tame inertia group. Canonically,

$$I/P \simeq \varprojlim_n \mu_n(k(\bar{s})) \stackrel{\mathrm{def}}{=} \widehat{\mathbb{Z}}(1)(k(\bar{s})).$$

If ϕ_n is the canonical map from I/P to $\mu_n(\bar{k})$ and u is an element of $\bar{V}$ of valuation $1/n$, then, for $\sigma \in I$,

$$\sigma(u) \equiv \phi_n(\sigma) u \quad (\text{modulo elements of valuation greater than } 1/n).$$

In summary,

$$\begin{aligned} & \mathrm{Gal}(\bar{\eta}/\eta) \supset I \supset P, \\ & \mathrm{Gal}(\bar{\eta}/\eta)/I \simeq \mathrm{Gal}(\bar{s}/s), \\ (0.3.1) \quad & I/P \simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ & P : \text{a pro-}p \text{ group} \quad (\{e\} \text{ if } p = 1), \end{aligned}$$

and the action (by inner automorphisms in $\mathrm{Gal}(\bar{\eta}/\eta)$) of $\mathrm{Gal}(\bar{\eta}/\eta)/I$ on I/P is identified with the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.4. More generally, let S be strictly local and regular, let D be a regular divisor in S , and let p' be the characteristic exponent of the generic point of D . Let $\bar{\eta}$ be a geometric generic point of S , and let $\bar{s}$ be the resulting geometric point centered at the closed point s . By Abhyankar’s lemma, the fundamental group $\pi_1(S - D, \bar{\eta})$ has a filtration analogous to (0.3.1):

$$\begin{aligned} & \pi_1(S - D, \bar{\eta}) \supset I \supset P, \\ (0.4.1) \quad & \pi_1(S - D, \bar{\eta})/I \simeq \mathrm{Gal}(\bar{s}/s), \\ & I/P \simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ & P : \text{with no quotient of order prime to } p'. \end{aligned}$$

The action by inner automorphisms of $\pi_1(S - D, \bar{\eta})/I$ on I/P is identified with the natural action of $\text{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.5. Néron desingularization

The method of Néron desingularization gives the following result.

Lemma 0.5.1. Let $S \rightarrow S_o$ be a morphism of henselian traits, with $S = \text{Spec}(V)$ and $S_o = \text{Spec}(V_o)$, and let π be a uniformizer of V_o . Suppose that π is a uniformizer of V , and that the extensions $k(s)/k(s_o)$ and $k(\eta)/k(\eta_o)$ are separable. Then V is a filtered direct limit of smooth henselian V_o -algebras B_i essentially of finite type over V_o .

To construct the desired V_o -algebras, one takes $V_o \subset B \subset V$, with B of finite type, and desingularizes $\text{Spec}(B)$ by Néron's method ([1], § 4).

Let $S = \text{Spec}(V)$ be a henselian trait.

(0.5.2) If S is of equal characteristic, with uniformizer π , one may apply (0.5.1) with $S_o = \mathbf{F}_p[\pi]_{(\pi)}$. The extension s/s_o is separable because $\mathbf{F}_p$ is perfect; the extension η/η_o is separable because π , being a uniformizer, is not a p -th power (if $p \neq 1$), and hence belongs to a p -basis. It follows that S is a limit of spectra of smooth henselian $\mathbf{F}_p[\pi]_{(\pi)}$ -algebras essentially of finite type.

(0.5.3) If S is complete and of mixed characteristic, with perfect residue field k , then V is an Eisenstein extension of the ring of Witt vectors (that is, the Cohen ring) $W(k)$:

$$V \simeq W(k)[\pi] / \left(\pi^n + \sum_{i=0}^{n-1} a_i \pi^i \right).$$

If the a_i are perturbed slightly, one obtains an isomorphic extension. We may therefore suppose that the a_i belong to $W(k')$, where k' is of finite type over $\mathbf{F}_p$. Let k_o be the perfect closure of k' in k , and set

$$V_o = W(k_o)[\pi] / \left(\pi^n + \sum_{i=0}^{n-1} a_i \pi^i \right).$$

Lemma 0.5.1 applies to V/V_o , and the residue field of V_o is a radicial extension of a field of finite type over $\mathbf{F}_p$.

1. Arithmetic proof of the monodromy theorem

The reader will find in [5], Appendix, a proof of the following result of A. Grothendieck.

Proposition 1.1. Let S be a henselian trait, let ℓ be a prime different from the characteristic exponent p of $k(s)$, and let ρ be a continuous representation of $\text{Gal}(\bar{\eta}/\eta)$ in $\text{GL}(n, \mathbb{Q}_\ell)$. Suppose that the following condition holds:

$(*_\ell)$ No finite extension of $k(s)$ contains all roots of unity of ℓ -power order.

Then there exists an open subgroup I_1 of the inertia group I such that $\rho(\sigma)$ is unipotent for every $\sigma \in I_1$.

Condition $(*_\ell)$ is automatically satisfied when $k(s)$ is of finite type over the prime field (or is radicial over such a field).

The following theorem follows immediately from 1.1.

Monodromy theorem 1.2. With the preceding notation, let X be a scheme of finite type over η , and let H be a cohomology space of $X_{\bar{\eta}}$ with coefficients in $\mathbb{Q}_\ell$, of one of the types (0.1.1)–(0.1.5). If condition $(*_\ell)$ is satisfied, the action on H of every element of I is quasi-unipotent.

The limit arguments of 0.5 allow us to dispense with hypothesis $(*_\ell)$.

Variant 1.3. Condition $(*_\ell)$ in 1.2 is superfluous.

Let H'_o be a $\mathbb{Z}_\ell$ -cohomology space of $X_{\bar{\eta}}$ giving rise to H , and put $H_o = H'_o/\text{torsion}$. By hypothesis, $H = H_o \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, and the action of $\text{Gal}(\bar{\eta}/\eta)$ is induced by

$$\rho : \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{GL}(H_o).$$

Let $I_{\ell'}$ be the largest prime-to- ℓ subgroup of I . Its image in $\text{GL}(H_o)$ is an ℓ -adic Lie group of order prime to ℓ , and is therefore finite.

Make an extension of traits $S' \rightarrow S$. This does not change H_o (SGA 4 XVI 6.6), and $\rho(I')$ has finite index in $\rho(I)$, because I'/P' has finite index in I/P , by what precedes. It therefore suffices to prove the theorem over S' ; we may suppose that

- a) $\text{Gal}(\bar{\eta}/\eta)$ acts trivially on $H_o/\ell H_o$;
- b) if S is of mixed characteristic, then S is complete with perfect residue field (EGA O_{III} 10.3.1).

Put $S = \text{Spec}(V)$, define V_o as in (0.5.2) and (0.5.3), and apply (0.5.1). We have

$$V = \varinjlim_i B_i.$$

Here $S_i = \text{Spec}(B_i)$ is essentially of finite type and smooth over V_o . Let $D_i \subset S_i$ be defined by the equation $\pi = 0$. For sufficiently large i , H_o comes from a twisted constant $\mathbb{Z}_\ell$ -sheaf $\mathcal{H}_i$ on $S_i - D_i$ (by “generic constructibility,” (0.1)), and $\mathcal{H}_i/\ell\mathcal{H}_i$ is constant.

The filtrations (0.3.1) and (0.4.1) give rise to commutative diagrams:

$$(1.3.1) \quad \begin{array}{ccccc} \text{Gal}(\bar{\eta}/\eta) & \supset & I & \supset & P \\ \downarrow & & \downarrow & & \downarrow \\ \pi_1(S_i - D_i, \bar{\eta}) & \supset & I_i & \supset & P_i \end{array} \quad \text{and} \quad \begin{array}{ccc} I/P & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \\ \downarrow & & \parallel \\ I_i/P_i & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \end{array}$$

The representation on H_o of $\text{Gal}(\bar{\eta}/\eta)$ is induced by an action of $\pi_1(S_i - D_i, \bar{\eta})$ on H_o that is trivial on $H_o/\ell H_o$. The group

$$\text{Ker}(\text{GL}(H_o) \rightarrow \text{GL}(H_o/\ell H_o))$$

is a pro- ℓ group. Thus P_i acts trivially, and we apply the following variant of 1.1.

Variant 1.4. Retain the notation of 0.4, and let ρ be a continuous representation of $\text{Gal}(\bar{\eta}/\eta)$ in $\text{GL}(n, \mathbb{Q}_\ell)$. Suppose that $k(s)$ satisfies $(*_\ell)$ and that $\rho(P)$ is finite. Then the action of I is quasi-unipotent.

2. Vanishing cycles

2.1. Let S be a strictly local trait, and retain the notation of 0.3. Thus, by hypothesis, $s = \bar{s}$. Let $f : X \rightarrow S$ be a scheme of finite type over S . The formalism of vanishing cycles is a tool for studying the relation between the cohomology of X_s and that of $X_{\bar{\eta}}$, as well as the action of the inertia group on the latter, using information about f that is local on X .

A key result of this kind is the specialization theorem (SGA 4 XVI 2.2), according to which X_s and $X_{\bar{\eta}}$ have the same cohomology when f is proper and smooth.

In the transcendental theory, if $f : X \rightarrow D$ is a proper morphism from an analytic space X to the disk D , vanishing cycles arise as follows:

- a) After shrinking D if necessary, one may suppose that X is a topological fibre bundle over the punctured disk D^* (Thom’s isotopy theorem). In particular, the fibres

X_t ($t \in D^*$) are all isomorphic.

- b) It is not unreasonable to picture the special fibre X_s as obtained from “the” general fibre X_t by contracting polyhedra in X_t ; one would then have a map $\pi : X_t \rightarrow X_s$. The method of vanishing cycles consists in studying the difference between the cohomology of X_s and that of X_t by means of the Leray spectral sequence for π .

The geometric theory explained below is very close to this transcendental theory: the general point $t \in D$ must be replaced by $\bar{\eta}$, the geometric generic point of S . Conversely, the transcendental theory can be modeled on the geometric theory by replacing t or $\bar{\eta}$ with the universal cover $\tilde{D}^*$ of D^* , and X_t or $X_{\bar{\eta}}$ with $X \times_D \tilde{D}^*$. This eliminates many ε ’s and η ’s, but sometimes loses information.

2.2. Fix a torsion ring Λ in which p is invertible (for example, $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$, with ℓ prime to p). Consider the morphisms

$$\begin{array}{ccccc}
X_s & \xrightarrow{i} & X & \xleftarrow{\bar{j}} & X_{\bar{\eta}} \\
\downarrow f_s & & \downarrow f & & \downarrow f_{\bar{\eta}} \\
s & \longrightarrow & S & \longleftarrow & \eta \longleftarrow \bar{\eta}
\end{array}$$

The sheaves of vanishing cycles ψ^i on X_s are

$$(2.2.1) \quad \psi^i = i^* R^i \bar{j}_* \Lambda.$$

If f is proper, then (SGA 4 XII 5.5)

$$(2.2.2) \quad i^* : H^p(X, R^q \bar{j}_* \Lambda) \xrightarrow{\sim} H^p(X_s, i^* R^q \bar{j}_* \Lambda),$$

and the Leray spectral sequence for $\bar{j}$ becomes

$$(2.2.3) \quad H^p(X_s, \psi^q) \implies H^{p+q}(X_{\bar{\eta}}, \Lambda).$$

When nothing happens at infinity, this spectral sequence can sometimes be obtained without a properness hypothesis. That some hypothesis at infinity is necessary is already clear from the trivial case $X_s = \emptyset$.

The definitions immediately give:

Proposition 2.3. Let $\bar{x}$ be a geometric point of X_s (for example, a closed point), let $X_{(\bar{x})}$ be the strict henselization of X at $\bar{x}$, and let $X_{(\bar{x})\bar{\eta}}$ be the geometric generic fibre of the projection $X_{(\bar{x})} \rightarrow S$. Then

$$\psi_{\bar{x}}^i = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda).$$

Corollary 2.4. Wherever f is smooth, $\psi^i = 0$ for $i > 0$ and $\psi^0 = \Lambda$.

This is SGA 4 XV 2.1.

The group $\text{Gal}(\bar{\eta}/\eta)$ (which, by hypothesis, equals the inertia group) acts on the ψ^i by transport of structure, and

Scholium 2.5. The spectral sequence (2.2.3) is $\text{Gal}(\bar{\eta}/\eta)$ -equivariant.

Variant 2.6. Put $\bar{S} = \text{Spec}(\bar{V})$ and $\bar{X} = X \times_S \bar{S}$. The scheme $X_{\bar{\eta}}$ is the complement of $X_s = X_{\bar{s}}$ in $\bar{X}$. Put

$$\phi^i = \underline{H}_{X_s}^{i+1}(\bar{X}, \Lambda)$$

(the local-cohomology sheaves). There is an exact sequence of sheaves

$$0 \longrightarrow \phi^{-1} \longrightarrow \Lambda \longrightarrow \psi^0 \longrightarrow \phi^0 \longrightarrow 0,$$

and, for $i > 0$, isomorphisms $\psi^i \xrightarrow{\sim} \phi^i$. If f is smooth, the ϕ^i vanish.

Put $\phi_{\text{gl}}^i = H_{X_s}^i(\bar{X}, \Lambda)$, the global analogue of ϕ^i . There is a spectral sequence

$$(2.6.1) \quad H^p(X_s, \phi^q) \implies \phi_{\text{gl}}^{p+q},$$

and a long exact cohomology sequence which, when f is proper, becomes

$$(2.6.2) \quad \cdots \longrightarrow H^i(X_s, \Lambda) \longrightarrow H^i(X_{\bar{\eta}}, \Lambda) \longrightarrow \phi_{\text{gl}}^i \longrightarrow \cdots.$$

Variant 2.7. Let K_t be the maximal tamely ramified extension of K in $\bar{K}$. Put $X_t = X \times_S \text{Spec}(K_t)$, and let $\bar{j}_t$ be the projection from X_t to X . Define tame variants of the ψ^i by

$$\psi_t^i = R^i \bar{j}_{t*} \Lambda.$$

Since P is a pro- p group and p is invertible in Λ , one obtains

$$(2.7.1) \quad H^*(X_t, \Lambda) = H^*(X_{\bar{\eta}}, \Lambda)^P,$$

$$(2.7.2) \quad \psi_t^i = (\psi^i)^P.$$

When f is proper, the Leray spectral sequence for $\bar{j}_t$ is obtained from (2.2.3) by passing to P -invariants:

$$(2.7.3) \quad H^p(X_s, \psi_t^q) \implies H^{p+q}(X_t, \Lambda) = H^{p+q}(X_{\bar{\eta}}, \Lambda)^P.$$

3. Geometric proof of the monodromy theorem

Let Λ be as in 2.2.

Conjecture 3.1 (purity). Let A be an excellent regular strictly local ring (0.0.2) and let D be a regular divisor in $\text{Spec}(A)$ defined by a parameter x . Put $U = \text{Spec}(A) - D = \text{Spec}(A[1/x])$. Then

$$(3.1.1) \quad H^q(U, \Lambda) = \begin{cases} \Lambda & \text{if } q = 0, \\ \Lambda(-1) & \text{if } q = 1 \quad (\text{notation (0.0.6)}), \\ 0 & \text{if } q > 1. \end{cases}$$

Here are some cases in which this conjecture is known (SGA 4 XIX): for $q = 0$ or 1 ; when A has characteristic 0 ; when A has equal characteristic p and resolution of singularities is available in characteristic p in dimensions $\leq \dim(A)$; or when $(\text{Spec}(A), D)$ is a smooth pair (the relative purity theorem, SGA 4 XVI 3.7).

When purity is available and $D = \sum_{i \in B} D_i$ is a normal-crossings divisor in $\text{Spec}(A)$ defined by part of a regular system of parameters, the cohomology of $U = \text{Spec}(A) - D$ is given by

$$(3.1.2) \quad \begin{aligned} H^0(U, \Lambda) &= \Lambda, \\ H^1(U, \Lambda) &= \Lambda(-1)^B, \\ H^q(U, \Lambda) &= \bigwedge^q H^1(U, \Lambda) \quad (\text{via the cup product}). \end{aligned}$$

These formulas are identical to those which, over $\mathbf{C}$, give the cohomology of $D^{*B} \times D^k$.

3.2. Let S be a strictly local trait, and retain the notation of (0.0.1) and (0.0.3). Let $f : X \rightarrow S$ be a morphism of finite type. Suppose that X is regular, that X_{η} is smooth, and that X_s is a normal-crossings divisor in X . We shall suppose that this divisor is, globally and not merely locally for the étale topology, a sum of regular divisors:

$$(X_s)_{\text{red}} = \sum_{i \in B} D_i$$

(the general case can be recovered by localization). Denote their multiplicities by m_i , so that $X_s = \sum_{i \in B} m_i D_i$.

Let x be a point of X_s , let $\bar{x}$ be a geometric point of X lying over x (for example, x may be closed and equal to $\bar{x}$), put $C = \{i \mid x \in D_i\}$, and let K be the complex concentrated in degrees 0 and -1

$$K : \mathbf{Z}^C \xrightarrow{d} \mathbf{Z}, \quad d((n_j)) = \sum_j m_j n_j.$$

Theorem 3.3 (uses purity). The groups of tame vanishing cycles $\psi_{t\bar{x}}^i$ are as follows:

(i) There is a canonical isomorphism of graded Λ -algebras

$$\psi_{t\bar{x}}^* \simeq \bigwedge^* (H_1(K) \otimes \Lambda(-1)) \otimes_{\Lambda} \psi_{t\bar{x}}^0.$$

(ii) $\psi_{t\bar{x}}^0$ is the Λ -algebra Λ^E , where E is a set on which the tame inertia group $I_t = I/P = \widehat{\mathbf{Z}}(1)(k(s))$ acts transitively, and whose cardinality is the largest prime-to- p divisor of $\#H_0(K)$.

Here is the method of proof.

a) Let $X_{(\bar{x})}$ be the strict localization of X at $\bar{x}$, let $U = X_{(\bar{x})} - X_{(\bar{x})s}$ (the complement of a normal-crossings divisor), let t_j be a local equation for D_j , put

$$X_n = X_{(\bar{x})}[(t_j^{1/n})] \quad ((n, p) = 1),$$

which is a regular scheme, and let U_n be the inverse image of U in X_n . By (3.1.2),

$$H^q(U_n, \Lambda) = \bigwedge^q (\Lambda(-1)^C).$$

For $m = nd$, the pullback map from $H^q(U_n, \Lambda)$ to $H^q(U_m, \Lambda)$ is multiplication by d^q . Putting $\tilde{U} = \varprojlim_{(n,p)=1} U_n$, one therefore has

$$(3.3.1) \quad \begin{cases} H^0(\tilde{U}, \Lambda) = \Lambda, \\ H^q(\tilde{U}, \Lambda) = \varinjlim_n H^q(U_n, \Lambda) = 0 \quad \text{if } q \neq 0. \end{cases}$$

b) $\tilde{U}$ is a pro-covering of U with group $\widehat{\mathbf{Z}}(1)(k(s))^C$. By definition,

$$\psi_{t\bar{x}}^* = H^*(U_{\eta_t}, \Lambda).$$

$\tilde{U}$ is a pro-covering, with group $H_1(K) \otimes \widehat{\mathbf{Z}}(1)(k(s))$, of each connected component of U_{η_t} , and these components are permuted transitively by the tame inertia group. Formula (3.3.1) and the Hochschild–Serre spectral sequence now give the formulas in 3.3.

Pictorially, one may say that $U_{\bar{\eta}}$ is the fibre of a morphism δ from the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s))^C, 1)$ represented by U to the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s)), 1)$ represented by η ; the morphism δ induces

$$d : (n_j) \mapsto \sum_j m_j n_j : \widehat{\mathbf{Z}}(1)(k(s))^C \longrightarrow \widehat{\mathbf{Z}}(1)(k(s)).$$

Corollary 3.4 (uses purity). Suppose that f is proper. Let N be the least common multiple of the multiplicities m_j ($j \in B$). Let $q \geq 0$, and let q' be the minimum of q and the greatest number of divisors D_j passing through one point. For T in the tame inertia group $I_t = I/P$, one has

$$(T^N - 1)^{q'+1} = 0 \quad \text{on} \quad H^q(X_{\bar{\eta}}, \Lambda)^P.$$

This follows at once from 3.3, Scholium 2.5, and Variant 2.7.

The required purity results are available for H^1 , in characteristic 0, or when S is essentially of finite type over a field.

Theorem 3.5. Let C be a smooth projective curve over the fraction field $k(\eta)$ of a discrete valuation ring V . The inertia group I acts on $H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell)$ by quasi-unipotent automorphisms of level 2: there exists N such that, for $T \in I$,

$$(T^N - 1)^2 = 0 \quad \text{on} \quad H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell).$$

We reduce to the case in which V is strictly local. The assertion is that there exists an open subgroup I_1 of I such that $(T - 1)^2 = 0$ on H^1 for $T \in I_1$.

We may extend the scalars of $k(\eta)$ to a finite extension, replacing V by its normalization in that extension. Since the image of the pro- p group P in an ℓ -adic Lie group $\mathrm{GL}(n, \mathbf{Q}_\ell)$ is finite, we may in particular suppose that P acts trivially.

Let C_o be a minimal model of C over V (for the existence of C_o , which is based on Abhyankar's theorem, see the discussion in [3], § 2, p. 87; one could instead reduce to the case where V is excellent by passing to its completion and invoking SGA 4 XVI 1.6).

Let C_1 be a regular model of C , obtained from C_o by blowups, whose special fibre is a normal-crossings divisor. The arguments of 3.3 and 3.4 apply to C_1 , ψ_t^0 , ψ_t^1 , and

$$H^1(C_{\bar{\eta}}, \mathbb{Z}/\ell^n)^P = H^1(C_{\bar{\eta}}, \mathbb{Z}/\ell^n),$$

with N independent of n ; Theorem 3.5 follows.

Remark 3.6. Let A be an abelian variety over $k(\eta)$. Since A is a quotient of a Jacobian, it follows again that the action of I on $H^1(A_{\bar{\eta}}, \mathbf{Q}_\ell)$, or on $T_\ell(A)$, is quasi-unipotent of level 2.

Remark 3.7. The action of I on $H^1(X_{\bar{\eta}}, \mathbf{Q}_\ell)$ is in fact quasi-unipotent of level 2 for every scheme X of finite type over η .

Theorem 3.5 (or Remark 3.6) is the basis for the proof of the stable reduction theorem for abelian varieties that will be given in Exposé IX, § 3.

3.8. It is clear that whenever purity and resolution of singularities are available (for example, in characteristic 0), the preceding method proves the monodromy theorem and gives bounds for the nilpotence exponent. Thus, in equal characteristic 0, if X is proper and smooth over η and T belongs to an open subgroup I_1 of I , one has

$$(T - 1)^{q+1} = 0 \quad \text{on} \quad H^q(X_{\bar{\eta}}, \mathbb{Z}_\ell) \quad (T \in I_1 \subset I).$$

4. Vanishing criteria for the sheaves of vanishing cycles

4.1. Let S be a strictly local trait and let $f : X \rightarrow S$ be a scheme of finite type over S . Put $n = \dim(X_\eta)$.

If X' is the scheme-theoretic closure of X_η in X , then:

- a) On $X_s - X'_s$, one has $\phi^{-1} = \Lambda$, and the other ϕ^i vanish;
- b) on X'_s , the ϕ^i and ψ^i relative to X/S or to X'/S coincide, and $\phi^{-1} = 0$.

Since X' is flat over S , this often permits a reduction to the case in which X is flat over S .

Theorem 4.2.

- (i) One has $\psi^i = 0$ for $i > n$.
- (ii) More precisely, let x be a point of X_s whose closure has dimension k ($k = \mathrm{trdeg}_{k(s)} k(x)$), and let $\bar{x}$ be a geometric point lying over x . Then

$$\psi_{\bar{x}}^i = 0 \quad \text{for} \quad i > n - k, \quad \text{i.e.} \quad d(\psi^i) \leq n - i \quad (\text{notation of SGA 4 XIV 2.1}).$$

Indeed,

$$\psi_{\bar{x}}^i = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda),$$

and $X_{(\bar{x})\bar{\eta}}$ is an inverse limit of affine schemes of dimension at most n over $k(\bar{\eta})$. Assertion (i) therefore follows from the affine Lefschetz theorem (SGA 4 XIV 3.2) by passage to the limit.

To prove (ii), we may suppose that X is flat over S (4.1) and that there is a quasi-finite flat S -morphism

$$g : X \rightarrow \mathbb{A}_S^k$$

such that $g(x)$ is the generic point of $\mathbb{A}_S^k$. Let S' be the strict localization of $\mathbb{A}_S^k$ at $\bar{x}$. Then:

- a) S'_η is the spectrum of a field, and $\text{Gal}(\bar{\eta}'/s'_\eta)$ is a p -group Q ;
- b) by (i), applied to g ,

$$H^i(X_{(\bar{x})\bar{\eta}'}, \Lambda) = 0 \quad \text{for } i > n - k;$$

- c) one has

$$H^i(X_{(\bar{x})\bar{\eta}}, \Lambda) = H^i(X_{(\bar{x})\bar{\eta}'}, \Lambda)^Q,$$

which proves the theorem.

Corollary 4.3. If f is proper and flat and the non-smooth locus of f in X_s has dimension at most d , the specialization morphism

$$H^i(X_s) \longrightarrow H^i(X_{\bar{\eta}})$$

is an isomorphism for $i > n + d + 1$ and an epimorphism for $i = n + d + 1$.

4.4. To prove, in certain cases, that the ϕ^i vanish for small i , we shall use local duality, either directly or through the local Lefschetz theorems of SGA 2 XIV. This tool is available only in characteristic 0, or in equal characteristic p when resolution of singularities à la Hironaka is available in the dimensions under consideration.

Theorem 4.5. Suppose that S has characteristic 0. Let $f : X \longrightarrow S$ be a morphism of finite type and let x be a closed point of X_s . Suppose that:

- (a) x is an isolated non-smooth point in its fibre;

- (b) X has relative geometric depth at least n at x : in a neighbourhood of x , the scheme X is identified with a subscheme defined by k equations in a smooth scheme of relative dimension N at x , with $n \leq N - k$.

Then $\phi_x^i = 0$ for $i < n$.

Let X_x be the strict localization of X at x , and put $V = X_x - \{x\}$. For every trait S' finite over S , put similarly

$$X'_x = X_x \times_S S', \quad V' = V \times_S S'.$$

Let $\bar{S}$ be the normalization of S in $\bar{\eta}$, and put

$$\bar{X}_x = X_x \times_S \bar{S}, \quad \bar{V} = V \times_S \bar{S}.$$

We have:

- (1) If $\phi^i = 0$ on $(X_x)_s$ (i.e. at every generization of x in X_s) for $i \leq m$, then

$$H_{\{x\}}^{i-1}(\bar{X}_x) = \tilde{H}^{i-1}(\bar{V}) \xrightarrow{\sim} \tilde{H}^i(\bar{V}_{\bar{\eta}}) = \phi^i \quad \text{for } i \leq m.$$

The ϕ^i are indeed the cohomology sheaves of $\bar{V}$ with support in V_s ; under the hypotheses of 4.5, condition (a) makes (1) applicable with $m = \infty$ (SGA 4 XV 2.1).

- (2) Hypothesis (b) of 4.5 is stable under base change. By SGA 2 XIV 5.6, it implies

$$\tilde{H}^i(V') = H_{\{x\}}^{i-1}(X'_x) = 0 \quad \text{for } i < n.$$

We conclude by observing that $\tilde{H}^{i-1}(\bar{V})$ is the filtered colimit of the $\tilde{H}^i(V')$.

Combining 4.2 and 4.5 gives:

Corollary 4.6. Let S be a henselian trait of characteristic 0, let X/S be a relative complete intersection of relative dimension n , and let $x \in X_s$ be an isolated non-smooth point in its fibre. Then $\phi_x^i = 0$ for $i \neq n$.

Corollary 4.7. Under the hypotheses of 4.6, ϕ_x^n is a flat Λ -module.

Indeed, if Λ' is a finite Λ -algebra, a general nonsense (SGA 4 XVII 5.2.11) gives, for the $\phi_{\Lambda'}^i$,

$$\phi_{\Lambda'}^{n-i} = \text{Tor}_i^{\Lambda}(\Lambda', \phi_{\Lambda}^n),$$

and we apply 4.6 to Λ' .

Variant 4.8. In 4.5, omit condition (a), and let Z be the non-smooth locus of f . One may still conclude that

$$\phi_x^i = 0 \quad \text{for } i < n - \dim Z_s.$$

We proceed by descending induction on $\dim \overline{\{x\}}$. The induction hypothesis permits us to apply 4.5 (1) with $m = n - \dim Z_s$, and the proof is completed as before.

4.9. It is possible to obtain a vanishing criterion using only information about the geometric fibres of f . For complete intersections, however, the result is one degree weaker than 4.5.

Proposition 4.10. Under the general hypotheses of 4.5, suppose that:

- (a) x is an isolated non-smooth point in its fibre;
- (b) for every point y of $X_{\bar{\eta}}$ whose closure in $X_{\bar{\eta}}$ has dimension $d(y)$ and which is a generization of x , the étale depth satisfies

$$\text{depth}_{\text{ét},y}(X_{\bar{\eta}}) \geq n - d(y), \quad \text{i.e.} \quad H_{\{y\}}^i(X_{\bar{\eta}}) = 0 \quad \text{for } i \leq n - d(y);$$

- (c) $\text{depth}_{\text{ét},x}(X_s) \geq n$.

Then $\phi_x^i = 0$ for $i < n - 1$.

Retain the notation of 4.5. We may apply 4.5 (1) with $m = \infty$. By the local Lefschetz theorem (SGA 2 XIV), hypothesis (b) gives

$$H^i(V') \xrightarrow{\sim} H^i(V_s) \quad (i < n-1)$$

$$H^i(V') \hookrightarrow H^i(V_s) \quad (i = n-1)$$

Condition (c) then gives

$$H^i(\bar{V}) = \varinjlim H^i(V') = 0 \quad \text{for } i \leq n - 1,$$

which proves 4.10.

4.11. As in 4.8, one can give a variant of 4.10 in which (a) is not assumed.

5. Action of monodromy on π_1

In this section (Grothendieck's exposé of 19 November 1968), we shall use the semistable reduction theorem for curves. Artin and Winters [2] gave a direct proof of this theorem; it can also be deduced [3] from the analogous theorem for abelian varieties (IX § 3, using 3.6). We shall also use Schlessinger's theory (Rim's Exposé VI).

5.1. Let k be an algebraically closed field, let C be a projective curve over k , and let $x_1, \dots, x_n$ be a finite set of closed points of C . The pointed curve

$$X = (C; x_1, \dots, x_n)$$

is called stable if:

- (a) C is reduced, and its only singularities are double points with distinct tangents;
- (b) the x_i are distinct smooth points of C ;
- (c) if C_1 is an irreducible component of C , and E is the set of points of the normalization C'_1 of C_1 lying above a singular point of C or above one of the x_i , then, if C'_1 has genus 0 (respectively 1), the set E has at least 3 (respectively 1) elements.

Condition (c) is equivalent to each of the following:

- (c') X has no nontrivial infinitesimal automorphisms.
- (c'') If ω is the invertible dualizing sheaf, the invertible sheaf

$$\omega' = \omega \left(\sum_i x_i \right)$$

is ample.

An S -scheme C equipped with sections $x_1, \dots, x_n$ is a stable curve over S if its geometric fibres are stable curves. In [3], §§ 1–2, only the case $n = 0$ is considered. The extension of the results of loc. cit. to the general case is immediate:

- (1) If X/k is stable, then $H^1(C, \omega'^{\otimes n}) = 0$ for $n \geq 2$, and $\omega'^{\otimes n}$ is very ample for $n \geq 3$ (cf. [3] 1.2).
- (2) The formal moduli scheme M of X is smooth, and C remains singular over a normal-crossings divisor of M .
- (3) If X and Y are stable over S , then $\text{Isom}_S(X, Y)$ is finite and unramified over S (isomorphisms are required to respect the marked points).
- (4) If X/k is stable, the map

$$\text{Aut}_k(X) \longrightarrow \text{Aut}_k \text{Pic}^\circ(C)$$

is injective.

Finally, we have:

Proposition 5.1.1. Let $X_\eta = (C_\eta; x_1, \dots, x_n)$ be stable over the generic point η of a trait S , with C_η smooth. There exist a finite separable extension η' of η and a stable curve X' over the normalization S' of S in η' such that

$$X' \otimes_{S'} \eta' = X_\eta \otimes_\eta \eta'.$$

This follows from the usual semistable reduction theorem, which permits one to choose η' so that the geometric special fibre of the minimal model C'' of $C_{\eta'}$ over S' satisfies (a). One then blows up, iteratively if necessary, smooth points of C'' so that (b) is satisfied. Next one contracts the chains of smooth rational components of self-intersection -2 which contain none of the $\bar{x}_i$ in the special fibre; the result is C' (cf. [3], proof of 2.3, p. 88).

As in loc. cit., one can check that C' exists as soon as the Jacobian of C_η has semistable reduction. The smoothness hypothesis on C_η is moreover unnecessary.

5.2. Let X_η be a geometrically connected scheme of finite type over the fraction field of a strictly local trait S . For a geometric point ξ of $X_{\bar{\eta}}$, there is an exact sequence

$$(5.2.1) \quad 0 \longrightarrow \pi_1(X_{\bar{\eta}}, \xi) \longrightarrow \pi_1(X, \xi) \longrightarrow \text{Gal}(\bar{\eta}/\eta) \longrightarrow 0.$$

It induces a homomorphism from $\text{Gal}(\bar{\eta}/\eta)$ to the group of outer automorphisms of $\pi_1(X_{\bar{\eta}})$. Passing to the largest prime-to- p quotient $\pi_1^{(p)}(X_{\bar{\eta}})$ of $\pi_1(X_{\bar{\eta}})$, one obtains

$$(5.2.2) \quad \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{Aut}_{\text{ext}}(\pi_1^{(p)}(X_{\bar{\eta}})).$$

Suppose that X_η has a rational point x , and let $\bar{x}$ be the geometric point

$$\bar{\eta} \longrightarrow \eta \xrightarrow{x} X_\eta.$$

For $\xi = \bar{x}$, the exact sequence (5.2.1) splits canonically, giving

$$(5.2.3) \quad [x] : \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{Aut}(\pi_1(X_{\bar{\eta}}, \bar{x})) \longrightarrow \text{Aut}(\pi_1^{(p)}(X_{\bar{\eta}}, \bar{x})).$$

Theorem 5.3. The image of P under (5.2.2) (or under (5.2.3), which amounts to the same thing) is finite.

5.3.1. Reduction to the case in which X_η is geometrically normal and integral.

Let X'_η be the normalization of X_η , let $(X'_{i\eta})_i$ be the family of connected components of X'_η , put

$$X''_\eta = X'_\eta \times_{X_\eta} X'_\eta,$$

and let $(X''_{j\eta})_j$ be the family of connected components of X''_η . After a finite extension of $k(\eta)$, we may suppose that the $X'_{i\eta}$ are geometrically normal and integral and have rational points x_i , and that the $X''_{j\eta}$ are geometrically connected and have rational points y_j .

For $k = 1$ or 2 , and whenever $p_k(y_j) \in X'_{i\eta}$, choose a class of paths ℓ_{kij} from $\text{pr}_k(\bar{y}_j)$ to $\bar{x}_i$. It is an element of a torsor under $\pi_1(X'_{i\eta}, \bar{x}_i)$. Let $\bar{\ell}_{kij}$ be the corresponding prime-to- p path (the same path modulo $\text{Ker}(\pi \rightarrow \pi^{(p)})$). We may choose $\bar{\ell}_{kij}$ to be invariant under P : the set of prime-to- p paths is an inverse limit of finite sets of cardinality prime to p on which the pro- p group P acts.

Consequently, if a finite-index subgroup P' of P acts trivially on the groups $\pi_1^{(p)}(X'_{i\eta}, \bar{x}_i)$, the Van Kampen formulas—equivalently, the fact that the normalization map $X'_\eta \rightarrow X_\eta$ is effective descent for étale coverings—show that P' acts trivially on $\pi_1^{(p)}(X_\eta, \bar{x})$.

5.3.2. If X_η is normal and U is an open subset, then $\pi_1(U)$ maps onto $\pi_1(X_\eta)$; we may therefore suppose that X_η is smooth and affine. By cutting with generic hyperplane sections (which replaces S by its strict localization at the generic point of the special fibre of an $\mathbb{A}_S^n$) and applying Bertini, we reduce to the case in which X_η is smooth, geometrically connected, and of dimension one.

After a finite extension of $k(\eta)$, we may suppose that X_η is the complement in a smooth projective curve $\bar{X}_\eta$ of a finite set of rational points x_i , whose number we are free to increase. By the semistable reduction theorem (5.1.1), we reduce to the case in which there is a stable curve $X = (C; x_1, \dots, x_n)$ over S such that

$$X_\eta = C_\eta - \{x_1, \dots, x_n\}.$$

We then apply the following result.

Theorem 5.4. Let S be a strictly local scheme, let $f : \bar{X} \rightarrow S$ be a proper flat morphism whose special fibre is a curve with ordinary double points as its only non-smooth points, and let Y be a subscheme of $\bar{X}$ which is the union of a finite number of disjoint sections contained in the smooth locus. Let x be a rational point of $X = \bar{X} - Y$. Let U be the open subset of S over which X is smooth, and let ξ be a geometric point of U . Then the image of $\pi_1(U, \xi)$ in $\text{Aut}(\pi_1^{(p)}(X_\xi, x\xi))$ is abelian and prime to p .

Observe that $\pi_1(U, \xi)$ can act on $\pi_1^{(p)}(X_\xi, x\xi)$ only because the groups $\pi_1^{(p)}$ satisfy the specialization theorem.

We reduce first to the case in which S is complete, and then to the universal case in which S is a formal moduli scheme for the special fibre equipped with its sections. Then S is the spectrum of a formal power-series ring over a Cohen ring:

$$S = \text{Spec}(W[[t_1, \dots, t_n]]),$$

and U is the complement of a normal-crossings divisor with equation $t_1 \cdots t_m = 0$. In this universal case, Abhyankar's lemma shows that $\pi_1(U, \xi)$ is abelian and prime to p , which proves the theorem.

Remark 5.5. One can prove a result analogous to 5.4 in arbitrary relative dimension by taking a generic plane section of dimension one and applying Bertini.

6. Appendix: arithmetic proof of the stable reduction theorem (by P. Deligne)

Let S be a trait and let A_η be an abelian variety over $k(\eta)$, of dimension $d \neq 0$. For a local morphism of traits $S' \rightarrow S$, denote by $A_{\eta'}$ the abelian variety over $k(\eta')$ obtained from A_η by extension of scalars. Let $\mathcal{A}_{S'}$ be the Néron model of $A_{\eta'}$ over S' , let $\mathcal{A}_{S', s'}$ be its special fibre, and let $\mathcal{A}_{S', s'}^\circ$ be the identity component of $\mathcal{A}_{S', s'}$. The stable reduction theorem is as follows.

Theorem 6.1. For a suitable S' , $A_{\eta'}$ has stable reduction, i.e. $\mathcal{A}_{S', s'}^\circ$ is an extension of an abelian variety by a torus.

It follows fairly readily from 6.1 that one may choose S' such that $k(\eta')$ is a finite separable extension of $k(\eta)$. Suppose first that the residue field of S is of finite type over the prime field (the same argument would work if $k(s)$ were radicial over such a field).

Let ℓ be a prime different from the residual characteristic exponent p , let $T_\ell(A)$ be the Tate module, let $V_\ell(A) = T_\ell(A) \otimes \mathbf{Q}_\ell$, and let

$$\rho : \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{GL}(T_\ell(A))$$

be the natural representation. After a preliminary extension of scalars, we are reduced to the case in which the inertia group I acts unipotently (1.2), and hence through its largest pro- ℓ quotient $\mathbf{Z}_\ell(1)$ (0.3). Let T be the image in $\text{GL}(T_\ell(A))$ of a generator of $\mathbf{Z}_\ell(1)$, and let r be the largest integer ≥ 0 such that $(T - 1)^r \neq 0$. We have:

Lemma 6.2. The map

$$(\sigma, v) \in I \times V_\ell(A) \longmapsto (\rho(\sigma) - 1)^r(v)$$

induces the morphism and the isomorphism in the commutative diagram

$$(6.2.1) \quad \begin{array}{ccc} V_\ell(A)_I \otimes \mathbf{Q}_\ell(r) & \xrightarrow{M} & V_\ell(A)^I \\ \downarrow & & \uparrow \\ (V_\ell(A)/\text{Ker}(T - 1)^r) \otimes \mathbf{Q}_\ell(r) & \xrightarrow{\sim} & \text{Im}(T - 1)^r \end{array}$$

$V_\ell(A)_I$ and $V_\ell(A)^I$ are $\text{Gal}(\bar{s}/s)$ -modules, and M is Galois-equivariant.

6.3. An ℓ -adic representation

$$\tau : \text{Gal}(\bar{s}/s) \longrightarrow \text{GL}(V)$$

is said to be of weight n ($n \in \mathbf{Z}$) if the following condition is satisfied:

(*) For a suitable model X of $k(s)$ (that is, an irreducible variety of finite type over the prime field such that $k(s) = k(X)$), τ factors through $\pi_1(X, \bar{s})$ and, for every closed point x of X , the eigenvalues of $\tau(F_x)$ are algebraic numbers all of whose complex conjugates have absolute value $|k(x)|^{n/2}$ (here F_x denotes the geometric Frobenius ϕ_x^{-1}).

If V (respectively W) has weight n (respectively m) and $n \neq m$, then $\text{Hom}_{\text{Gal}}(V, W) = 0$.

6.4. It follows from the universal property of the Néron model that $V_\ell(A)^I = V_\ell(\mathcal{A}_{S,s}^\circ)$. If a is the dimension of the largest abelian quotient of $\mathcal{A}_{S,s}^\circ$ and m is the dimension of its multiplicative part, then, by Weil, $V_\ell(A)^I$ is an extension of a representation of dimension $2a$ and weight -1 by a representation of dimension m and weight -2 .

6.5. Let A^* be the abelian variety dual to A . Recall that $V_\ell(A^*)$ and $V_\ell(A)$ (and hence $V_\ell(A^*)^I$ and $V_\ell(A)_I$) are in duality with values in $\mathbf{Q}_\ell(1)$. If a^* and m^* are defined for A^* as a and m are for A , it follows from 6.4 that $V_\ell(A)_I$ is an extension of a representation of dimension m^* and weight 0 by a representation of dimension $2a^*$ and weight -1 .

6.6. The Galois module $\mathbf{Q}_\ell(r)$ has weight $-2r$. Since $\text{Im}(T - 1)^r \neq 0$, it follows from 6.4, 6.5, and the isomorphism in 6.2 that $r \leq 1$. If $r = 0$, then

$$\begin{cases} a + m \leq d, \\ 2a + m = 2d, \end{cases}$$

so $m = 0$, $a = d$, and A_η has good reduction. Suppose that $r = 1$. It then follows from 6.2 that $\text{Im}(T - 1)$ has weight -2 and that $V_\ell(A)/V_\ell(A)^I$ has weight 0 ; these spaces have the same dimension $m_1 \leq m$.

We then have

$$\dim V_\ell(A)^I = 2d - m_1 = 2a + m,$$

and

$$2 \dim \mathcal{A}_{S,s} \geq 2(a + m) \geq 2a + m + m_1 = 2d = 2 \dim \mathcal{A}_{S,s},$$

so A_η has stable reduction ($\dim \mathcal{A}_{S,s} = a + m$). At the same time, we have obtained $m_1 = m$; that is,

$$\mathrm{Im}(T - 1) \subset V_\ell(A)^I \simeq V_\ell(\mathcal{A}_{S,s}^\circ)$$

is the V_ℓ of the multiplicative part of $\mathcal{A}_{S,s}^\circ$.

6.7. To treat the general case, we shall use (0.5) (cf. 1.3). We reduce to the case in which:

- (a) the conditions of (0.5.2) or (0.5.3) are satisfied;
- (b) $\mathrm{Gal}(\bar{\eta}/\eta)$ acts trivially on A_ℓ ;
- (c) the action of I on $T_\ell(A)$ is unipotent.

Under these hypotheses, we shall prove that A_η has stable reduction. Hypothesis (a) allows us to apply (0.5.1) and write $S = \varprojlim S_i$ as in 1.3, whose notation we retain. Let s_i be the closed point of S_i .

For sufficiently large i , the identity component $\mathcal{A}^\circ$ of the Néron model $\mathcal{A}_S$ descends to a smooth group scheme $\mathcal{A}_i$ with connected fibres over S_i , and $(\mathcal{A}_i)_\ell$ is constant over $S_i - D_i$. The group $\pi_1(S_i - D_i, \bar{\eta})$ acts on $T_\ell(A)$ and acts trivially on $T_\ell(A)/\ell T_\ell(A) \simeq A_\ell$. As in 1.3, one sees that P_i acts trivially. We then have, by (1.3.1),

$$T_\ell(A)_I = T_\ell(A)_{I_i}, \quad T_\ell(\mathcal{A}_{i,s_i}) = T_\ell(\mathcal{A}_s^\circ) = T_\ell(A)^I = T_\ell(A)^{I_i},$$

and (6.2.1) is a diagram of $\mathrm{Gal}(\bar{s}_i/s_i)$ -modules to which the arguments of 6.4–6.6 may be applied.

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Exposé II

Finiteness properties of the fundamental group

by Michèle Raynaud

Let k be a separably closed field of characteristic $p \geq 0$, let X be a connected scheme of finite type over k , and let $\Pi_1^{(p)}(X)$ be the largest “prime-to- p ” quotient of $\Pi_1(X)$. In this exposé we prove that, when X is a surface, $\Pi_1^{(p)}(X)$ is topologically finitely presented, and that the same holds for arbitrary X if resolution of singularities is assumed.

For a surface, the proof method, due to J.-P. Murre, consists in reducing to a fibration over P^1 having properties (i), (ii), and (iii) of the following proposition.

Proposition 2.1. Let k be an algebraically closed field and let S be a projective, irreducible, smooth surface over k . Then there exists a blow-up

$$g : S' \longrightarrow S,$$

where S' is a regular surface equipped with a fibration

$$f : S' \longrightarrow P^1,$$

such that f has a section σ and the following conditions hold:

- (i) f is smooth at the points of $\sigma(P^1)$.
- (ii) The fibres of f are geometrically integral curves having only ordinary singularities.
- (iii) There exists a nonempty open subset U of P^1 such that the fibres of f over the points of U are smooth.

The proposition follows from the existence of a projective embedding $S \longrightarrow P^r$ for which a Lefschetz pencil exists (XVII 2.5). Indeed, let $D = \{H_t\}_{t \in P^1}$ be such a pencil of hyperplanes. Recall that it has the following properties (loc. cit. 2.2):

- (a) The axis Δ of D meets S transversely; the closed subscheme $\Delta \cap S$ is therefore reduced and consists of finitely many closed points $x_1, \dots, x_n$ of S .
- (b) There exists a nonempty open subset U of P^1 such that, for every $t \in U$, H_t meets S transversely.
- (c) For every $t \in P^1 - U$, H_t meets S transversely except at one point, which is an ordinary quadratic singularity.

Through every point x of S not belonging to Δ there passes a unique hyperplane H_t . The map assigning $t \in P^1$ to x is a rational map $a : S \dashrightarrow P^1$ with domain of definition $S - (\Delta \cap S)$.

Consider the blow-up $g : S' \longrightarrow S$ of the closed subscheme $S \cap \Delta$. The scheme S' is regular, and the fibre S'_{x_i} over x_i is the projective bundle defined by the tangent space of S at x_i , hence is isomorphic to P^1 (EGA IV 19.4.2, 19.4.4). Moreover, the rational map $f = ag$ is everywhere defined (XVIII 3.1). Finally, to each x_i one can associate the section of f that sends $t \in P^1$ to the tangent vector at x_i of $H_t \cap S$.

Let us verify conditions (i), (ii), and (iii). For each $t \in P^1$, the curves S'_t and $S_t = H_t \cap S$ are isomorphic. Indeed, the morphism $S'_t \longrightarrow S_t$ is an isomorphism except possibly at the points x_i , where S_t is regular; moreover, since S'_t is a complete intersection in S' , it is integral, and is therefore isomorphic to S_t .

The asserted properties of the fibres follow from (a)–(c). Finally, f is smooth at the points of σ , since f is flat and the fibres S'_t are geometrically regular at $\sigma(t)$.

Corollary 2.2. With the notation of 2.1, let Y be a normal-crossings divisor in S , and let $Y' = Y \times_S S'$. Then f , g , and U may be chosen so that $Y' \times_{P^1} U$ is a divisor with normal crossings relative to U .

It is known that, if a Lefschetz pencil exists for an embedding $S \longrightarrow P^r$, the Lefschetz pencils form an open subset of the variety of lines in P^r (XVII 3.2.1). The same is true of the set of

Lefschetz pencils whose axis does not meet Y and for which there is a nonempty open subset U_1 of P^1 such that $Y \cap H_t$ is smooth for $t \in U_1$. It is then enough to construct f and g from a pencil satisfying these two conditions and replace U by $U \cap U_1$.

2.3.0. Let G be a profinite group. Recall that G is said to be topologically finitely generated if there is an integer n such that G is isomorphic to a quotient of the free pro-group $F(n)$ on n generators. Put $K = \text{Ker}(F(n) \rightarrow G)$. The group G is said to be topologically finitely presented if, in addition, there are finitely many elements $k_1, \dots, k_r$ of K such that the smallest closed normal subgroup of K containing $k_1, \dots, k_r$ is K .

Theorem 2.3.1. Let k be a separably closed field of characteristic $p \geq 0$, and let X be a connected scheme of finite type over k . Suppose that schemes of finite type and dimension at most $\dim X$ over an algebraic closure of k admit strong resolution of singularities (SGA 5 I 3.1.5). This condition is satisfied when $\dim X = 2$, by [1], or when $\text{char } k = 0$, by [2]. Then the fundamental group $\Pi_1^{(p)}(X)$ is topologically finitely presented.

By SGA 1 IX 4.10, we may suppose that k is algebraically closed.

1) Reduction to the case in which X is a regular surface.

If $f : X' \rightarrow X$ is a morphism of finite type, put $X'' = X' \times_X X'$, and let $(X'_a)_{a \in A}$ and $(X''_b)_{b \in B}$ be the families of connected components of X' and X'' , respectively. When f is an effective descent morphism for the category of finite étale coverings, the fundamental group $\Pi_1(X)$ can be computed explicitly

in terms of the fundamental groups $\Pi_1(X'_a)$ and $\Pi_1(X''_b)$. It follows (SGA 1 IX 5.2, 5.3) that if the $\Pi_1(X'_a)$ are topologically finitely generated, then so is $\Pi_1(X)$; and if the $\Pi_1(X'_a)$ are topologically finitely presented while the $\Pi_1(X''_b)$ are topologically finitely generated, then $\Pi_1(X)$ is topologically finitely presented.

Since X admits resolution of singularities, there is a regular scheme X' and a proper birational morphism $f : X' \rightarrow X$. Since f is an effective descent morphism for finite étale coverings, it suffices to prove the theorem under the additional assumption that X is regular. One first deduces that $\Pi_1(X)$ is topologically finitely generated; knowing that $\Pi_1(X')$ is topologically finitely presented and that the $\Pi_1(X''_b)$ are topologically finitely generated then shows that $\Pi_1(X)$ is topologically finitely presented.

Now let $(U_i)_{i \in I}$ be a finite affine open covering of X , and let $f : \bigsqcup U_i \rightarrow X$ be the canonical morphism. Since f is an effective descent morphism for finite étale coverings, it is enough to prove the theorem for the U_i . We are thus reduced to the case in which X is affine and smooth.

Assume henceforth that X is affine and smooth over k , with $\dim X > 2$. The Lefschetz theorem for quasi-projective schemes ([3] V 7.4) applies: if Y is a sufficiently general hyperplane section of X , then

$$\Pi_1(Y) \simeq \Pi_1(X).$$

It therefore suffices to prove the theorem for Y , reducing us to the case $\dim X = 2$.

2) The case in which X is a smooth surface over k .

By the strong resolution theorem for surfaces, there exist a smooth, integral, projective surface S and an open immersion $i : X \rightarrow S$ such that $Y = S - X$ is a normal-crossings divisor. We apply Corollary 2.2 and retain its notation.

We show that it is enough to prove that the fundamental group of $X' = X \times_S S'$ is topologically finitely presented. Since X and X' are regular and

$$\text{codim} \left(\bigcup_{1 \leq i \leq n} \{x_i\}, X \right) \geq 2,$$

every étale covering of X' induces on $X - \bigcup_{1 \leq i \leq n} \{x_i\}$ an étale covering which extends uniquely to an étale covering of X (SGA 2 1.11). This gives an isomorphism

$$\Pi_1(X') \simeq \Pi_1(X).$$

Consider the cartesian diagram

$$\begin{array}{ccc} X' & \longleftarrow & X'_U \\ \downarrow h & & \downarrow h_U \\ P^1 & \longleftarrow & U \end{array}$$

where $h = f|_{X'}$. Let L be the set of primes different from p , and let $\bar{\eta}$ be a geometric point over the generic point η of P^1 . We verify the hypotheses of SGA 1 XIII 4.8. The morphism h has a section σ , because f does. To prove condition (a) of SGA 1 XIII 4.7, note first that, since the fibres of h are geometrically integral, h is 0-acyclic and locally 0-acyclic (SGA 4 XV 4.1, 1.16). It remains to show that, for every locally constant sheaf of sets F on X' , (F, h) is cohomologically proper in dimensions ≤ 0 . If ξ is a closed point of P^1 and Z denotes the integral closure of $\text{Spec } \mathcal{O}_{P^1, \xi}$ in $\bar{\eta}$, then $X'_Z = X' \times_{P^1} Z$ satisfies condition (S_2) at its closed points and is regular at every other point, hence is normal. Consequently, an étale covering of X'_Z whose restriction to $X'_{\bar{\eta}}$ is connected is necessarily connected. This proves the assertion, and hence condition (a).

Since X'_U is the complement in S'_U of a divisor with normal crossings relative to U , h_U is locally 1-aspherical for L (SGA 4 XV 2.1); and, for every finite constant sheaf of L -groups F on X'_U , (F, h_U) is cohomologically proper in dimensions ≤ 1 (SGA 1 XIII 2.4), which gives condition (b). Since P^1 is normal and $T = P^1 - U$ has codimension 1, we have

$$\text{prof ét}_T(P^1) \geq 2,$$

which gives condition (c). The condition

$$\text{prof hop}_x(X') \geq 3$$

holds at every closed point of X' by the purity theorem, which gives condition (e). Finally, for every $t \in P^1$, h is smooth at $\sigma(t)$, and hence is locally 1-aspherical for L at $\sigma(t)$, which gives condition (d).

By SGA 1 XIII 4.7, if a is a geometric point of $X'_{\bar{\eta}}$, there is an exact sequence

$$1 \longrightarrow \Pi_1^{(p)}(X'_{\bar{\eta}}, a) \longrightarrow \Pi_1'(X'_U, a) \longrightarrow \Pi_1(U, a) \longrightarrow 1,$$

and the choice of the section σ defines a splitting of this exact sequence. Let K be the image of $\Pi_1(U, a)$ in $\text{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$. The group of coinvariants under K ,

$$\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K,$$

is the quotient of $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ by the closed normal subgroup generated by the elements $x^{-1}q(x)$, where $x \in \Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ and $q \in K$. By loc. cit. (4.7.4), there is an isomorphism

$$\Pi_1^{(p)}(X', a) \simeq \Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K.$$

Write $P^1 - U = \coprod_{1 \leq i \leq n} \{t_i\}$. For each i , let $\bar{T}_i$ be the strict localization of P^1 at a geometric point $\bar{t}_i$ over t_i , and let s_i be the generic point of $\bar{T}_i$. An étale covering of U whose restriction to every s_i is trivial extends to an étale covering of P^1 , and is therefore trivial. For each i , let $\bar{k}(s_i)$ be an algebraic closure of $k(s_i)$. The preceding assertion says that $\Pi_1(U, a)$ is the smallest closed normal subgroup containing the images of the Galois groups $\text{Gal}(\bar{k}(s_i)/k(s_i))$ in $\Pi_1(U, a)$. By I 5.3, the image K_i of $\text{Gal}(\bar{k}(s_i)/k(s_i))$ in $\text{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$ is finite. Since K is the smallest closed normal subgroup containing all the K_i , and since $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ is topologically finitely presented (SGA 1 XIII 3.2), this proves that $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K$, and hence also the isomorphic group $\Pi_1^{(p)}(X)$, is topologically finitely presented.

Remark 2.3.2. Resolution of singularities is not needed to prove Theorem 2.3.1 when X is the open complement of a normal-crossings divisor in a projective scheme smooth over k .

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Exposé VI

FORMAL DEFORMATION THEORY

by D. S. Rim

1. Formal existence theorem

We recall the notion of a cofibered category (S.G.A. 1 IV). Let $\underline{C}$ be a fixed category. A cofibered category $\underline{A}$ over $\underline{C}$ is, by definition, a category $\underline{A}$ together with a functor $P : \underline{A} \rightarrow \underline{C}$ such that:

Ax. 1. For every map f in $\underline{C}$ and every object a in $\underline{A}$ for which $P(a)$ is the source of f , there is a cocartesian f -morphism $\alpha : a \rightarrow b$. Thus, for every f -morphism $\alpha' : a \rightarrow b'$, there is a unique T -morphism $\tau : b \rightarrow b'$ in $\underline{A}$ such that $\alpha' = \tau\alpha$, where T is the target of f .

Ax. 2. A composite of cocartesian morphisms in $\underline{A}$ is again cocartesian.

The following properties of a cofibered category $P : \underline{A} \rightarrow \underline{C}$ follow immediately from the definitions.

(Cancellation) Let $a \xrightarrow[\beta]{\alpha} a'$ be two cocartesian maps such that $P(\alpha) = P(\beta)$. If there is a cocartesian map $b \xrightarrow{\gamma} a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

(Factorization) Given cocartesian maps $a \xrightarrow{\alpha} a'$ and $a \xrightarrow{\beta} a''$, there is a cocartesian map $a' \xrightarrow{\gamma} a''$ such that $\gamma\alpha = \beta$ if and only if there is a map $f : P(a') \rightarrow P(a'')$ in $\underline{C}$ such that $fP(\alpha) = P(\beta)$. Moreover, γ is unique.

Let $\underline{A}$ be a cofibered category over $\underline{C}$. For each object S of $\underline{C}$, denote by $\underline{A}(S)$ the subcategory of $\underline{A}$ whose objects are the objects a for which $P(a) = S$, and in which

$$\mathrm{Hom}_{\underline{A}(S)}(a, a') = \{\alpha \in \mathrm{Hom}_{\underline{A}}(a, a') \mid P(\alpha) = \mathrm{id}_S\}.$$

A cofibered groupoid over $\underline{C}$ is a cofibered category $\underline{A}$ over $\underline{C}$ such that $\underline{A}(S)$

is a groupoid for every object S of $\underline{C}$. Recall that a groupoid is a category in which every morphism is an isomorphism. Thus a cofibered groupoid over $\underline{C}$ is a cofibered category $P : \underline{A} \rightarrow \underline{C}$ in which every map of $\underline{A}$ is cocartesian. Furthermore, for every $\alpha \in \mathrm{Fl}(\underline{A})$, α is an isomorphism in $\underline{A}$ if and only if $P(\alpha)$ is an isomorphism.

Throughout the rest of this section, fix a complete noetherian local ring Λ with residue field k , and let $\underline{C}_\Lambda$ be the category of artinian local Λ -algebras with residue field k , whose maps are local homomorphisms over Λ . From 1.2 onward, a cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is always assumed to satisfy $\underline{A}(k) = \{e\}$, where $\{e\}$ denotes the category in which $\mathrm{Hom}_{\{e\}}(a, b)$ consists of exactly one element for every two objects a, b of $\{e\}$. A map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will simply be called a $\underline{C}_\Lambda$ -map if $P(a') = P(a)$ and $P(\alpha) = 1_{P(a)}$. By abuse of language, a map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called surjective if $P(\alpha) : P(a') \rightarrow P(a)$ is surjective in $\underline{C}_\Lambda$.

Ex. 1.1(a). Let $F : \underline{C} \rightarrow (\text{categories})$ be a functor. Define the category $\underline{F}$ as follows. An object of $\underline{F}$ is a pair (S, a) with $S \in \mathrm{ob}(\underline{C})$ and $a \in \mathrm{ob}F(S)$; a map $(S, a) \rightarrow (S', a')$ is a pair (f, u') , where $f : S \rightarrow S'$ is a map in $\underline{C}$ and $u' : F(f)a \rightarrow a'$ is a map in $F(S')$. The composite of

$$(S, a) \xrightarrow{(f, u')} (S', a') \xrightarrow{(g, u'')} (S'', a'')$$

is

$$(gf, u'' \cdot F(g)u') : (S, a) \rightarrow (S'', a'').$$

With respect to the functor $P : \underline{F} \rightarrow \underline{C}$ given by $(S, a) \mapsto S$, $\underline{F}$ becomes a cofibered category over $\underline{C}$. If $F(S)$ is a groupoid for every $S \in \mathrm{ob}(\underline{C})$, then $\underline{F}$ is a cofibered groupoid over $\underline{C}$. In particular, a functor $F : \underline{C} \rightarrow (\text{groups})$ yields a cofibered group, and a functor $F : \underline{C} \rightarrow (\text{Ens})$ yields a cofibered discrete groupoid. A natural transformation $\varphi : F \rightarrow G$, where $F, G : \underline{C} \rightarrow (\text{categories})$, induces a cocartesian functor $\varphi : \underline{F} \rightarrow \underline{G}$ over $\underline{C}$.

Ex. 1.1(b). Let X and Y be schemes over Λ together with a fixed

isomorphism

$$X \otimes_{\Lambda} k \xleftarrow{\tau} Y \otimes_{\Lambda} k.$$

Then the functor $\text{Isom}_{X,Y} : \underline{C}_{\Lambda} \longrightarrow (\text{groupoids})$ given by

$$\text{Isom}_{X,Y}(S) = \left\{ X \otimes_{\Lambda} S \xleftarrow{\sigma} Y \otimes_{\Lambda} S \mid \sigma \otimes_{\Lambda} k = \tau \right\}$$

yields a cofibered groupoid $P : \underline{\text{Isom}}_{X,Y} \longrightarrow \underline{C}_{\Lambda}$.

Ex. 1.1(c). Let X_0 be a fixed scheme over k , and let $\underline{M}_{X_0}$ be the category of deformations of X_0 over $\underline{C}_{\Lambda}$ (see §4). Thus an object of $\underline{M}_{X_0}$ is a pair (R, X) , where $R \in \underline{C}_{\Lambda}$ and X is a deformation of X_0 to R . Then the functor $P : \underline{M}_{X_0} \longrightarrow \underline{C}_{\Lambda}$ given by $(R, X) \longmapsto R$ defines a cofibered groupoid over $\underline{C}_{\Lambda}$.

Ex. 1.1(d). For any groupoid X , denote by $[X]$ the discrete groupoid consisting of the isomorphism classes of objects of X . If $P : \underline{F} \longrightarrow \underline{C}_{\Lambda}$ is a cofibered groupoid, we obtain a functor $[F] : \underline{C}_{\Lambda} \longrightarrow (\text{Ens})$ given by $[F](S) = [F(S)]$, and hence a cofibered discrete groupoid which, by abuse of notation, is again denoted by $[F]$. Definition 1.2. A cofibered groupoid $\underline{A}$ over $\underline{C}_{\Lambda}$ is called semi-homogeneous if the following properties hold:

(1) Every diagram

$$\begin{array}{ccc} a'' & \text{-----} & a' \\ \vdots & & \downarrow \\ a_1 & \text{-----} & a \end{array}$$

of solid arrows in $\underline{A}$, where $a' \longrightarrow a$ is surjective, can be completed to a commutative diagram by including the dotted arrows.

(2) Let

$$\begin{array}{ccc} & & R \\ & \nearrow \pi_1 & \\ R \times_k k[\varepsilon] & & \\ & \searrow \pi_2 & \\ & & k[\varepsilon] \end{array}$$

be the projection maps, where $k[\varepsilon]$ is the ring of dual numbers over k , and let

$$\begin{array}{ccc} a' & & \\ & \searrow \alpha & \\ & & a \\ & \nearrow \beta & \\ b' & & \end{array}$$

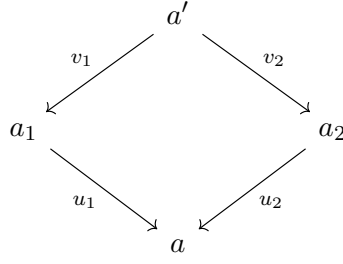
be π_1 -maps in $\underline{A}$. Then there is a $\underline{C}_{\Lambda}$ -map $\gamma : a' \longrightarrow b'$ such that $\alpha = \beta\gamma$ if and only if $(\pi_2)_*(a') \simeq (\pi_2)_*(b')$.

Remark 1.3(a). A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. In particular, 1.2(2) implies that the canonical map

$$[\underline{A}(R \times_k k[\varepsilon])] \longrightarrow [\underline{A}(R)] \times [\underline{A}(k[\varepsilon])]$$

is bijective.

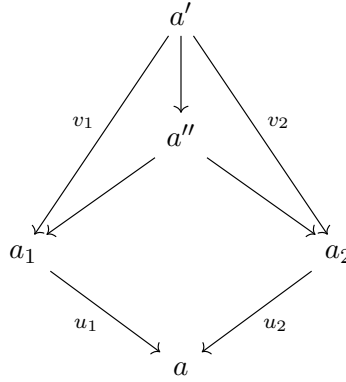
Remark 1.3(b). Consider a commutative diagram



in a cofibered category $\underline{A}$ over $\underline{C}_\Lambda$. Put $R = P(a)$, $R' = P(a')$, and $R_i = P(a_i)$ for $i = 1, 2$. If a'' is the direct image under the map

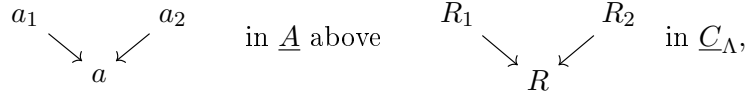
$$(P(v_1), P(v_2)) : R' \longrightarrow R_1 \times_R R_2,$$

we obtain a commutative diagram

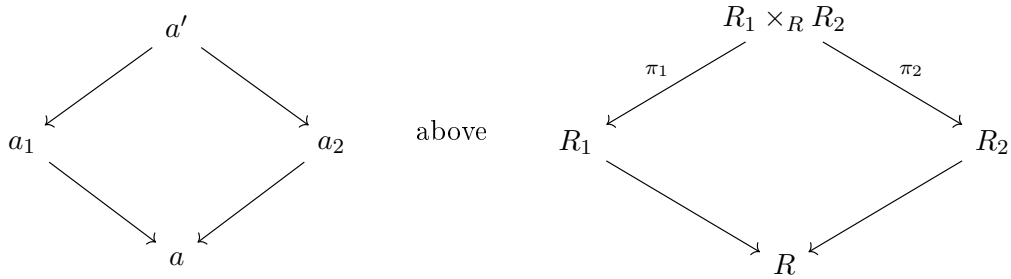


Thus 1.2(1) may be replaced by:

(1)' Every diagram



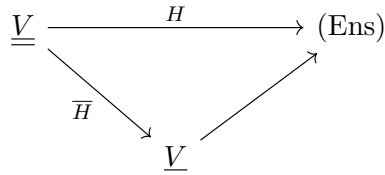
where $R_2 \longrightarrow R$ is surjective, can be completed to a commutative diagram



where π_i ($i = 1, 2$) are the projections.

Remark 1.3(c). If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, then $[\underline{A}(k[\varepsilon])]$ is canonically endowed with a vector space structure over k , where $k[\varepsilon]$ is the ring of dual numbers over k . Indeed, let $\underline{V}$ be the category of

finite-dimensional vector spaces over k . Every product-preserving functor $H : \underline{V} \longrightarrow (\text{Ens})$ factors canonically as



where $\underline{\underline{V}} \rightarrow (\text{Ens})$ is the forgetful functor. On the other hand, the functor

$$G : \underline{\underline{V}} \rightarrow \underline{\underline{C}}_\Lambda, \quad W \mapsto k[W] = k \oplus W, \quad W^2 = 0,$$

takes products in $\underline{\underline{V}}$ to fibre products over k in $\underline{\underline{C}}_\Lambda$. If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{\underline{C}}_\Lambda$, then

$$[\underline{A}(k[V] \times_k k[W])] \rightarrow [\underline{A}(k[V])] \times [\underline{A}(k[W])]$$

is bijective by 1.3(a). Hence the functor $V \mapsto [\underline{A}(k[V])]$ commutes with products, proving the assertion.

Definition 1.4. (1) Let R be an object of $\underline{\underline{C}}_\Lambda$. A small extension of R in $\underline{\underline{C}}_\Lambda$ is a surjective map $R' \xrightarrow{f} R$ in $\underline{\underline{C}}_\Lambda$ such that $(\ker f)^2 = 0$ and $\ker f \simeq k$. In other words, $R' \xrightarrow{f} R$ is a small extension of R if

$$0 \rightarrow k \xrightarrow{t} R' \xrightarrow{f} R \rightarrow 0$$

is exact for some $t \in R'$ with $t^2 = 0$.

(2) Let $\underline{A}$ be a cofibered groupoid over $\underline{\underline{C}}_\Lambda$, and let a be an object of $\underline{A}$. A small prolongation of a is an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ such that $P(\alpha)$ is a small extension in $\underline{\underline{C}}_\Lambda$. An arrow $a' \xrightarrow{\alpha} a$ will be called versal for small prolongations of a if, for every small prolongation $a'_1 \rightarrow a$, there exists a commutative diagram

$$\begin{array}{ccc} a' & \xrightarrow{\quad} & a'_1 \\ & \searrow \alpha & \swarrow \\ & a & \end{array}$$

Proposition 1.5. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{\underline{C}}_\Lambda$.

(1) Let

$$\begin{array}{ccc} & & R \\ & \nearrow \pi_1 & \\ R \times_k k[\varepsilon] & & \\ & \searrow \pi_2 & \\ & & k[\varepsilon] \end{array}$$

be the projection maps in $\underline{\underline{C}}_\Lambda$, and let $a' \xrightarrow{\alpha} a$ be a π_1 -map. Then there exists a map $a \xrightarrow{\beta} a'$ such that $\alpha\beta = 1_a$ if and only if there exists a map $a \rightarrow (\pi_2)_* a'$.

(2) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{\underline{C}}_\Lambda$, and consider a commutative diagram

$$\begin{array}{ccc} & a'' & \\ \alpha' \swarrow & & \searrow \alpha'_1 \\ a' & & a' \\ \alpha \searrow & & \swarrow \alpha_1 \\ & a & \end{array} \quad \text{above} \quad \begin{array}{ccc} & R' \times_R R' & \\ \swarrow & & \searrow \\ R' & & R' \\ \searrow & & \swarrow \\ & R & \end{array}$$

Define the canonical map $\tilde{f} : R' \times_R R' \rightarrow k[\ker f]$ by

$$\tilde{f}(r'_1, r'_2) = r'_1(0) + (r'_1 - r'_2).$$

If there exists a map $a' \rightarrow (\tilde{f})_*(a'')$, then there exists a map $\gamma : a' \rightarrow a'$ such that $\alpha = \alpha_1\gamma$.

Proof. (1) The “only if” part is clear. Assume conversely that there exists a map $\varphi : a \longrightarrow (\pi_2)_* a'$. Set

$$h : R \longrightarrow R \times_k k[\varepsilon], \quad h = 1 \times P(\varphi).$$

The composite $\pi_1 h$ is the identity. We therefore obtain maps

$$a \xrightleftharpoons[\alpha_1]{\gamma} h_* a, \quad \alpha_1 \gamma = 1_a, \quad P(\gamma) = h, \quad P(\alpha_1) = \pi_1.$$

On the other hand, $\pi_2 h = P(\varphi)$, so there exists a π_2 -map

$$h_* a \longrightarrow (\pi_2)_* a'.$$

Thus $(\pi_2)_*(h_* a) = (\pi_2)_* a'$, and 1.2(2) gives a $\underline{C}_\Lambda$ -map $u : h_* a \longrightarrow a'$ such that $\alpha u = \alpha_1$. Define $\beta = u\gamma$. Then

$$\alpha\beta = \alpha u\gamma = \alpha_1 \gamma = 1_a.$$

(2) The map

$$1 \times \tilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\ker f]$$

is an isomorphism compatible with the projections onto the first factor, and $\pi_2(1 \times \tilde{f}) = \tilde{f}$, where $\pi_2 : R' \times_k k[\ker f] \longrightarrow k[\ker f]$ is the second projection. Since $R' \xrightarrow{f} R$ is a small extension, $k[\ker f] \simeq k[\varepsilon]$. Hence, if there exists a map

$$a' \longrightarrow (\tilde{f})_* a'' = (\pi_2)_*((1 \times \tilde{f})_* a''),$$

part (1) yields a map $\gamma' : a' \longrightarrow a''$ such that $\alpha'\gamma' = 1_{a'}$. Define $\gamma = \alpha'_1 \gamma' : a' \longrightarrow a'$. Then $\alpha_1 \gamma = \alpha$.

Corollary 1.6. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

$$\begin{array}{ccc} a' & & a_1 \\ & \searrow \alpha & \swarrow \alpha_1 \\ & a & \end{array}$$

in $\underline{A}$, where α_1 is a small prolongation. Assume that a is versal for small prolongations of e . Then there exists a map $\rho : a' \longrightarrow a_1$ such that $\alpha = \alpha_1 \rho$ if and only if there exists a map $f : P(a') \longrightarrow P(a_1)$ such that $P(\alpha) = P(\alpha_1)f$ in $\underline{C}_\Lambda$.

Proof. Assume that $P(\alpha) = P(\alpha_1)f$, where $f : P(a') \longrightarrow P(a_1)$. Replacing a' by its direct image under f , we may and shall assume that $P(\alpha) = P(\alpha_1)$. Put $R' = P(a_1)$ and $R = P(a)$. Since $\underline{A}$ is semi-homogeneous, we obtain a commutative diagram

$$\begin{array}{ccc} & a'' & \\ \beta \swarrow & & \searrow \beta_1 \\ a' & & a_1 \\ \alpha \swarrow & & \searrow \alpha_1 \\ & a & \end{array} \quad \text{above} \quad \begin{array}{ccc} & R' \times_R R' & \\ \swarrow & & \searrow \\ R' & & R' \\ \swarrow & & \searrow \\ & R & \end{array}$$

Since a is versal for small prolongations of e , so is a' ; the assertion therefore follows from 1.5(2).

Definition 1.7. For each object R of $\underline{C}_\Lambda$, put

$$\bar{\Omega}_R = M/(m_\Lambda R + M^2),$$

where m_Λ and M are the maximal ideals of Λ and R , respectively. This is a finite-dimensional vector space over k , and $R \mapsto \overline{\Omega}_R$ defines a functor

$$\overline{\Omega} : \underline{C}_\Lambda \longrightarrow \underline{V},$$

where $\underline{V}$ is the category of finite-dimensional vector spaces over k . If $\underline{A}$ is a cofibered category over $\underline{C}_\Lambda$, the composite functor $\underline{A} \xrightarrow{P} \underline{C}_\Lambda \longrightarrow \underline{V}$ is again denoted $\overline{\Omega}$, by abuse of notation.

Proposition 1.8. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$. Given an object a of $\underline{A}$ that is versal for small prolongations of e , there exists an arrow $a' \longrightarrow a$ such that:

- (i) a' is versal for small prolongations of a ;
- (ii) $P(a') \longrightarrow P(a)$ is surjective, and its kernel is annihilated by the maximal ideal of $P(a)$;
- (iii) $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

Proof. Put $R = P(a)$, and consider an exact sequence

$$0 \longrightarrow J \longrightarrow S \longrightarrow R \longrightarrow 0,$$

where S is a formal power-series ring over Λ in finitely many variables and $J \subset M^2$, with M the maximal ideal of S . Let Σ be the set of ideals J_1 of S such that

- (i) $J \supset J_1 \supset MJ$, and
- (ii) there exists a π_1 -morphism $a_1 \longrightarrow a$ in $\underline{A}$, where $\pi_1 : S/J_1 \longrightarrow R$ is the canonical map.

Property 1.2(1) of $\underline{A}$, together with the fact that J/MJ is a finite-dimensional vector space over k , implies that Σ has a smallest ideal. Denote it by J' , and choose a π' -morphism $a' \xrightarrow{g} a$, where $\pi' : S/J' \longrightarrow R$ is the canonical map. Since $J' \subset J \subset M^2$, the map $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

We claim that $a' \xrightarrow{g} a$ is versal for small prolongations of a . Indeed, let $a_1 \longrightarrow a$ be a small prolongation, and put $R_1 = P(a_1)$. We obtain an exact commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & J/MJ & \longrightarrow & S/MJ & \longrightarrow & R \longrightarrow 0 \\ & & \downarrow h & & \downarrow h_1 & & \parallel \\ 0 & \longrightarrow & k & \longrightarrow & R_1 & \longrightarrow & R \longrightarrow 0. \end{array}$$

If $h = 0$, then $\pi_1 : R_1 \longrightarrow R$ admits a cross-section, so h_1 induces a map $f : S/J' \longrightarrow R_1$. If $h \neq 0$, then $h_1 : S/MJ \longrightarrow R_1$ is surjective and hence, by the definition of J' , again induces $f : S/J' \longrightarrow R_1$. In either case we obtain a commutative diagram

$$\begin{array}{ccc} S/J' & \xrightarrow{f} & R_1 \\ & \searrow \pi' & \swarrow \pi_1 \\ & R & \end{array}$$

Since a is versal for small prolongations of e , Corollary 1.6 gives a commutative diagram

$$\begin{array}{ccc} a' & \xrightarrow{\rho} & a_1 \\ & \searrow & \swarrow \\ & a & \end{array}$$

as required.

Let $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ be a cofibered category in groupoids. It then gives a cofibered category in groupoids

$$\text{pro}(P) : \text{pro}(\underline{A}) \longrightarrow \text{pro}(\underline{C}_\Lambda)$$

(for the definition of the category of pro-objects $\text{pro}(\underline{A})$, see A. Grothendieck, Séminaire Bourbaki 195), fitting into the commutative diagram

$$\begin{array}{ccc} \underline{A} & \longrightarrow & \text{pro}(\underline{A}) \\ P \downarrow & & \downarrow \text{pro}(P) \\ \underline{C}_\Lambda & \longrightarrow & \text{pro}(\underline{C}_\Lambda). \end{array}$$

Let $\widehat{\underline{C}}_\Lambda$ be the full subcategory of $\text{pro}(\underline{C}_\Lambda)$ whose objects are systems $(R_i)_{i \in \mathbb{N}}$ with $R_i \in \underline{C}_\Lambda$ such that each map $\varphi_i : R_i \rightarrow R_{i-1}$ is surjective and, for all sufficiently large i , induces an isomorphism

$$\frac{m_i}{m_\Lambda R_i + m_i^2} \xrightarrow{\sim} \frac{m_{i-1}}{m_\Lambda R_{i-1} + m_{i-1}^2},$$

where $\mathbb{N}$ denotes the ordered category of natural numbers and m_i is the maximal ideal of R_i .

Let $\widehat{\underline{A}}$ be the full subcategory of $\text{pro}(\underline{A})$ defined by

$$\text{ob}(\widehat{\underline{A}}) = \left\{ a \in \text{ob}(\text{pro}(\underline{A})) \mid \text{pro}(P)(a) \in \text{ob}(\widehat{\underline{C}}_\Lambda) \right\}.$$

We obtain a cofibered category in groupoids $\widehat{P} : \widehat{\underline{A}} \rightarrow \widehat{\underline{C}}_\Lambda$ and a commutative diagram

$$\begin{array}{ccc} \underline{A} & \hookrightarrow & \widehat{\underline{A}} \\ P \downarrow & & \downarrow \widehat{P} \\ \underline{C}_\Lambda & \hookrightarrow & \widehat{\underline{C}}_\Lambda. \end{array}$$

The category $\widehat{\underline{C}}_\Lambda$ is canonically equivalent to the category of complete local Λ -algebras of formally finite type over Λ with fixed residue field k ; we henceforth make this identification.

Definition 1.9. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. An object ξ of $\widehat{\underline{A}}$ is called quasi-initial if, for every object a of $\widehat{\underline{A}}$, there exists a map $\xi \rightarrow a$ in $\widehat{\underline{A}}$. A quasi-initial object ξ is called minimal if the morphism $\xi : h_R \rightarrow [\underline{A}]$ induces a bijection

$$\xi(k[\varepsilon]) : \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \xrightarrow{\sim} [\underline{A}(k[\varepsilon])], \quad R = P(\xi).$$

An object ξ of $\widehat{\underline{A}}$ is called a versal formal object of $\underline{A}$ if every surjective map $\alpha : a' \rightarrow a$ in $\underline{A}$ induces a surjective map

$$\text{Hom}_{\widehat{\underline{A}}}(\xi, a') \rightarrow \text{Hom}_{\widehat{\underline{A}}}(\xi, a).$$

A versal formal object ξ is called minimal if $\xi : h_R \rightarrow [\underline{A}]$ induces the above bijection on $k[\varepsilon]$, where $R = P(\xi)$.

Proposition 1.10. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$.

- (i) Every (minimally) versal formal object of $\underline{A}$ is a quasi-initial (minimal) object of $\widehat{\underline{A}}$.
- (ii) Suppose that ξ is an object of $\widehat{\underline{A}}$ for which

$$\widetilde{\xi}(k[\varepsilon]) : \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \rightarrow [\underline{A}(k[\varepsilon])]$$

is injective, where $P(\xi) = R$. Then every endomorphism of ξ in $\widehat{\underline{A}}$ is an automorphism. In particular, a quasi-initial minimal object of $\widehat{\underline{A}}$, if it exists, is unique up to noncanonical isomorphism.

Proof. (i) Let ξ be a versal formal object of $\underline{A}$. For an object a of $\widehat{\underline{A}}$, choose a presentation by surjective maps in $\underline{A}$,

$$\cdots \rightarrow a_n \xrightarrow{\alpha_n} a_{n-1} \rightarrow \cdots \rightarrow a_1 \xrightarrow{\alpha_1} a_0, \quad a = \varprojlim_n a_n.$$

Suppose maps $\varphi_i : \xi \rightarrow a_i$ have been chosen with $\alpha_i \varphi_i = \varphi_{i-1}$ for all $i < n$. Since ξ is versal and α_n is surjective, there is a map $\varphi_n : \xi \rightarrow a_n$ with $\alpha_n \varphi_n = \varphi_{n-1}$. Thus $\varphi = \varprojlim_n \varphi_n$ defines a map $\xi \rightarrow a$.

(ii) Let $\alpha : \xi \rightarrow \xi$ be an endomorphism in $\widehat{\underline{A}}$. For every $\lambda \in \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon])$, we have

$$\lambda_* \xi \simeq (\lambda \circ P(\alpha))_* \xi.$$

Injectivity therefore gives $\lambda = \lambda \circ P(\alpha)$ for every λ . It follows that $P(\alpha) : R \rightarrow R$ is surjective. Every surjective endomorphism of a noetherian ring is an automorphism, so $P(\alpha)$, and hence α , is an automorphism.

Theorem 1.11 (Schlessinger). Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then $\underline{A}$ admits a minimally versal formal object if and only if:

(i) $\underline{A}$ is semi-homogeneous;

(ii) $[\underline{A}(k[\varepsilon])]$ is a finite-dimensional vector space over k (cf. 1.3(c)).

Proof. Necessity. Let $\xi \in \widehat{\underline{A}}$ be a minimally versal formal object of $\underline{A}$, and consider a diagram

$$\begin{array}{ccc} a_1 & & a_2 \\ & \searrow & \swarrow \alpha \\ & a & \end{array}$$

in $\underline{A}$, where α is surjective. Since ξ is quasi-initial, there is a map $\xi \rightarrow a_1$. Since α is surjective, the composite $\xi \rightarrow a_1 \rightarrow a$ lifts to a_2 ; hence we obtain a commutative diagram

$$\begin{array}{ccc} & \xi & \\ \swarrow & & \searrow \\ a_1 & & a_2 \\ \searrow & & \swarrow \alpha \\ & a & \end{array}$$

which proves condition 1.2(1).

We now prove condition 1.2(2). Let

$$\begin{array}{ccc} & & S \\ & \nearrow \pi_1 & \\ S \times_k k[\varepsilon] & & \\ & \searrow \pi_2 & \\ & & k[\varepsilon] \end{array}$$

be the projections, and let

$$\begin{array}{ccc} a' & & b' \\ & \searrow \alpha & \swarrow \beta \\ & a & \end{array}$$

be π_1 -maps in $\underline{A}$. By the definition of ξ , there is a commutative diagram

$$\begin{array}{ccc} & \xi & \\ \swarrow u & & \searrow v \\ a' & & b' \\ \searrow \alpha & & \swarrow \beta \\ & a & \end{array}$$

and therefore $\pi_1 P(u) = \pi_1 P(v)$. Suppose now that $(\pi_2)_* a' \simeq (\pi_2)_* b'$. Minimality of ξ gives $\pi_2 P(u) = \pi_2 P(v)$, whence

$$P(u) = \pi_1 P(u) \times \pi_2 P(u) = \pi_1 P(v) \times \pi_2 P(v) = P(v).$$

Since $\underline{A}$ is a cofibered groupoid, there is a unique isomorphism $\sigma : a' \xrightarrow{\sim} b'$ in $\underline{A}(S \times_k k[\varepsilon])$ such that $\sigma u = v$. The equality $\beta \sigma u = \alpha u$ then implies $\beta \sigma = \alpha$.

Sufficiency. Let $c_1, \dots, c_n$ be objects of $\underline{A}(k[\varepsilon])$ whose classes form a k -basis of $[\underline{A}(k[\varepsilon])]$. Choose

$$a_1 \in \underline{A}(k[\varepsilon] \times_k \cdots \times_k k[\varepsilon])$$

such that $(\pi_j)_* a_1 \simeq c_j$. Then

$$\tilde{a}_1(k[\varepsilon]) : \text{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \longrightarrow [\underline{A}(k[\varepsilon])]$$

is bijective, where $R_1 = P(a_1)$. In particular, $a_1 \longrightarrow e$ is versal for small prolongations of e . Choose $a_2 \longrightarrow a_1$ versal for small prolongations of a_1 such that $\overline{\Omega}_{a_2} \longrightarrow \overline{\Omega}_{a_1}$ is bijective. Continue inductively, choosing $a_{n+1} \longrightarrow a_n$ versal for small prolongations and inducing a bijection on $\overline{\Omega}$, and set

$$\xi = \varprojlim_n a_n.$$

Then $\overline{\Omega}_\xi \longrightarrow \overline{\Omega}_{a_1}$ is bijective, so

$$\xi(k[\varepsilon]) : \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \xrightarrow{\sim} [\underline{A}(k[\varepsilon])]$$

is bijective.

It remains to prove versality. Consider a diagram

$$\begin{array}{ccc} & a' & \\ & \downarrow \alpha & \\ \xi & \xrightarrow{h} & a \end{array}$$

where α is surjective. We must lift h to a' . By induction we may assume that a' is a small prolongation of a . Since $P(a)$ is artinian, h factors through some a_q ,

$$\xi \longrightarrow a_q \longrightarrow a.$$

Semi-homogeneity gives a commutative diagram

$$\begin{array}{ccccc} & a'' & \longrightarrow & a' & \\ & \downarrow \alpha' & & \downarrow \alpha & \\ \xi & \longrightarrow & a_q & \longrightarrow & a, \end{array}$$

where α' is also a small prolongation. Since $a_{q+1} \longrightarrow a_q$ is versal for small prolongations of a_q , there is a commutative diagram

$$\begin{array}{ccccc} & & & a'' & \\ & & \nearrow & \downarrow \alpha' & \\ \xi & \longrightarrow & a_{q+1} & \longrightarrow & a_q, \end{array}$$

which supplies the required lift.

Remark 1.12. Let

$$\cdots \longrightarrow a_n \longrightarrow a_{n-1} \longrightarrow \cdots \longrightarrow a_2 \longrightarrow a_1 \longrightarrow a_0 = e$$

be an infinite sequence of arrows in $\underline{A}$ such that

$$\begin{cases} a_i \text{ is versal for small prolongations of } a_{i-1}, \\ P(a_i) \longrightarrow P(a_{i-1}) \text{ is surjective,} \\ \overline{\Omega}_{a_i} \longrightarrow \overline{\Omega}_{a_{i-1}} \text{ is bijective for } i \gg 0. \end{cases}$$

The preceding proof shows that $\varprojlim_i a_i$ is a versal formal object of $\underline{A}$.

Remark 1.13. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. If $\underline{A}$ satisfies 1.11(i) and (ii), then so does $[\underline{A}]$; moreover, $\xi \in \text{ob}(\widehat{\underline{A}})$ is a minimally versal formal object of $\underline{A}$ if and only if it is so for $[\underline{A}]$, by 1.10(ii).

Remark 1.14. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a minimally versal formal object ξ . Any invariant of $R = P(\xi)$, a complete local Λ -algebra of formally finite type over Λ , is automatically an invariant of $\underline{A}$ over $\underline{C}_\Lambda$. Thus, for example, we say that $\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$ if R is (formally) smooth over Λ , that is, a formal power-series algebra over Λ ; and we say that $\underline{A}$ is irreducible over $\underline{C}_\Lambda$ if $\text{Spec}(R)$ is irreducible.

In the next section we introduce smoothness for a functor $\underline{A} \longrightarrow \underline{B}$ over $\underline{C}_\Lambda$, where $\underline{A}$ and $\underline{B}$ are cofibered categories in groupoids over $\underline{C}_\Lambda$. It follows from 2.4 that $\underline{A}$ being (formally) smooth over $\underline{C}_\Lambda$ is equivalent to the functor $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ being smooth, so no ambiguity arises. We define

$$\dim_{\underline{C}_\Lambda} \underline{A} = \dim(k \otimes_\Lambda R).$$

For this, let $\underline{C}_k$ be the category of artinian local k -algebras with residue field k . The functor $\underline{C}_k \longrightarrow \underline{C}_\Lambda$ is fully faithful, and its left adjoint is

$$\underline{C}_\Lambda \longrightarrow \underline{C}_k, \quad R \longmapsto k \otimes_\Lambda R.$$

Let $\underline{A}|_{\underline{C}_k}$ be the pullback of $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ along $\underline{C}_k \longrightarrow \underline{C}_\Lambda$. If ξ is a minimally versal formal object for $\underline{A}$ over $\underline{C}_\Lambda$, then $k \otimes_\Lambda \xi$ is minimally versal for $\underline{A}|_{\underline{C}_k}$ over $\underline{C}_k$; the arrow $\xi \longrightarrow k \otimes_\Lambda \xi$ is cocartesian over $R \longrightarrow k \otimes_\Lambda R$. Consequently,

$$\dim_{\underline{C}_\Lambda} \underline{A} = \dim_{\underline{C}_k} (\underline{A}|_{\underline{C}_k}).$$

1.15. In the remainder of this paragraph, we briefly explain how to modify the preceding theory to allow residue-field extensions.

Let Λ and K be complete noetherian local rings, with K a Λ -algebra. Assume that:

- (a) the natural map $\Lambda \longrightarrow K$ is a local homomorphism;
- (b) the residue field K' of K is an extension of finite type of the residue field k of Λ .

The most interesting case is $K = K'$ with K a finite inseparable extension of k .

Denote by $\widehat{C}_{\Lambda, K}$ the category whose objects are complete noetherian local rings R equipped with a Λ -algebra structure and a Λ -epimorphism $s : R \longrightarrow K$. Let $C_{\Lambda, K}$ be its full subcategory consisting of pairs (R, s) for which $\ker(s)$ has finite length as an R -module. The category $\widehat{C}_{\Lambda, K}$ identifies with a full subcategory of $\text{Pro}(C_{\Lambda, K})$. If K is artinian, then every object of $C_{\Lambda, K}$ is artinian.

For a finite-dimensional vector space V over K' , write $D_V(K)$ for the algebra of dual numbers over K defined by V . Its elements have the form $x + v\varepsilon$, where $x \in K$, $v \in V$, and $\varepsilon^2 = 0$, with multiplication

$$(x + v\varepsilon)(x' + v'\varepsilon) = xx' + (xv' + x'v)\varepsilon.$$

The map

$$s : D_V(K) \longrightarrow K, \quad s(x + v\varepsilon) = x,$$

makes $D_V(K)$ an object of $C_{\Lambda, K}$. The role previously played by $k[\varepsilon]$ is now played by $D_{K'}(K)$.

As in 1.8, every cofibered groupoid $\underline{A}$ over $C_{\Lambda, K}$ extends naturally to a cofibered category $\widehat{\underline{A}}$ over $\widehat{C}_{\Lambda, K}$. A map in $\underline{A}$ or $\widehat{\underline{A}}$ is called surjective when the corresponding algebra homomorphism in $C_{\Lambda, K}$ or $\widehat{C}_{\Lambda, K}$ is surjective.

We assume that $\underline{A}(K) = \{e\}$.

Definition 1.16. A cofibered groupoid $\underline{A}$ over $C_{\Lambda, K}$ is called semi-homogeneous if the following properties hold:

(I) Every diagram of solid arrows

$$\begin{array}{ccc} a'' & \xrightarrow{\quad\quad\quad} & a' \\ \downarrow & & \downarrow \\ a_1 & \xrightarrow{\quad\quad\quad} & a \end{array}$$

in $\underline{A}$, where $a' \rightarrow a$ is surjective, can be completed to a commutative diagram by including the dotted arrows.

(II) For every R , the natural map

$$[\underline{A}(R \times_K D_{K'}(K))] \longrightarrow [\underline{A}(R)] \times [\underline{A}(D_{K'}(K))]$$

is bijective.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$.

1.17. First restrict $\underline{A}$ to the full subcategory of $C_{\Lambda, K}$ whose objects are the $D_V(K)$. Put

$$\Omega = \Omega_{K/\Lambda}^1 \otimes_K K', \quad d : K \longrightarrow \Omega.$$

A map $f : D_V(K) \longrightarrow D_W(K)$ can be written

$$f(x + v\varepsilon) = x + (u(v) + D(dx))\varepsilon,$$

with $u \in \text{Hom}_{K'}(V, W)$ and $D \in \text{Hom}_{K'}(\Omega, W)$. Identifying f with (u, D) , composition is given by

$$(u, D) \circ (u', D') = (uu', D + uD').$$

Condition (II) implies:

(II') the functor from K' -vector spaces to sets

$$V \longmapsto [\underline{A}(D_V(K))]$$

commutes with products.

Lemma 1.18. Assume that condition (II') is satisfied by $\underline{A}$. Then:

(1) $[\underline{A}(D_{K'}(K))]$ carries a unique vector-space structure such that, functorially in V ,

$$[\underline{A}(D_V(K))] \simeq V \otimes_{K'} [\underline{A}(D_{K'}(K))].$$

(2) There is a linear map

$$\gamma : \text{Hom}(\Omega, K') \longrightarrow [\underline{A}(D_{K'}(K))]$$

such that, for every $D \in \text{Hom}(\Omega, K')$, the map

$$[\underline{A}((1, D))] : [\underline{A}(D_{K'}(K))] \longrightarrow [\underline{A}(D_{K'}(K))]$$

is $x \longmapsto x + \gamma(D)$.

(3) For every $(u, D) : D_V(K) \longrightarrow D_W(K)$, the induced map

$$(u, D) : [\underline{A}(D_V(K))] \longrightarrow [\underline{A}(D_W(K))],$$

that is,

$$(u, D) : V \otimes [\underline{A}(D_{K'}(K))] \longrightarrow W \otimes [\underline{A}(D_{K'}(K))],$$

is given by

$$x \longmapsto u(x) + \gamma(D).$$

Introduce the notation

$$\begin{aligned} F(V) &= [\underline{A}(D_V(K))], \\ A &= F(K'), \\ F(u, D) : F(V) &\longrightarrow F(W) \quad \text{for the map induced by } (u, D) : D_V(K) \longrightarrow D_W(K), \\ F(D) : F(V) &\longrightarrow F(V), \quad F(D) = F(1, D). \end{aligned}$$

By (II'), and because K' is a K' -vector-space object in the category of K' -vector spaces, $F(K')$ is a K' -vector-space object in the category of sets, hence a K' -vector space. This proves (1). We henceforth identify $F(V)$ with $V \otimes A$.

The identity

$$(u, D) = (1, D) \circ (u, 0)$$

gives

$$F(u, D) = F(D) \circ (u \otimes 1_A).$$

The functoriality of F amounts to the identities

$$\begin{aligned} F(0) &= \text{Id}, \\ F(D) \circ F(D') &= F(D + D'), \\ (u \otimes 1) \circ F(D) &= F(uD) \circ (u \otimes 1_A). \end{aligned}$$

For any $D : \Omega \longrightarrow V$ and any $n \in \mathbb{N}$, let

$$i : V \hookrightarrow V \oplus K'^n, \quad p : V \oplus K'^n \longrightarrow K'^n$$

be the canonical inclusion and projection. Then $piD = 0$, and the following diagram is commutative:

$$\begin{array}{ccccc} V \otimes A & \longrightarrow & (V \oplus K'^n) \otimes A & \longrightarrow & K'^n \otimes A \\ \downarrow F(D) & & \downarrow F(i, D) & & \parallel \\ V \otimes A & \longrightarrow & (V \oplus K'^n) \otimes A & \longrightarrow & K'^n \otimes A. \end{array}$$

Let $x : K'^n \longrightarrow V$ be a map, put $x_i = x(e_i)$, and let (a_i) be a family of elements of A . We have

$$(1_V + x) \circ i \circ D = D.$$

Consequently, if

$$F(i, D) \left(\sum_i e_i a_i \right) = \sum_i e_i a_i + a, \quad a \in V \otimes A,$$

then

$$F(D) \left(\sum_i x_i a_i \right) = \sum_i x_i a_i + a,$$

and a is independent of the x_i . Hence

$$\begin{aligned} F(D)(y) &= y + G(D), \\ G(D + D') &= G(D) + G(D'), \\ uG(D) &= G(uD) \quad \text{for every } u : V \longrightarrow W. \end{aligned}$$

Every such G is given, for some $\gamma \in \Omega \otimes A$, by

$$G(D) = (D \otimes 1_A)(\gamma).$$

This proves (2) and (3).

Definition 1.19. (1) An object $\xi \in \widehat{\underline{A}}$ is called a versal formal object if every surjective map $\alpha : a' \rightarrow a$ in $\underline{A}$ induces a surjective map

$$\mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a') \rightarrow \mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a).$$

(2) A versal formal object $\xi \in \widehat{\underline{A}}$, with image (R, s) in $\widehat{C}_{\Lambda, K}$, is called minimal if the following condition holds. For any two maps

$$\varphi_1, \varphi_2 : R \rightrightarrows D_{K'}(K),$$

if $\varphi_1(\xi)$ is isomorphic to $\varphi_2(\xi)$, then there is a derivation $D : K \rightarrow K'$ such that

$$(1, D)(\varphi_1) = \varphi_2, \quad \text{that is,} \quad \varphi_2 - \varphi_1 = Ds.$$

Theorem 1.20. If $\underline{A}$ is semi-homogeneous and

$$\dim_{K'}[\underline{A}(D_{K'}(K))] < \infty,$$

then $\underline{A}$ admits a minimally versal formal object, and any two such objects are isomorphic.

With the notation of 1.18, let H^* be a subspace of $[\underline{A}(D_{K'}(K))]$ complementary to the image of γ , and let H be the dual of H^* . Set

$$R_0 = D_H(K).$$

Choose $\xi_0 \in \underline{A}(R_0)$ whose class, under the identifications

$$[\underline{A}(R_0)] \simeq H \otimes [\underline{A}(D_{K'}(K))] \simeq \mathrm{Hom}(H^*, [\underline{A}(D_{K'}(K))]),$$

is the inclusion $H^* \hookrightarrow [\underline{A}(D_{K'}(K))]$. Then:

(1) for every V , the maps

$$\mathrm{Hom}_{C_{\Lambda, K}}(D_H(K), D_V(K)) \rightarrow [\underline{A}(D_V(K))]$$

are surjective;

(2) the minimality condition of 1.19(2) holds for (R_0, ξ_0) .

Choose $S \in \widehat{C}_{\Lambda, K}$ formally smooth over Λ , together with an epimorphism $t : S \rightarrow R_0$, such that

$$\Omega_{R_0/\Lambda} \otimes K' \xrightarrow{\sim} \Omega_{S/\Lambda} \otimes K'.$$

If M is the maximal ideal of S , define inductively, for $n \geq 1$,

$$R_n = S/I_n, \quad \xi_n \in \mathrm{Ob}(\underline{A}(R_n)),$$

by the following conditions:

(a) I_{n+1} is the intersection of the ideals I such that

$$I_n M \subset I \subset I_n$$

and ξ_n extends to an object $\xi \in \underline{A}(S/I)$;

(b) ξ_{n+1} is an extension of ξ_n .

If

$$R = \varprojlim_n R_n, \quad \xi = \varprojlim_n \xi_n,$$

then (R, ξ) is a minimally versal formal object, and it is unique up to isomorphism.

2. Prorepresentable cofibered groupoids

Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a minimally versal formal object ξ . In general, ξ alone does not determine $\underline{A}$ up to $\underline{C}_\Lambda$ -equivalence. One must also know when two maps

$$R \begin{smallmatrix} \xrightarrow{f} \\ \xrightarrow{g} \end{smallmatrix} S$$

give rise to a $\underline{C}_\Lambda$ -isomorphism $\xi(f) \rightarrow \xi(g)$, where $R = P(\xi)$. As an application of 1.11 and 1.12, we now consider the problem of recovering $\underline{A}$, up to $\underline{C}_\Lambda$ -equivalence. We obtain a satisfactory answer for a broad class of cofibered groupoids.

Many cofibered groupoids that arise in practice satisfy a property slightly stronger than semi-homogeneity; it will be defined explicitly in 2.5. To investigate them, it is convenient to recast the preceding discussion in terms of “smooth functors.” All the notation and terminology of the preceding section remain in force throughout this section.

Definition 2.1. Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We say that φ is smooth if every surjective map $\beta : b' \rightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram

$$(*) \quad \begin{array}{ccc} \varphi(a') & \xrightarrow{\quad} & b' \\ & \searrow \varphi(\alpha) & \downarrow \beta \\ & & \varphi(a) \end{array}$$

for some $\alpha : a' \rightarrow a$ in $\underline{A}$. We say that φ is minimally smooth if it is smooth and

$$\varphi(k[\varepsilon]) : [\underline{A}(k[\varepsilon])] \rightarrow [\underline{B}(k[\varepsilon])]$$

is bijective.

Remark 2.2(a). Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. The following conditions are equivalent:

- (i) φ is smooth;
- (ii) every surjective map $\beta : b' \rightarrow \varphi(a)$ in $\underline{B}$ can be completed to a diagram $(*)$ in which $\varphi(a') \rightarrow b'$ is a $\underline{C}_\Lambda$ -isomorphism;
- (iii) every small prolongation $\beta : b' \rightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram $(*)$.

Remark 2.2(b). For every cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$, the natural functor

$$\underline{A} \rightarrow [\underline{A}]$$

is minimally smooth.

Remark 2.2(c). For each $R \in \widehat{\underline{C}}_\Lambda$, let

$$h_R : \underline{C}_\Lambda \rightarrow (\text{Ens})$$

be the functor defined by

$$h_R(S) = \text{Hom}_{\Lambda\text{-alg}}(R, S) \quad (S \in \underline{C}_\Lambda).$$

It determines a cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$. A map $R \rightarrow S$ in $\widehat{\underline{C}}_\Lambda$ induces a functor

$$\underline{h}_S \rightarrow \underline{h}_R.$$

This functor is smooth if and only if S is a formal power-series ring over R .

Remark 2.2(d). Given $R \in \widehat{\underline{C}}_\Lambda$, denote by $(R, \underline{C}_\Lambda)$ the category of arrows in $\widehat{\underline{C}}_\Lambda$ with source R and target in $\underline{C}_\Lambda$. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Given $\xi \in \text{Ob } \widehat{\underline{A}}$ with $P(\xi) = R$, let

$$\underline{A}' \rightarrow (R, \underline{C}_\Lambda)$$

be the pullback of $\underline{A} \rightarrow \underline{C}_\Lambda$ along the canonical functor $(R, \underline{C}_\Lambda) \rightarrow \underline{C}_\Lambda$. Choose a section $\xi^\#$ such that $\xi^\#(1_R) = \xi$. Thus, for every map $f : R \rightarrow S$, we choose an f -map $\xi \rightarrow \xi(f)$.

We obtain a functor

$$\xi^\# : \underline{h}_R \rightarrow \underline{A}$$

over $\underline{C}_\Lambda$ by sending $(R \xrightarrow{f} S)$ to $\xi(f)$. Up to natural equivalence, this functor does not depend on the choice of $\xi^\#$.

The assertion that $\xi^\# : \underline{h}_R \rightarrow \underline{A}$ is (minimally) smooth means that every surjective map $a' \xrightarrow{\alpha} \xi(f)$ in $\underline{A}$ can be completed to a commutative diagram

$$\begin{array}{ccc} \xi(f') & \xrightarrow{\quad} & a' \\ & \searrow \xi(\quad) & \downarrow \alpha \\ & & \xi(f) \end{array}$$

(and, in the minimal case, that $\underline{h}_R(k[\varepsilon]) \rightarrow \underline{A}(k[\varepsilon])$ is bijective). This means precisely that ξ is a (minimally) versal formal object of $\underline{A}$ over $\underline{C}_\Lambda$.

Thus the existence of a (minimally) versal formal object of $\underline{A}$ is equivalent to the existence of a (minimally) smooth functor

$$\underline{h}_R \rightarrow \underline{A}$$

over $\underline{C}_\Lambda$ for some $R \in \text{Ob}(\widehat{\underline{C}}_\Lambda)$.

The following general properties of smoothness are immediate.

Proposition 2.3. Let

$$\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$$

be functors of cofibered groupoids over $\underline{C}_\Lambda$.

(a) If φ and ψ are both smooth, then so is $\psi \circ \varphi$.

(b) If $\psi \circ \varphi$ is smooth and φ is surjective on isomorphism classes of objects, then ψ is smooth.

Remark 2.4. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a versal formal object ξ . By 2.2(d), this means that

$$\xi : \underline{h}_R \rightarrow \underline{A}$$

is smooth, where $R = P(\xi)$. Therefore the following assertions are equivalent:

- $\underline{A}$ is formally smooth over $\underline{C}_\Lambda$ (cf. 1.13), that is, R is a formal power-series algebra over Λ ;
- the composite functor

$$\underline{h}_R \xrightarrow{\xi} \underline{A} \xrightarrow{P} \underline{C}_\Lambda (= \underline{h}_\Lambda)$$

is smooth;

- $P : \underline{A} \rightarrow \underline{C}_\Lambda$ is smooth.

The last equivalence follows from Proposition 2.3(b).

We now consider a stronger property than semi-homogeneity.

Definition 2.5. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called homogeneous if every diagram

$$\begin{array}{ccc} & & a' \\ & & \downarrow \\ b' & \longrightarrow & a \end{array}$$

in which $a' \rightarrow a$ is surjective admits a fibre product $a' \times_a b'$ in $\underline{A}$ such that

$$P(a' \times_a b') = P(a') \times_{P(a)} P(b').$$

Remark 2.6(a). Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

$$(*) \quad \begin{array}{ccc} & a' & \\ \swarrow & & \searrow \\ a_1 & & a_2 \\ \searrow & & \swarrow \\ & a & \end{array} \quad \text{in } \underline{A} \text{ above the diagram} \quad \begin{array}{ccc} & R_1 \times_R R_2 & \\ \swarrow \pi_1 & & \searrow \pi_2 \\ R_1 & & R_2 \\ \searrow & & \swarrow \\ & R & \end{array}$$

in $\underline{C}_\Lambda$, where the π_i are the projections. If a fibre product $a_1 \times_a a_2$ exists in $\underline{A}$, there is a unique map

$$\varphi : a' \longrightarrow a_1 \times_a a_2$$

compatible with the projections. If

$$P(a_1 \times_a a_2) = R_1 \times_R R_2,$$

then $P(\varphi)$ is the identity map of $R_1 \times_R R_2$; hence φ is a $\underline{C}_\Lambda$ -isomorphism, that is, $a' = a_1 \times_a a_2$. Thus, if $\underline{A}$ is homogeneous, every diagram $(*)$ defines a fibre product $a_1 \times_a a_2$ whenever $R_1 \longrightarrow R$ is surjective. It follows at once

that every homogeneous cofibered groupoid is, a fortiori, semi-homogeneous.

Remark 2.6(b). Consider a diagram of categories and functors

$$\begin{array}{ccc} X & & Y \\ & \searrow \varphi & \swarrow \psi \\ & Z & \end{array}$$

The ordinary fibre product $X \times_Z Y$ is the category whose objects are pairs (x, y) satisfying $\varphi(x) = \psi(y)$ and whose maps $(x, y) \longrightarrow (x', y')$ are pairs

$$\alpha : x \longrightarrow x', \quad \beta : y \longrightarrow y'$$

such that $\varphi(\alpha) = \psi(\beta)$. The fibre product $X \times_Z^2 Y$ in the sense of 2-categories is the category whose objects are triples $(x, y; \gamma)$, where

$$\varphi(x) \xrightarrow{\gamma} \psi(y)$$

is an isomorphism. A map $(x, y; \gamma) \longrightarrow (x', y'; \gamma')$ is a pair $\alpha : x \longrightarrow x'$ and $\beta : y \longrightarrow y'$ for which the diagram

$$\begin{array}{ccc} \varphi(x) & \xrightarrow{\varphi(\alpha)} & \varphi(x') \\ \gamma \downarrow & & \downarrow \gamma' \\ \psi(y) & \xrightarrow{\psi(\beta)} & \psi(y') \end{array}$$

commutes.

The notions of a (semi-)homogeneous cofibered groupoid and of a smooth functor can be rephrased in terms of fibre products of categories in this 2-categorical sense. Let $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ be a cofibered category, and consider $R_1 \times_R R_2$ in $\underline{C}_\Lambda$. Choose direct-image functors

$$(\pi_i)_* : \underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_i), \quad i = 1, 2.$$

They induce a functor

$$\underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_1) \times_{\underline{A}(R)}^2 \underline{A}(R_2).$$

Up to canonical natural equivalence, it is independent of the chosen direct-image functors. The groupoid $\underline{A}$ is homogeneous if and only if this functor is an equivalence whenever $R_1 \rightarrow R$ is surjective; it is semi-homogeneous if and only if

$$[\underline{A}(R_1 \times_R R_2)] \rightarrow [\underline{A}(R_1)] \times_{[\underline{A}(R)]} [\underline{A}(R_2)]$$

is surjective whenever $R_1 \rightarrow R$ is surjective, and is bijective whenever $R = k$ and $R_1 = k[\varepsilon]$.

Now let $\varphi : \underline{A} \rightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. For every map $R' \rightarrow R$ in $\underline{C}_\Lambda$, φ induces, up to canonical natural equivalence, a functor

$$\underline{A}(R') \rightarrow \underline{A}(R) \times_{\underline{B}(R)}^2 \underline{B}(R').$$

The functor φ is smooth if and only if this functor is surjective on isomorphism classes of objects whenever $R' \rightarrow R$ is surjective.

Remark 2.6(c). If $\underline{A}$ is homogeneous, then for any finite-dimensional k -vector spaces V and W the functor

$$\underline{A}(k[V \times W]) \rightarrow \underline{A}(k[V]) \times \underline{A}(k[W])$$

is an equivalence of categories. Recall our convention $\underline{A}(k) = \{e\}$. In particular,

$$\text{Aut}_{\underline{A}(k[V \times W])}(1_{V \times W}) \xrightarrow{\sim} \text{Aut}_{\underline{A}(k[V])}(1_V) \times \text{Aut}_{\underline{A}(k[W])}(1_W),$$

where 1_V is the direct image of e under the unique map $k \rightarrow k[V]$ in $\underline{C}_\Lambda$. Consequently, if $\underline{A}$ is homogeneous, both $\text{Aut}(1_\varepsilon)$ and $[\underline{A}(k[\varepsilon])]$ carry canonical k -vector-space structures.

Remark 2.6(d). For every $R \in \widehat{\underline{C}}_\Lambda$, the cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$ is homogeneous.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. Homogeneity of $\underline{A}$, however, does not in general imply homogeneity of $[\underline{A}]$.

Proposition 2.7. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$. The following conditions are equivalent.

(i) $[\underline{A}]$ is homogeneous; that is,

$$[\underline{A}(R' \times_R S)] \rightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$$

is bijective for every small extension $R' \rightarrow R$.

(ii) For every small prolongation $a' \rightarrow a$ in $\underline{A}$, the map

$$\text{Aut}_{\underline{A}(R')}(a') \rightarrow \text{Aut}_{\underline{A}(R)}(a)$$

is surjective, where $R' = P(a')$ and $R = P(a)$.

Proof. (i) $\Rightarrow$ (ii). Let $f : R' \rightarrow R$ be a small extension in $\underline{C}_\Lambda$, and let $a' \rightarrow a$ be an f -map in $\underline{A}$. Given $\tau \in \text{Aut}_{\underline{A}(R)}(a)$, consider $a' \times_a a'$ and $a' \times_a a'_\tau$, where $a'_\tau \rightarrow a$ is the composite

$$a' \rightarrow a \xrightarrow{\tau} a.$$

Since

$$[\underline{A}(R' \times_R R')] \rightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(R')]$$

is bijective, there is an isomorphism

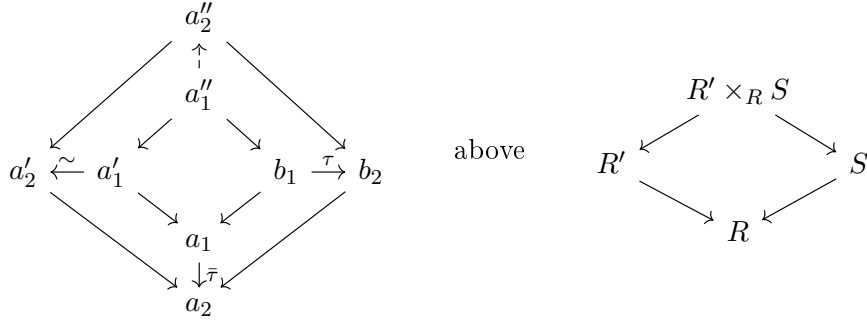
$$a' \times_a a' \xrightarrow{u} a' \times_a a'_\tau.$$

Projection onto the second factor then gives an isomorphism $\tau' : a' \rightarrow a'$ lifting τ .

(ii) $\Rightarrow$ (i). Let a''_i ($i = 1, 2$) be objects of $\underline{A}(R' \times_R S)$ with the same image under

$$[\underline{A}(R' \times_R S)] \rightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)].$$

We obtain a diagram of solid arrows



where $\bar{\tau} : a_1 \longrightarrow a_2$ is the isomorphism induced by τ , so the lower-right square commutes. Under condition (ii), the given isomorphism $a'_1 \xrightarrow{\sim} a'_2$ can be replaced so that the left square also commutes. Since

$$a''_i = a'_i \times_{a_i} b_i \quad (i = 1, 2)$$

by Remark 2.6(a), we obtain an isomorphism $a''_1 \xrightarrow{\sim} a''_2$. Hence

$$[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$$

is injective, and therefore bijective because $\underline{A}$ is homogeneous by hypothesis. $\square$

Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let $f : R' \longrightarrow R$ be a small extension in $\underline{C}_\Lambda$. There is a canonical map

$$\tilde{f} : R' \times_R R' \longrightarrow k[\text{Ker } f], \quad (r'_1, r'_2) \longmapsto r'_1(0) + (r'_1 - r'_2),$$

and

$$1 \times \tilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\text{Ker } f]$$

is an isomorphism. Given two f -maps $a_i \longrightarrow a$ ($i = 1, 2$) in $\underline{A}$, the fibre product $a_1 \times_a a_2$ is an object over $R' \times_R R'$. We may therefore form

$$\tau(a_1, a_2) = (\tilde{f})_*(a_1 \times_a a_2) \in \underline{A}(k[\text{Ker } f]).$$

Since $1 \times \tilde{f}$ is an isomorphism, there is an isomorphism

$$a_1 \times_a a_2 \simeq a_1 \times \tau(a_1, a_2)$$

compatible with projection onto the first factor. Statement 1.5(2) now takes the following more definite form.

Lemma 2.8. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let

$$a_i \xrightarrow{\alpha_i} a \quad (i = 1, 2)$$

be f -maps in $\underline{A}$.

(1) There exists $\rho : a_1 \longrightarrow a_2$ in $\underline{A}$ with $\alpha_2 \circ \rho = \alpha_1$ if and only if there is an arrow $a_1 \longrightarrow \tau(a_1, a_2)$ in $\underline{A}$.

(2) There exists such a ρ with $P(\rho) = \text{id}_{R'}$ if and only if

$$\tau(a_1, a_2) = 0 \quad \text{in} \quad [\underline{A}(k[\text{Ker } f])].$$

Proof. We prove both assertions simultaneously. The existence of $\rho : a_1 \longrightarrow a_2$ satisfying $\alpha_2 \circ \rho = \alpha_1$ (and $P(\rho) = \text{id}_{R'}$) is equivalent successively to the existence of

$$h : a_1 \longrightarrow a_1 \times_a a_2, \quad \text{pr}_1 \circ h = \text{id}_{a_1},$$

with $P(h)$ the diagonal map; to the existence of

$$g : a_1 \longrightarrow a_1 \times \tau(a_1, a_2), \quad \text{pr}_1 \circ g = \text{id}_{a_1},$$

with

$$P(g) : R' \longrightarrow R' \times_k k[\text{Ker } f], \quad r' \longmapsto (r', r'(0));$$

and to the existence of an arrow $\gamma : a_1 \longrightarrow \tau(a_1, a_2)$ in $\underline{A}$, with

$$P(\gamma) : R' \longrightarrow k[\text{Ker } f], \quad r' \longmapsto r'(0).$$

Such an arrow factors through $R' \longrightarrow k \longrightarrow k[\text{Ker } f]$ if and only if $\tau(a_1, a_2) = 0$. $\square$

Proposition 2.9. (a) Let

$$\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$$

be functors of cofibered groupoids over $\underline{C}_\Lambda$. Assume that

- (i) $\underline{B}$ is semi-homogeneous and $\underline{C}$ is homogeneous;
- (ii) the map

$$\psi(k[\varepsilon]) : [\underline{B}(k[\varepsilon])] \longrightarrow [\underline{C}(k[\varepsilon])]$$

is injective.

Then $\psi \circ \varphi$ is smooth if and only if φ and ψ are both smooth.

(b) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. If $\underline{A}$ is semi-homogeneous and $\underline{B}$ is homogeneous, then

$$\widehat{\varphi} : \widehat{\underline{A}} \longrightarrow \widehat{\underline{B}}$$

is smooth. Explicitly, for every surjective map $b' \xrightarrow{\beta} \widehat{\varphi}(a)$ in $\widehat{\underline{B}}$, there are a map $\alpha : a' \longrightarrow a$ in $\widehat{\underline{A}}$ and a $\widehat{\underline{C}}_\Lambda$ -isomorphism

$$\rho : \widehat{\varphi}(a') \xrightarrow{\sim} b'$$

such that $\beta \circ \rho = \varphi(\alpha)$.

Proof. (a) Suppose $\psi \circ \varphi$ is smooth. If φ is smooth, then ψ is smooth by Proposition 2.3(b), so it remains to prove that φ is smooth. Let $b' \xrightarrow{\beta} \varphi(a)$ be a small prolongation in $\underline{B}$. Since $\psi \circ \varphi$ is smooth, there are an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ and a $\underline{C}_\Lambda$ -isomorphism

$$\rho : \psi(b') \longrightarrow \psi\varphi(a')$$

for which

$$\begin{array}{ccc} \psi(b') & \xrightarrow{\rho} & \psi\varphi(a') \\ & \searrow \psi(\beta) & \swarrow \psi\varphi(\alpha) \\ & \psi\varphi(a) & \end{array}$$

commutes. Since $\underline{B}$ is semi-homogeneous, we can choose a commutative diagram

$$\begin{array}{ccc} & b'' & \\ b' \swarrow & & \searrow \varphi(a') \\ & \varphi(a) & \swarrow \varphi(\alpha) \\ & \beta \searrow & \end{array} \quad \text{above} \quad \begin{array}{ccc} & R' \times_R R' & \\ R' \swarrow & & \searrow R' \\ & R & \end{array}$$

where $R = P(a)$ and $R' = P(b') = P(a')$. Homogeneity of $\underline{C}$ gives

$$\psi(b'') = \psi(b') \times_{\psi\varphi(a)} \psi\varphi(a')$$

$$0 = \tau(\psi\varphi(a'), \psi(b')) = (\widetilde{f})_*\psi(b'').$$

(b) Let a surjective map $\beta : b' \longrightarrow \widehat{\varphi}(a)$ in $\widehat{\underline{B}}$ be given. It is represented by a sequence of commutative diagrams

in $\underline{B}$, where every α_v , β_v , and β'_v is surjective and $\beta = \varprojlim_v \beta_v$. It is enough to show inductively that every diagram of solid arrows below, with α_n , β_n , β_{n+1} , and β'_n surjective, can be completed by the dotted arrows:

By hypothesis $\underline{A}$ is semi-homogeneous and $\alpha_n : a'_n \rightarrow a_n$ is surjective. Hence there is a commutative diagram

Homogeneity of $\underline{B}$ implies that

is a fibre-product diagram in $\underline{B}$. Thus there is a unique map $\delta : b'_{n+1} \rightarrow \varphi(a')$ with the required commutativity. Since β_{n+1} and β'_n are surjective, so is δ . Smoothness of φ therefore gives a map $a'_{n+1} \xrightarrow{\alpha'} a'$ in $\underline{A}$ and a $\underline{C}_\Lambda$ -isomorphism

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such that $\delta \circ \rho = \varphi(\alpha')$. We obtain the commutative diagram

$$\begin{array}{ccccc}
\varphi(a'_{n+1}) & \xrightarrow{\varphi(\gamma' \circ \alpha')} & & \xrightarrow{\quad} & \varphi(a'_n) \\
& \searrow \rho & & & \nearrow \sim \\
& & b'_{n+1} & \xrightarrow{\quad} & b'_n \\
& \searrow \varphi(\gamma \circ \alpha') & \downarrow & & \downarrow \\
& & \varphi(a_{n+1}) & \xrightarrow{\quad} & \varphi(a_n)
\end{array}$$

which completes the induction and proves (b). $\square$

Corollary 2.10. (a) Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. Let ξ be a versal formal object of $\underline{A}$, and let η be a minimally versal formal object of $\underline{B}$. If $\underline{B}$ is homogeneous, then for every map $\beta : \eta \rightarrow \varphi(\xi)$ in $\widehat{\underline{B}}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\beta) : P(\eta) \rightarrow P(\varphi(\xi)) = P(\xi).$$

(b) Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let ξ and η be versal formal objects of $\underline{A}$. If η is minimal, then for every map $\alpha : \eta \rightarrow \xi$ in $\widehat{\underline{A}}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\alpha) : P(\eta) \rightarrow P(\xi).$$

Proof. (a) A map $\beta : \eta \rightarrow \varphi(\xi)$ induces a commutative diagram

$$\begin{array}{ccc}
\underline{h}_{P(\xi)} & \xrightarrow{\tilde{\beta}} & \underline{h}_{P(\eta)} \\
\downarrow \tilde{\xi} & & \downarrow \tilde{\eta} \\
\underline{A} & \xrightarrow{\varphi} & \underline{B}.
\end{array}$$

Since $\tilde{\xi}$ and φ are smooth, the composite $\tilde{\eta} \circ \tilde{\beta} = \varphi \circ \tilde{\xi}$ is smooth. On the other hand,

$$\tilde{\eta}(k[\varepsilon]) : \underline{h}_{P(\eta)}(k[\varepsilon]) \rightarrow [\underline{B}(k[\varepsilon])]$$

is an isomorphism. Proposition 2.9(a) therefore shows that $\tilde{\beta}$ is smooth, which means precisely that $P(\xi)$ is a formal power-series ring over $P(\eta)$.

(b) Apply (a) with $\underline{A} = \underline{B}$. $\square$ We conclude by considering the “representability” of a cofibered groupoid. Let $\underline{C}$ be a fixed category. Every object R of $\underline{C}$ defines a functor

$$h_R : \underline{C} \rightarrow (\text{Sets}), \quad h_R(S) = \text{Hom}_{\underline{C}}(R, S),$$

and hence a fully faithful functor

$$h : \underline{C}^{\text{op}} \rightarrow \underline{\text{Hom}}(\underline{C}, (\text{Sets})).$$

A *cocategory* in $\underline{C}$ is a diagram

$$h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1} \begin{array}{c} \xrightarrow{s_1} \\ \xleftarrow{\epsilon} \\ \xrightarrow{s_2} \end{array} h_{R_0} \quad \text{with } s_i \circ \epsilon = \text{id} \quad (i = 1, 2),$$

where the first arrow is denoted by c and

$$h_{R_1} \times_{h_{R_0}} h_{R_1} = (h_{R_1})_{s_1} \times_{h_{R_0}} s_2(h_{R_1}),$$

such that

- (i) c is associative and compatible with s_i for $i = 1, 2$;
- (ii) both composites

$$\begin{aligned} h_{R_1} &\xrightarrow{\sim} h_{R_0} \times_{h_{R_0}} h_{R_1} \xrightarrow{\epsilon \times 1} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}, \\ h_{R_1} &\xrightarrow{\sim} h_{R_1} \times_{h_{R_0}} h_{R_0} \xrightarrow{1 \times \epsilon} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1} \end{aligned}$$

are identities.

If $(R_0, R_1, s_1, s_2, \epsilon, c)$ is a cocategory in $\underline{C}$, then for every $S \in \underline{C}$ we define a category $h_{(R_0, R_1, \dots)}(S)$ by

$$\text{ob } h_{(R_0, R_1, \dots)}(S) = \text{Hom}_{\underline{C}}(R_0, S), \quad \text{Fl } h_{(R_0, R_1, \dots)}(S) = \text{Hom}_{\underline{C}}(R_1, S),$$

with composition supplied by

$$c : h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1}.$$

The categories so obtained depend functorially on S and define a cofibered category $\underline{h}_{(R_0, R_1, \dots)}$ over $\underline{C}$. For example, every object R of $\underline{C}$ defines a discrete cocategory

$$\underline{h}_R = \underline{h}_{(R, R, s_1, s_2, \epsilon, c)}$$

by taking s_1, s_2, ϵ , and c all to be identity maps. A homomorphism

$$(R_0, R_1, \dots) \longrightarrow (R'_0, R'_1, \dots)$$

of cocategories is a pair of maps of represented functors $h_{R_0} \longrightarrow h_{R'_0}$ and $h_{R_1} \longrightarrow h_{R'_1}$ whose induced map

$$h_{(R_0, R_1, \dots)}(S) \longrightarrow h_{(R'_0, R'_1, \dots)}(S)$$

is a functor for every $S \in \underline{C}$. Every cocategory $(R_0, R_1, \dots)$ has, in particular, a homomorphism

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0, R_1, \dots)}.$$

A *cogroupoid* in $\underline{C}$ is a cocategory $(R_0, R_1, \dots)$ for which $h_{(R_0, R_1, \dots)}(S)$ is a groupoid for every $S \in \underline{C}$; equivalently, $\underline{h}_{(R_0, R_1, \dots)}$ is a category cofibered in groupoids over $\underline{C}$.

In practice, cocategories and cogroupoids arise as follows. Let $P : \underline{F} \longrightarrow \underline{C}$ be a cofibered category, and fix an object ξ of $\underline{F}$. Put $R_0 = P(\xi)$, and let $(R_0, \underline{C})$ denote the category of arrows in $\underline{C}$ with source R_0 . Let $P : \underline{F}' \longrightarrow (R_0, \underline{C})$ be the pullback of $\underline{F}$ along the canonical functor $(R_0, \underline{C}) \longrightarrow \underline{C}$. Choose a cocartesian section $\xi^\#$ of $\underline{F}'$ over $(R_0, \underline{C})$ such that $\xi^\#(\text{id}_{R_0}) = \xi$. Thus, for every arrow $R_0 \xrightarrow{f} S$ in $\underline{C}$, we choose a cocartesian f -map $\xi \longrightarrow \xi^\#(f)$.

Define

$$\text{Hom}_{\xi^\#} : (R_0 \amalg R_0, \underline{C}) \longrightarrow (\text{Sets})$$

by

$$X = (f_1, f_2) \longmapsto \text{Hom}_{\underline{F}(S)}(\xi^\#(f_1), \xi^\#(f_2)),$$

where S is the common target of f_1 and f_2 . Up to canonical natural equivalence, this functor is independent of the choice of $\xi^\#$; henceforth we write simply ξ for $\xi^\#$.

Suppose the functor is represented by an object

$$X_0 = \left(R_0 \begin{array}{c} \xrightarrow{s_1} \\ \xrightarrow{s_2} \end{array} R_1 \right)$$

of $(R_0 \amalg R_0, \underline{C})$. Thus there is an R_0 -map $\varphi : \xi(s_1) \longrightarrow \xi(s_2)$ in $\underline{F}$ such that

$$\tilde{\varphi} : \text{Hom}_{(R_0 \amalg R_0, \underline{C})}(X_0, X) \longrightarrow \text{Hom}_{\xi}(X)$$

is bijective for every X in $(R_0 \amalg R_0, \underline{C})$. Then X_0 naturally defines a cocategory $(R_0, R_1, \dots)$ in $\underline{C}$. Indeed, a triple of arrows

$$R_0 \begin{array}{c} \xrightarrow{u} \\ \xrightarrow{v} \\ \xrightarrow{w} \end{array} S$$

in $\underline{C}$ defines a composition

$$\text{Hom}_{\underline{F}(S)}(\xi(v), \xi(w)) \times \text{Hom}_{\underline{F}(S)}(\xi(u), \xi(v)) \longrightarrow \text{Hom}_{\underline{F}(S)}(\xi(u), \xi(w)),$$

and hence a composition rule

$$h_{R_1}(S) \times_{h_{R_0}(S)} h_{R_1}(S) \longrightarrow h_{R_1}(S),$$

functorial in S . Consequently, Hom_{ξ} defines the cocategory $(R_0, R_1, \dots)$ in $\underline{C}$ and the cofibered category $\underline{h}_{(R_0, R_1, \dots)}$ over $\underline{C}$. Explicitly,

$$\text{ob } \underline{h}_{(R_0, R_1, \dots)} = \coprod_{S \in \text{ob}(\underline{C})} \text{Hom}_{\underline{C}}(R_0, S).$$

An arrow

$$(R_0 \xrightarrow{f} S) \longrightarrow (R_0 \xrightarrow{f'} S')$$

is a pair of arrows $S \xrightarrow{h} S'$ and $R_1 \xrightarrow{u} S'$ in $\underline{C}$ such that

$$u \circ s_1 = h \circ f, \quad u \circ s_2 = f'.$$

The structural functor $P : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{C}$ sends $(R_0 \xrightarrow{f} S)$ to S .

The pair (ξ, φ) now defines a functor

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{F} \quad \text{over } \underline{C}$$

by

$$(R_0 \xrightarrow{f} S) \longmapsto \xi(f)$$

on objects and by

$$\left[(R_0 \xrightarrow{f} S) \xrightarrow{(h, u)} (R_0 \xrightarrow{f'} S') \right] \longmapsto \left[\xi(f) \xrightarrow{h_*} \xi(hf) \xrightarrow{u(\varphi)} \xi(f') \right]$$

on arrows. For every $S \in \underline{C}$, the functor

$$(\xi, \varphi)(S) : \underline{h}_{(R_0, R_1, \dots)}(S) \longrightarrow \underline{F}(S)$$

is fully faithful by the representability of Hom_{ξ} . It follows that

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{F}$$

is a $\underline{C}$ -equivalence if and only if ξ is quasi-initial in $\underline{F}$ over $\underline{C}$, that is, if every object a of $\underline{F}$ admits a cocartesian arrow $\xi \longrightarrow a$.

One may instead consider the functor

$$\text{Isom}_{\xi} : (R_0 \amalg R_0, \underline{C}) \longrightarrow (\text{Sets}),$$

given by

$$X = (f_1, f_2) \mapsto \text{Isom}_{\underline{F}(S)}(\xi(f_1), \xi(f_2)),$$

where S is the common target of f_1 and f_2 . If this functor is represented by an object $(R_0 \rightrightarrows R_1)$ of $(R_0 \amalg R_0, \underline{C})$, then it defines a cogroupoid $(R_0, R_1, \dots)$ in $\underline{C}$, together with a functor

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{F}.$$

When $\underline{F}$ is a cofibered groupoid, this functor is a $\underline{C}$ -equivalence if and only if ξ is quasi-initial in $\underline{F}$ over $\underline{C}$. Definition and Proposition 2.11. A cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$ is called smooth if either, and hence both, of the following equivalent conditions holds:

- (i) R_1 is a formal power-series ring over R_0 via s_1 (and via s_2);
- (ii) the functor

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0, R_1, \dots)}$$

is smooth over $\underline{C}_\Lambda$.

A minimally smooth cogroupoid, also called a normalized smooth cogroupoid, in $\widehat{\underline{C}}_\Lambda$ is a smooth cogroupoid $(R_0, R_1, \dots)$ for which $\underline{h}_{(R_0, R_1, \dots)}(k[\varepsilon])$ is totally disconnected; equivalently,

$$s_1(k[\varepsilon]) = s_2(k[\varepsilon]).$$

Proof. Smoothness of $\epsilon : \underline{h}_{R_0} \rightarrow \underline{h}_{(R_0, R_1, \dots)}$ means that, for every surjective map

$$(S', f') \xrightarrow{(h, u)} (S, f)$$

in $\underline{h}_{(R_0, R_1, \dots)}$, there are a lift f'' of f to S' and an isomorphism

$$(S', f'') \xrightarrow{(1, u')} (S', f').$$

Equivalently, given $u \in h_{R_1}(S)$ and a lift f' of $s_1(u)$ to S' , there is a lift u' of u to S' such that $s_1(u') = f'$, for every surjection $S' \rightarrow S$ in $\underline{C}_\Lambda$. This is precisely the smoothness of

$$s_1 : \underline{h}_{R_1} \longrightarrow \underline{h}_{R_0}$$

over $\underline{C}_\Lambda$, and that in turn says that R_1 is a formal power-series ring over R_0 via s_1 . □

Proposition 2.12. Let $(R_0, R_1, \dots)$ be a cogroupoid in $\widehat{\underline{C}}_\Lambda$.

(a) If $(R_0, R_1, \dots)$ is smooth, then $\underline{h}_{(R_0, R_1, \dots)}$ is homogeneous.

(b) If $s_1(k[\varepsilon]) = s_2(k[\varepsilon])$, then the following conditions are equivalent:

- (i) $(R_0, R_1, \dots)$ is smooth;
- (ii) $\underline{h}_{(R_0, R_1, \dots)}$ is homogeneous;
- (iii) $\underline{h}_{(R_0, R_1, \dots)}$ is semi-homogeneous.

(c) If $\underline{h}_{(R_0, R_1, \dots)}$ is semi-homogeneous, then it is equivalent to $\underline{h}_{(S_0, S_1, \dots)}$ for some minimally smooth cogroupoid S_* .

Proof. (a) Consider a diagram

$$\begin{array}{ccc} (S_1, f_1) & & (S_2, f_2) \\ & \searrow (h_1, u_1) & \swarrow (h_2, u_2) \\ & (S, f) & \end{array}$$

in $\underline{h}_{(R_0, R_1, \dots)}$, where $h_2 : S_2 \rightarrow S$ is surjective in $\underline{C}_\Lambda$. Since $s_1 : \underline{h}_{R_1} \rightarrow \underline{h}_{R_0}$ is smooth, $u_1^{-1}u_2$ lifts to an element $v \in h_{R_1}(S_2)$ with $s_1(v) = f_2$. We have

$$h_2 \circ s_2(v) = h_1 \circ f_1,$$

so $f_1 \times s_2(v)$ is an element of $h_{R_0}(S_1 \times_S S_2)$ and gives a commutative diagram

$$\begin{array}{ccc}
 & (S_1 \times_S S_2, f_1 \times s_2(v)) & \\
 \pi_1 \swarrow & & \searrow (\pi_2, v^{-1}) \\
 (S_1, f_1) & & (S_2, f_2) \\
 \searrow (h_1, u_1) & & \swarrow (h_2, u_2) \\
 & (S, f) &
 \end{array}$$

We claim that its top object is the fibre product

$$(S_1, f_1) \times_{(S, f)} (S_2, f_2)$$

in $\underline{h}_{(R_0, R_1, \dots)}$. Indeed, for every object (T, g) , a map to the top object is equivalently a tuple $(h'_1 \times h'_2, w_1, w_2)$ satisfying

$$\begin{aligned}
 h_1 w_1 &= h_2 w_2, & s_1(w_1) &= h'_1 g, & s_2(w_1) &= f_1, \\
 s_1(w_2) &= h'_2 g, & s_2(w_2) &= s_2(v).
 \end{aligned}$$

Putting $w'_2 = v^{-1} w_2$ gives a bijection with the tuples $(h'_1 \times h'_2, w_1, w'_2)$ satisfying

$$\begin{aligned}
 u_2 h_2 w'_2 &= u_1 h_1 w_1, & s_1(w_1) &= h'_1 g, & s_2(w_1) &= f_1, \\
 s_1(w'_2) &= h'_2 g, & s_2(w'_2) &= f_2.
 \end{aligned}$$

These are exactly the pairs of arrows from (T, g) to (S_1, f_1) and (S_2, f_2) whose composites to (S, f) coincide. Thus

$$\begin{aligned}
 \text{Hom}((T, g), (S_1 \times_S S_2, f_1 \times s_2(v))) \\
 \simeq \text{Hom}((T, g), (S_1, f_1)) \times_{\text{Hom}((T, g), (S, f))} \text{Hom}((T, g), (S_2, f_2)),
 \end{aligned}$$

which proves (a).

(b) It remains only to prove (iii) $\Rightarrow$ (i). If $\underline{h}_{(R_0, R_1, \dots)}$ is semi-homogeneous, then

$$\dim[\underline{h}_{(R_0, R_1, \dots)}(k[\varepsilon])] = \dim h_{R_0}(k[\varepsilon]) < \infty,$$

so it admits a minimally versal formal object. On the other hand, $(R_0, 1_{R_0})$ is a quasi-initial minimal object of $\underline{h}_{(R_0, R_1, \dots)}$, because

$$h_{R_0}(k[\varepsilon]) \longrightarrow [\underline{h}_{(R_0, R_1, \dots)}(k[\varepsilon])]$$

is bijective. Proposition 1.10(ii) therefore makes $(R_0, 1_{R_0})$ a minimally versal formal object of $\underline{h}_{(R_0, R_1, \dots)}$. Equivalently,

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0, R_1, \dots)}$$

is minimally smooth. This proves (i). (c) Let (S_0, ξ_0) be a minimally versal formal object of $\underline{h}_{(R_0, R_1, \dots)}$ (Theorem 1.11). For any object

$$\xi : R_0 \longrightarrow T \quad \text{of } \underline{h}_{(R_0, R_1, \dots)}(T),$$

consider the functor

$$\text{Isom}(\xi, \xi) : \underline{C}_\Lambda / (T \hat{\otimes}_\Lambda T) \longrightarrow (\text{Sets}).$$

It assigns to R the set of isomorphisms between the two images of ξ on R obtained from

$$R_0 \xrightarrow{\xi} T \rightrightarrows T \hat{\otimes}_\Lambda T \longrightarrow R.$$

This functor is represented by

$$R_1 \otimes_{R_0 \widehat{\otimes}_\Lambda R_0} (T \widehat{\otimes}_\Lambda T).$$

Passing to the inverse limit shows that $\text{Isom}(\xi_0, \xi_0)$ is prorepresented by an object S_1 , and that

$$\underline{h}_{(R_0, R_1, \dots)} \simeq \underline{h}_{(S_0, S_1, \dots)}.$$

By (b), the cogroupoid S_* is minimally smooth. $\square$

Definition 2.13. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called (minimally) prorepresentable if it is $\underline{C}_\Lambda$ -equivalent to $\underline{h}_{(R_0, R_1, \dots)}$ for some (normalized) cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$.

Proposition 2.14. If $(R_0, R_1, \dots)$ is a normalized cogroupoid in $\widehat{\underline{C}}_\Lambda$, then every homomorphism

$$(\alpha_0, \alpha_1) : (R_0, R_1, \dots) \longrightarrow (R_0, R_1, \dots)$$

which induces a $\underline{C}_\Lambda$ -equivalence

$$\underline{h}_{(\alpha_0, \alpha_1)} : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{h}_{(R_0, R_1, \dots)}$$

is an isomorphism. In

particular, a normalized prorepresentation of a cofibered groupoid over $\underline{C}_\Lambda$, if it exists, is unique up to isomorphism.

Proof. Consider the commutative diagram

$$\begin{array}{ccc} \underline{h}_{(R_0, R_1, \dots)} & \xrightarrow{\underline{h}_{(\alpha_0, \alpha_1)}} & \underline{h}_{(R_0, R_1, \dots)} \\ \uparrow \epsilon & & \uparrow \epsilon \\ \underline{h}_{R_0} & \xrightarrow{\underline{h}_{\alpha_0}} & \underline{h}_{R_0} \end{array}$$

in which $\underline{h}_{(\alpha_0, \alpha_1)}$ is a $\underline{C}_\Lambda$ -equivalence. Since $\underline{h}_{(R_0, R_1, \dots)}(k[\varepsilon])$ is totally disconnected, both

$$[\underline{h}_{(\alpha_0, \alpha_1)}(k[\varepsilon])] \quad \text{and} \quad [\epsilon(k[\varepsilon])]$$

are bijections, and hence so is $\underline{h}_{\alpha_0}(k[\varepsilon])$. Thus $\alpha_0 : R_0 \longrightarrow R_0$ is a surjective endomorphism and therefore an automorphism. It follows that α_1 is also an automorphism of R_1 . $\square$ Lemma 2.15. Let A and B be discrete groupoids over $\underline{C}_\Lambda$, and let $\varphi : A \longrightarrow B$ be a minimally smooth functor. If A and B are homogeneous, then φ is a $\underline{C}_\Lambda$ -equivalence.

Proof. We must show that $\varphi(S) : A(S) \longrightarrow B(S)$ is bijective for every $S \in \text{ob}(\underline{C}_\Lambda)$. It is surjective for every S and bijective for $S = k[\varepsilon]$. We argue by induction on the length of S . Let $S' \xrightarrow{f} S$ be a small extension in $\underline{C}_\Lambda$. Since A and B are homogeneous discrete groupoids, the canonical isomorphism

$$1 \times \tilde{f} : S' \times_S S' \xrightarrow{\sim} S' \times_k k[\text{Ker } f]$$

induces a commutative diagram

$$\begin{array}{ccc} A(S') \times_{A(S)} A(S') & \xrightarrow{\sim} & A(S') \times A(k[\text{Ker } f]) \\ \downarrow & & \downarrow \\ B(S') \times_{B(S)} B(S') & \xrightarrow{\sim} & B(S') \times B(k[\text{Ker } f]). \end{array}$$

Thus $A(S')$ and $B(S')$ are principal homogeneous sets over $A(S)$ and $B(S)$, with structural groups identified by

$$A(k[\text{Ker } f]) \xrightarrow{\sim} B(k[\text{Ker } f]).$$

If $\varphi(S)$ is bijective, it follows that $\varphi(S')$ is injective, and hence bijective. This completes the induction. $\square$

Corollary 2.16. Let $\underline{A}$ be a homogeneous groupoid over $\underline{C}_\Lambda$ such that

$$\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$$

is finite-dimensional. For every $\xi \in \mathrm{ob}(\widehat{\underline{A}})$, the functor

$$\mathrm{Isom}_\xi : \widehat{\underline{C}}_{R \widehat{\otimes}_\Lambda R} \longrightarrow (\mathrm{Sets})$$

is prorepresentable, where $R = P(\xi)$.

Proof. If Isom_ξ is homogeneous, Theorem 1.11 supplies a minimally smooth functor

$$\Phi : \underline{h}_{R_1} \longrightarrow \underline{\mathrm{Isom}}_\xi.$$

Lemma 2.15 makes Φ a $\widehat{\underline{C}}_{R \widehat{\otimes}_\Lambda R}$ -equivalence, and hence Isom_ξ is prorepresentable. It remains to prove homogeneity. Consider a diagram

$$\begin{array}{ccc} T_1 & & T_2 \\ & \searrow & \swarrow \\ & T & \end{array}$$

in $\widehat{\underline{C}}_{R \widehat{\otimes}_\Lambda R}$, with $T_2 \longrightarrow T$ surjective. Since $\underline{A}$ is homogeneous, the fibre products

$$\xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2)$$

exist in $\underline{A}$, and there are canonical isomorphisms

$$\xi_i(T_1 \times_T T_2) \xrightarrow{\sim} \xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2) \quad (i = 1, 2)$$

(cf. 2.6(a)). Therefore

$$\mathrm{Isom}_\xi(T_1 \times_T T_2) \longrightarrow \mathrm{Isom}_\xi(T_1) \times_{\mathrm{Isom}_\xi(T)} \mathrm{Isom}_\xi(T_2)$$

is an isomorphism. $\square$

Theorem 2.17. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then it is prorepresentable by a (normalized) smooth cogroupoid if and only if

- (1) $\underline{A}$ is homogeneous;
- (2) $[\underline{A}(k[\varepsilon])]$ and $\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$ are finite-dimensional vector spaces over k .

Proof. Suppose first that $\underline{A}$ is prorepresented by a smooth cogroupoid. Thus $\underline{A}$ is $\underline{C}_\Lambda$ -equivalent to

$$\underline{h} = \underline{h}_{(R_0, R_1, \dots)}$$

for a smooth cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$. Proposition 2.12(a) shows that $\underline{h}$ is homogeneous. Moreover,

$$\begin{aligned} [\underline{h}(k[\varepsilon])] &= \mathrm{Coker} \left(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow[s_2]{s_1} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon]) \right), \\ \mathrm{Aut}_{\underline{h}}(1_\varepsilon) &= \mathrm{Ker} \left(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow{s_1} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon]) \right) \\ &= \mathrm{Ker} \left(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow{s_2} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon]) \right). \end{aligned}$$

Both are finite-dimensional vector spaces over k , so the same is true for $\underline{A}$.

Conversely, assume (1) and (2). Since $\underline{A}$ is homogeneous and $[\underline{A}(k[\varepsilon])]$ is finite-dimensional, it has a minimally versal formal object ξ . Put $P(\xi) = R_0$ and consider

$$\mathrm{Isom}_\xi : \widehat{\underline{C}}_{R_0 \widehat{\otimes}_\Lambda R_0} \longrightarrow (\mathrm{Sets}),$$

given by

$$T \longmapsto \mathrm{Isom}_{\underline{A}(T)}(\xi_1(T), \xi_2(T)),$$

where $\xi_i(T) = \xi(f_i)$ and $f_i : R_0 \longrightarrow T$ are the composite maps in

$$R_0 \begin{array}{c} \xrightarrow{s_1} \\ \xrightarrow{s_2} \end{array} R_0 \widehat{\otimes}_\Lambda R_0 \longrightarrow T.$$

Corollary 2.16 prorepresents this functor by an object

$$R_1 \in \mathrm{ob}(\widehat{\underline{C}}_{R_0 \widehat{\otimes}_\Lambda R_0})$$

and an isomorphism $\xi_1(R_1) \xrightarrow{\varphi} \xi_2(R_1)$ for which

$$\Phi : \underline{h}_{R_1} \longrightarrow \underline{\mathrm{Isom}}_\xi$$

is an isomorphism over $\widehat{\underline{C}}_{R_0 \widehat{\otimes}_\Lambda R_0}$. The pair (R_0, R_1) therefore defines a cogroupoid, and

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{A}$$

is a $\underline{C}_\Lambda$ -equivalence. Because ξ is minimally versal, the composite

$$\underline{h}_{(R_0, R_0, \dots)} \xrightarrow{\epsilon} \underline{h}_{(R_0, R_1, \dots)} \xrightarrow{(\xi, \varphi)} \underline{A}$$

is minimally smooth. Since (ξ, φ) is an equivalence, ϵ is minimally smooth as well. Thus $(R_0, R_1, \dots)$ is a normalized smooth cogroupoid in $\widehat{\underline{C}}_\Lambda$. $\square$

Corollary 2.18. A functor $F : \underline{C}_\Lambda \longrightarrow (\mathrm{Sets})$ is prorepresentable if and only if it is homogeneous and $\dim F(k[\varepsilon]) < \infty$.

Corollary 2.19. Every smooth cogroupoid in $\widehat{\underline{C}}_\Lambda$ can be normalized up to isomorphism.

Given a cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$, set $\bar{R}_1 = R_1/J$, where J is the ideal of R_1 generated by

$$\{s_1(r) - s_2(r) \mid r \in R_0\}.$$

Then $(R_0, \bar{R}_1, \dots)$ defines a formal group scheme over R_0 with the induced composition law.

Now let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \dots)$ via (ξ, φ) . Let

$$\underline{C}_k \hookrightarrow \underline{C}_{R_0} \hookrightarrow \underline{C}_{R_0 \widehat{\otimes}_\Lambda R_0}$$

be the embeddings corresponding to the surjections

$$R_0 \widehat{\otimes}_\Lambda R_0 \longrightarrow R_0 \longrightarrow k,$$

and let Aut_ξ and Aut^o be the restrictions of Isom_ξ to $\underline{C}_{R_0}$ and $\underline{C}_k$, respectively. Thus $\mathrm{Aut}_\xi(S)$ is the automorphism group of $\xi(S)$ for $S \in \mathrm{ob}(\underline{C}_{R_0})$, while $\mathrm{Aut}^o(S)$ is the automorphism group of 1_S for $S \in \mathrm{ob}(\underline{C}_k)$, where 1_S is the direct image of $1 \in \underline{A}(k)$ under $k \longrightarrow S$.

Since R_1 prorepresents Isom_ξ on $\underline{C}_{R_0 \widehat{\otimes}_\Lambda R_0}$, it follows that $\bar{R}_1$ prorepresents Aut_ξ on $\underline{C}_{R_0}$, and that $k \otimes_{R_0} \bar{R}_1$ prorepresents Aut^o on $\underline{C}_k$. Although $\underline{A}$ is prorepresented by the smooth cogroupoid $(R_0, R_1, \dots)$, neither $(R_0, \bar{R}_1, \dots)$ nor $(k, k \otimes_{R_0} \bar{R}_1, \dots)$ is smooth in general. More precisely:

Corollary 2.20. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \dots)$ via (ξ, φ) .

- (a) $(R_0, \bar{R}_1, \dots)$ and $(k, k \otimes_{R_0} \bar{R}_1, \dots)$ prorepresent Aut_ξ and Aut^o , respectively.

(b) The following statements are equivalent:

- (i) the formal group scheme $\underline{h}_{(R_0, \bar{R}_1, \dots)}$ is formally smooth over R_0 ;
- (ii) for every surjective arrow $a' \rightarrow a$ in $\underline{A}$, the map

$$\mathrm{Aut}_{P(a')}(a') \rightarrow \mathrm{Aut}_{P(a)}(a)$$

is surjective;

- (iii) R_0 prorepresents the functor $[\underline{A}]$ over $\underline{C}_\Lambda$.

Proof. We have already noted (a). For (b), the equivalence (i) $\Leftrightarrow$ (ii) is clear, whereas (ii) $\Leftrightarrow$ (iii) follows from 2.7 and 2.15, since

$$\tilde{\xi} : \underline{h}_{R_0} \rightarrow [\underline{A}]$$

is minimally smooth. $\square$

Remark 2.21. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let $\underline{A}|_{\underline{C}_k}$ be the pullback of $p : \underline{A} \rightarrow \underline{C}_\Lambda$ along $\underline{C}_k \hookrightarrow \underline{C}_\Lambda$. If $\underline{A}$ is prorepresented by $(R_0, R_1, \dots)$ via (ξ, φ) , then $\underline{A}|_{\underline{C}_k}$ is prorepresented by

$$(k \otimes_\Lambda R_0, k \otimes_\Lambda R_1, \dots)$$

via $(k \otimes_\Lambda \xi, k \otimes_\Lambda \varphi)$. Here $k \otimes_\Lambda \xi$ is the direct image of ξ under $R_0 \rightarrow k \otimes_\Lambda R_0$, and $k \otimes_\Lambda \varphi$ is the image of φ under

$$\mathrm{Isom}_{\underline{A}(R_0)}(\xi(s_1), \xi(s_2)) \rightarrow \mathrm{Isom}_{\underline{A}(k \otimes_\Lambda R_1)}(k \otimes_\Lambda \xi(s_1), k \otimes_\Lambda \xi(s_2)).$$

If $(R_0, R_1, \dots)$ is normalized, then so is $(k \otimes_\Lambda R_0, k \otimes_\Lambda R_1, \dots)$; moreover, the relative dimension of R_1 over R_0 equals that of $k \otimes_\Lambda R_1$ over $k \otimes_\Lambda R_0$, since R_1 is smooth over R_0 .

Remark 2.22. Suppose that $\underline{A}$ is prorepresented by smooth cogroupoids $(R_0, R_1, \dots)$ and $(R'_0, R'_1, \dots)$, with $(R_0, R_1, \dots)$ normalized. We obtain a homomorphism

$$(i_0, i_1) : (R_0, R_1, \dots) \rightarrow (R'_0, R'_1, \dots).$$

By 2.10(b), R'_0 is a formal power-series ring over R_0 , and (i_0, i_1) induces an isomorphism

$$R'_0 \otimes_{R_0} (R_0, R_1, \dots) \xrightarrow{\sim} (R'_0, R'_1, \dots).$$

3. The coherent sheaves $\underline{D}_X^i$

One important functor which is usually not prorepresentable, but nevertheless admits a formal versal object, is the deformation functor. With this in view, we recall some basic facts about extensions of commutative rings and their obstruction theory. This section contains nothing new, except perhaps for its systematic use of Grothendieck cohomology on a site. Its purpose is to make this exposé as self-contained as possible; in fact, we prove somewhat more than will be needed later. The basic results are stated in Theorem 3.12. Throughout, all rings are commutative and have a unit.

Given a ring R , let $\underline{C}_R$ denote the category of R -algebras. For every $A \in \mathrm{Ob}(\underline{C}_R)$, put

$$\mathrm{Cov}(A) = \{\text{surjective arrows } \varphi : A' \rightarrow A \text{ in } \underline{C}_R\}.$$

This defines a pretopology, and hence a site, on $\underline{C}_R$. For every $A \in \mathrm{Ob}(\underline{C}_R)$, let $\underline{C}_{A|R}$ be the category of arrows in $\underline{C}_R$ with target A . We endow $\underline{C}_{A|R}$ with the topology induced by the forgetful functor

$$j_A : \underline{C}_{A|R} \rightarrow \underline{C}_R.$$

The category $\underline{C}_{A|R}$ has a final object $(1_A : A \rightarrow A)$ and an initial object $(i_A : R \rightarrow A)$, and it admits both fibre products and coproducts. When there is no risk of confusion, we write simply B for an object $(B \rightarrow A)$ of $\underline{C}_{A|R}$.

Let $\mathcal{C}$ be an abelian category. A $\mathcal{C}$ -valued sheaf on $\underline{C}_{A|R}$ is, by definition, a functor

$$F : \underline{C}_{A|R}^{\text{op}} \longrightarrow \mathcal{C}$$

such that, for every surjective arrow $B' \longrightarrow B$ in $\underline{C}_{A|R}$, the sequence

$$0 \longrightarrow F(B) \longrightarrow F(B') \rightrightarrows F(B' \times_B B')$$

is exact. The additive presheaves are those which preserve coproducts; thus

$$F(B_1 \otimes_R B_2) = F(B_1) \oplus F(B_2).$$

For example, an A -module M defines the functor

$$\text{Der}_R(-, M) : \underline{C}_{A|R}^{\text{op}} \longrightarrow (A\text{-mod}),$$

which sends B to the R -linear derivations $\text{Der}_R(B, M)$, with M regarded as a B -module via the structural map $B \longrightarrow A$. This is plainly an additive sheaf on $\underline{C}_{A|R}$. Here $(A\text{-mod})$ denotes the category of A -modules.

A presheaf F on $\underline{C}_{A|R}$ with values in $(A\text{-mod})$ is said to be of finite type if $F(B)$ is an A -module of finite type whenever B is an R -algebra of finite type. For example, if A is noetherian and M is an A -module of finite type, then $\text{Der}_R(-, M)$ is an additive sheaf of finite type on $\underline{C}_{A|R}$.

We record a little more notation and a few basic facts from homological algebra.

3.1(a). Throughout the remainder of this section, $\tilde{\underline{C}}_{A|R}$ and $\check{\underline{C}}_{A|R}$ denote, respectively, the categories of sheaves and presheaves with values in $(A\text{-mod})$. There is a left-exact inclusion functor

$$i_{A|R} : \tilde{\underline{C}}_{A|R} \longrightarrow \check{\underline{C}}_{A|R}$$

and an exact associated-sheaf functor

$$G_{A|R} : \check{\underline{C}}_{A|R} \longrightarrow \tilde{\underline{C}}_{A|R},$$

left adjoint to $i_{A|R}$.

An object $B \in \text{Ob}(\underline{C}_{A|R})$ induces a continuous functor $j : \underline{C}_{B|R} \longrightarrow \underline{C}_{A|R}$ and hence a commutative diagram

$$(*) \quad \begin{array}{ccc} \tilde{\underline{C}}_{A|R} & \xrightarrow{\tilde{j}} & \tilde{\underline{C}}_{B|R} \\ i_{A|R} \downarrow & & \downarrow \\ \check{\underline{C}}_{A|R} & \xrightarrow{j^\vee} & \check{\underline{C}}_{B|R}. \end{array}$$

The restriction functor $\tilde{j}$, as well as $j^\vee$, is exact, because the site $\underline{C}_{B|R}$ is simply the category of arrows in $\underline{C}_{A|R}$ with target B , endowed with the induced topology.

A ring homomorphism $S \longrightarrow R$ induces a continuous functor $\beta : \underline{C}_{A|S} \longrightarrow \underline{C}_{A|R}$ and hence a commutative diagram

$$(**) \quad \begin{array}{ccc} \tilde{\underline{C}}_{A|S} & \xrightarrow{\tilde{\beta}} & \tilde{\underline{C}}_{A|R} \\ i_{A|S} \downarrow & & \downarrow \\ \check{\underline{C}}_{A|S} & \xrightarrow{\beta^\vee} & \check{\underline{C}}_{A|R}. \end{array}$$

Notice that $\widetilde{\beta}$ need not be exact.

3.1(b). The right-derived functor of the inclusion

$$i_{A|R} : \widetilde{\underline{C}}_{A|R} \longrightarrow \bigvee \underline{C}_{A|R}$$

will be denoted by $\underline{H}_{A|R}^\bullet$. For every sheaf F on $\underline{C}_{A|R}$, put

$$H^\bullet(A|R, F) = [\underline{H}_{A|R}^\bullet(F)](A).$$

For $B' \longrightarrow B$ in $\text{Cov}(B)$, define

$$H^0(\{B' \longrightarrow B\}, F) = \ker(F(B') \rightrightarrows F(B' \times_B B')).$$

We denote the right-derived functors of $H^0(\{B' \longrightarrow B\}, -)$ by $H^\bullet(\{B' \longrightarrow B\}, -)$. These are the cohomology groups of the Čech semisimplicial module

$$F(B') \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} F(B' \times_B B') \begin{array}{c} \longrightarrow \\ \longrightarrow \\ \longrightarrow \end{array} F(B' \times_B B' \times_B B') \begin{array}{c} \longrightarrow \\ \longrightarrow \\ \longrightarrow \\ \longrightarrow \end{array} \cdots$$

The right-derived functor of

$$\check{H}_{A|R}^0 : \bigvee \underline{C}_{A|R} \longrightarrow \bigvee \underline{C}_{A|R}$$

will be denoted by $\check{\underline{H}}_{A|R}^\bullet$, where

$$[\check{\underline{H}}_{A|R}^0(F)](B) = \varinjlim_{\{B' \rightarrow B\} \in \text{Cov}(B)} H^0(\{B' \longrightarrow B\}, F).$$

For every presheaf F on $\underline{C}_{A|R}$, put

$$\check{H}^\bullet(A|R, F) = [\check{\underline{H}}_{A|R}^\bullet(F)](A) = \varinjlim_{\{A' \rightarrow A\} \in \text{Cov}(A)} H^\bullet(\{A' \longrightarrow A\}, F).$$

For every sheaf F on $\underline{C}_{A|R}$ there is the usual spectral sequence

$$\check{H}^p(A|R, \underline{H}_{A|R}^q(F)) \Longrightarrow H^{p+q}(A|R, F).$$

Here $\check{H}^0(A|R, \underline{H}_{A|R}^q(F)) = 0$ for every $q > 0$. Consequently,

$$\check{H}^1(A|R, F) = H^1(A|R, F),$$

and the sequence

$$0 \longrightarrow \check{H}^2(A|R, F) \longrightarrow H^2(A|R, F) \longrightarrow \check{H}^1(A|R, \underline{H}_{A|R}^1(F)) \longrightarrow \check{H}^3(A|R, F)$$

is exact. 3.1(c). If B is an object of $\underline{C}_{A|R}$, diagram $(*)$ in 3.1(a), together with the exactness of $\widetilde{j}$ and $j^\vee$, gives a canonical isomorphism

$$H^\bullet(B|R, \widetilde{j}F) = [\underline{H}_{A|R}^\bullet(F)](B).$$

On the other hand, a ring homomorphism $S \longrightarrow R$ induces diagram $(**)$ in 3.1(a). The functor $\widetilde{\beta}$ need not be exact; however, β preserves fibre products, and we obtain a spectral sequence

$$\underline{H}_{A|R}^p(R^q \widetilde{\beta}F) \Longrightarrow \beta^\vee \underline{H}_{A|S}^{p+q}(F),$$

or, on evaluating at A ,

$$H^p(A|R, R^q \widetilde{\beta}F) \Longrightarrow H^{p+q}(A|S, F).$$

Since there is a canonical isomorphism

$$\tilde{\beta} = G_{A|R} \circ \beta^\vee \circ i_{A|S}$$

and the functor $\beta^\# = G_{A|R} \circ \beta^\vee$ is exact, we have

$$R^\bullet \tilde{\beta} F = \beta^\# \underline{H}_{A|S}^\bullet(F).$$

Thus the preceding spectral sequence may be written

$$H^p(A|R, \beta^\# \underline{H}_{A|S}^q(F)) \implies H^{p+q}(A|S, F).$$

Lemma 3.2. Let $B \in \text{Ob}(\underline{C}_{A|R})$, and choose a surjective arrow $P \longrightarrow B$ in $\underline{C}_{B|R}$, where P is a polynomial algebra over R . Then, for every $F \in \text{Ob}(\underline{C}_{A|S}^\vee)$, the natural map

$$H^\bullet(\{P \longrightarrow B\}, F) \longrightarrow [\check{H}^\bullet(F)](B)$$

is an isomorphism.

Proof. It suffices to prove that the natural map

$$H^0(\{P \longrightarrow B\}, F) \longrightarrow [\check{H}^0(F)](B) = \varinjlim_{\{B' \rightarrow B\} \in \text{Cov}(B)} H^0(\{B' \longrightarrow B\}, F)$$

is an isomorphism. This is clear because, for every $\{B' \longrightarrow B\} \in \text{Cov}(B)$, there is a map of covers

$$\{P \longrightarrow B\} \longrightarrow \{B' \longrightarrow B\}.$$

□

Corollary 3.3.

- (a) Suppose that A is noetherian. If $F \in \text{Ob}(\tilde{\underline{C}}_{A|R})$ is of finite type, then $\underline{H}^n(F)$ is of finite type for every n .
- (b) If P is a polynomial algebra over R , then

$$H^n(P|R, F) = 0$$

for every $n > 0$ and every $F \in \text{Ob}(\tilde{\underline{C}}_{P|R})$.

- (c) A sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

in $\tilde{\underline{C}}_{A|R}$ is exact if and only if

$$0 \longrightarrow F'(P) \longrightarrow F(P) \longrightarrow F''(P) \longrightarrow 0$$

is exact for every polynomial R -algebra $P \in \text{Ob}(\underline{C}_{A|R})$.

Proof.

- (a) In view of the spectral sequence

$$\check{H}^p(B|R, \underline{H}^q(F)) \implies H^{p+q}(B|R, F)$$

and the equality $\check{H}^0(\underline{H}^q(F)) = 0$ for $q > 0$, it is enough to show that $\check{H}^n(G)$ is of finite type for every n and every presheaf G of finite type. By Lemma 3.2,

$$\check{H}^\bullet(B|R, G) = H^\bullet(\{P \longrightarrow B\}, G),$$

where P is a polynomial algebra over R and $\{P \longrightarrow B\} \in \text{Cov}(B)$. The latter is the cohomology of the complex

$$\cdots G(P \times_B P \times_B P) \xleftarrow{\quad} G(P \times_B P) \xleftarrow{\quad} G(P).$$

If B is an R -algebra of finite type, P may be chosen of finite type over R . All the iterated fibre products $P \times_B \cdots \times_B P$ are then R -algebras of finite type. If G is of finite type, their images under G are A -modules of finite type. Hence $H^\bullet(\{P \rightarrow B\}, G)$ is an A -module of finite type because A is noetherian.

(b) Let P be a polynomial algebra over R . Every surjective R -algebra map $B \rightarrow P$ admits a section, so

$$H^i(\{B \rightarrow P\}, G) = 0 \quad (i > 0).$$

It follows that $\check{H}^i(P|R, G) = 0$ for every $i > 0$ and every presheaf G on $\underline{C}_{P|R}$. Consequently the spectral sequence

$$\check{H}^p(P|R, \underline{H}_{P|R}^q(F)) \Rightarrow H^{p+q}(P|R, F)$$

degenerates, and

$$\check{H}^0(P|R, \underline{H}^n(F)) \simeq H^n(P|R, F)$$

for every n . For a sheaf F , however, $\check{H}^0(P|R, \underline{H}^n(F)) = 0$ whenever $n > 0$. Thus $H^n(P|R, F) = 0$ for every $n > 0$.

(c) If $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ is exact in $\tilde{\underline{C}}_{A|R}$, then the sequence

$$0 \rightarrow F'(B) \rightarrow F(B) \rightarrow F''(B) \rightarrow H^1(B|R, F') \rightarrow H^1(B|R, F) \rightarrow \cdots$$

is exact. If $P \in \text{Ob}(\underline{C}_{A|R})$ is a polynomial algebra over R , then $H^1(P|R, F') = 0$ by (b), and hence

$$0 \rightarrow F'(P) \rightarrow F(P) \rightarrow F''(P) \rightarrow 0$$

is exact.

Conversely, let $\underline{P}_{A|R}$ be the full subcategory of $\underline{C}_{A|R}$ with

$$\text{Ob}(\underline{P}_{A|R}) = \{P \in \text{Ob}(\underline{C}_{A|R}) \mid P \text{ is a polynomial algebra over } R\},$$

endowed with the induced topology, and let $p : \underline{P}_{A|R} \rightarrow \underline{C}_{A|R}$ be the inclusion. Every object of $\underline{C}_{A|R}$ is covered by an object of $\underline{P}_{A|R}$; the comparison lemma therefore shows that

$$\tilde{p} : \tilde{\underline{C}}_{A|R} \rightarrow \tilde{\underline{P}}_{A|R}$$

is an equivalence of categories. If the sequence obtained by evaluating $F' \rightarrow F \rightarrow F''$ at every $P \in \text{Ob}(\underline{P}_{A|R})$ is exact, then

$$0 \rightarrow \tilde{p}F' \rightarrow \tilde{p}F \rightarrow \tilde{p}F'' \rightarrow 0$$

is exact in $\tilde{\underline{P}}_{A|R}$, and therefore the original sequence is exact in $\tilde{\underline{C}}_{A|R}$. $\square$

Remark 3.4. Every surjective arrow in $\underline{P}_{A|R}$ admits a section. It follows at once that every presheaf on $\underline{P}_{A|R}$ is a sheaf; that is,

$$\tilde{\underline{P}}_{A|R} = \check{\underline{P}}_{A|R}.$$

Consequently, the restriction functor

$$\tilde{\underline{C}}_{A|R} \rightarrow \check{\underline{P}}_{A|R}$$

is an equivalence of categories. Lemma 3.5. Consider a diagram of R -algebras

$$\begin{array}{ccc} & C & \\ & \downarrow & \\ B' & \xrightarrow{\text{surjective}} & B \end{array}$$

and let S be an R -algebra.

(i) If $\text{Tor}_1^R(B, S) = 0$, then the canonical map

$$(B' \times_B C) \otimes_R S \longrightarrow (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S)$$

is an isomorphism.

(ii) Suppose that B' is flat over R . If

$$\text{Tor}_i^R(C, S) = 0 \quad (0 < i < n) \quad \text{and} \quad \text{Tor}_i^R(B, S) = 0 \quad (0 < i \leq n),$$

then

$$\text{Tor}_i^R(B' \times_B C, S) = 0 \quad (0 < i < n).$$

Proof. Let $0 \longrightarrow J \longrightarrow B' \longrightarrow B \longrightarrow 0$ be exact.

(i) If $\text{Tor}_1^R(B, S) = 0$, then

$$0 \longrightarrow J \otimes_R S \longrightarrow B' \otimes_R S \longrightarrow B \otimes_R S \longrightarrow 0$$

is exact, and hence so is

$$0 \longrightarrow J \otimes_R S \longrightarrow (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S) \longrightarrow C \otimes_R S \longrightarrow 0.$$

We therefore obtain a commutative diagram with exact rows:

$$\begin{array}{ccccccc} J \otimes_R S & \longrightarrow & (B' \times_B C) \otimes_R S & \longrightarrow & C \otimes_R S & \longrightarrow & 0 \\ \parallel & & \downarrow & & \parallel & & \\ 0 \longrightarrow J \otimes_R S & \longrightarrow & (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S) & \longrightarrow & C \otimes_R S & \longrightarrow & 0. \end{array}$$

It follows that

$$(B' \times_B C) \otimes_R S \longrightarrow (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S)$$

is an isomorphism.

(ii) Since B' is flat over R and $\text{Tor}_i^R(B, S) = 0$ for $0 < i \leq n$, the long exact Tor sequence gives

$$\text{Tor}_i^R(J, S) = 0 \quad (0 < i < n).$$

Applying the same sequence to

$$0 \longrightarrow J \longrightarrow B' \times_B C \longrightarrow C \longrightarrow 0$$

and using the hypothesis on C gives

$$\text{Tor}_i^R(B' \times_B C, S) = 0 \quad (0 < i < n).$$

□

Corollary 3.6.

(a) Let A and S be R -algebras. If $\text{Tor}_1^R(A, S) = 0$, then for every $G \in \text{Ob}(\underline{C}_{A \otimes_R S|S}^\vee)$ the canonical map

$$\check{H}^\bullet(A \otimes_R S|S, G) \longrightarrow \check{H}^\bullet(A|R, \check{S}G)$$

is an isomorphism. Here

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R S|S}, \quad X \longmapsto X \otimes_R S.$$

In particular, if S is flat over R , then

$$\check{S} \check{H}^\bullet(G) \longrightarrow \check{H}^\bullet(\check{S}G)$$

is an isomorphism for every such G .

(b) Let $P \longrightarrow B$ be a surjective map of R -algebras, where P is a polynomial algebra over R . If

$$\mathrm{Tor}_i^R(B, S) = 0 \quad (0 < i \leq n),$$

then, for every iterated fibre product of copies of P over B ,

$$\mathrm{Tor}_i^R(P \times_B P \times_B \cdots \times_B P, S) = 0 \quad (0 < i < n).$$

Proof.

(a) Given $B \in \mathrm{Ob}(\underline{\mathcal{C}}_{A|R})$, choose a cover $\{P \longrightarrow B\} \in \mathrm{Cov}(B)$ with P a polynomial algebra over R . If $\mathrm{Tor}_1^R(B, S) = 0$, Lemma 3.5(i) shows that the natural map

$$\left(\overbrace{P \times_B \cdots \times_B P}^n \right) \otimes_R S \longrightarrow \overbrace{(P \otimes_R S) \times_{B \otimes_R S} \cdots \times_{B \otimes_R S} (P \otimes_R S)}^n$$

is an isomorphism for every n . Consequently,

$$\begin{aligned} \check{H}^\bullet(B|R, \check{S}G) &= H^\bullet(\{P \longrightarrow B\}, \check{S}G) \\ &= H^\bullet(\{P \otimes_R S \longrightarrow B \otimes_R S\}, G) \\ &= \check{H}^\bullet(B \otimes_R S|S, G). \end{aligned}$$

If S is flat over R , then $\mathrm{Tor}_1^R(X, S) = 0$ for every $X \in \mathrm{Ob}(\underline{\mathcal{C}}_{A|R})$. The preceding isomorphism is therefore natural in X , and gives

$$\check{S} \check{H}^\bullet(G) \xrightarrow{\sim} \check{H}^\bullet(\check{S}G).$$

(b) Put

$$P^{(m)} = \overbrace{P \times_B \cdots \times_B P}^{m+1}.$$

For $m = 0$ there is nothing to prove, since P is flat over R . Suppose inductively that $\mathrm{Tor}_i^R(P^{(m-1)}, S) = 0$ for $0 < i < n$. Since $P \longrightarrow B$ is surjective and $P^{(m)} = P \times_B P^{(m-1)}$, Lemma 3.5(ii) gives

$$\mathrm{Tor}_i^R(P^{(m)}, S) = 0 \quad (0 < i < n).$$

□

Proposition 3.7.

(a) Let A and S be R -algebras. If

$$\mathrm{Tor}_i^R(A, S) = 0 \quad (0 < i \leq n),$$

then, for every sheaf F on $\underline{\mathcal{C}}_{A \otimes_R S|S}$, the canonical map

$$H^i(A \otimes_R S|S, F) \longrightarrow H^i(A|R, \tilde{S}F)$$

is an isomorphism for every $i \leq n$. Here

$$S : \underline{\mathcal{C}}_{A|R} \longrightarrow \underline{\mathcal{C}}_{A \otimes_R S|S}, \quad X \longmapsto X \otimes_R S.$$

(b) Let P be a polynomial algebra over R . Then, for every additive sheaf F on $\underline{\mathcal{C}}_{A \otimes_R P|R}$, the canonical map

$$H^n(A \otimes_R P|R, F) \longrightarrow H^n(A|R, F)$$

is an isomorphism for every $n > 0$.

(c) Let $S \longrightarrow R \longrightarrow A$ be ring homomorphisms. Then, for every additive sheaf F on $\underline{C}_{A|S}$, there is an exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(A|R, F_R) \longrightarrow H^0(A|S, F) \longrightarrow H^0(R|S, F) \longrightarrow H^1(A|R, F_R) \\ \longrightarrow H^1(A|S, F) \longrightarrow H^1(R|S, F) \longrightarrow \cdots \longrightarrow H^n(A|R, F_R) \\ \longrightarrow H^n(A|S, F) \longrightarrow H^n(R|S, F) \longrightarrow H^{n+1}(A|R, F_R) \longrightarrow \cdots, \end{aligned}$$

where F_R is the sheaf on $\underline{C}_{A|R}$ defined by

$$F_R(X) = \ker(F(X) \longrightarrow F(R))$$

for every $X \in \text{Ob}(\underline{C}_{A|R})$, via the forgetful functor $\underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$.

Proof.

(a) Consider the continuous functor

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R S|S}, \quad S(X) = X \otimes_R S.$$

Since S sends polynomial algebras over R to polynomial algebras over S , the functor

$$\tilde{S} : \underline{\tilde{C}}_{A \otimes_R S|S} \longrightarrow \underline{\tilde{C}}_{A|R}$$

is exact by 3.3(c). We therefore obtain the canonical map

$$\check{S} \underline{H}_{A \otimes_R S|S}^\bullet(F) \longrightarrow \underline{H}_{A|R}^\bullet(\tilde{S}F)$$

and the following commutative diagram of spectral sequences:

$$\begin{array}{ccc} E_S^{p,q}(S(X)) = \check{H}^p(X \otimes_R S|S, \underline{H}_{A \otimes_R S|S}^q(F)) & \Longrightarrow & H^{p+q}(X \otimes_R S|S, F) \\ \downarrow & & \downarrow \\ \check{H}^p(X|R, \check{S} \underline{H}_{A \otimes_R S|S}^q(F)) & & \\ \downarrow & & \\ E_R^{p,q}(X) = \check{H}^p(X|R, \underline{H}_{A|R}^q(\tilde{S}F)) & \Longrightarrow & H^{p+q}(X|R, \tilde{S}F). \end{array}$$

By 3.6(a), the map

$$E_S^{p,q}(S(X)) \longrightarrow \check{H}^p(X|R, \check{S} \underline{H}_{A \otimes_R S|S}^q(F))$$

is an isomorphism whenever $\text{Tor}_1^R(X, S) = 0$. Suppose inductively that

$$[\check{S} \underline{H}_{A \otimes_R S|S}^q(F)](X) \longrightarrow [\underline{H}_{A|R}^q(\tilde{S}F)](X)$$

is an isomorphism for every $q < n$ and every $X \in \text{Ob}(\underline{C}_{A|R})$ satisfying

$$\text{Tor}_i^R(X, S) = 0 \quad (0 < i \leq n).$$

It follows from 3.6(b) that $E_S^{p,q}(S(X)) \longrightarrow E_R^{p,q}(X)$ is an isomorphism for every $q < n$ and every p , under the same Tor hypothesis. Moreover, $E_S^{0,n}(S(X)) \longrightarrow E_R^{0,n}(X)$ is an isomorphism, since both sides are zero. Hence

$$H^n(X \otimes_R S|S, F) \longrightarrow H^n(X|R, \tilde{S}F)$$

is an isomorphism whenever $\text{Tor}_i^R(X, S) = 0$ for $0 < i \leq n$. Equivalently,

$$[\check{S} \underline{H}_{A \otimes_R S|S}^i(F)](X) \longrightarrow [\underline{H}_{A|R}^i(\tilde{S}F)](X)$$

is an isomorphism for every $i \leq n$ under that hypothesis.

(b) Let

$$f : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R P|R}, \quad f(X) = X \otimes_R P.$$

Choose a surjective arrow $P' \longrightarrow X$ in $\underline{C}_{A|R}$, with P' polynomial over R . Then

$$P' \otimes_R P \longrightarrow X \otimes_R P$$

is a surjective arrow in $\underline{C}_{A \otimes_R P|R}$, and $P' \otimes_R P$ is again polynomial over R . Since P is flat over R , 3.5(i) gives

$$\begin{aligned} \check{H}^\bullet(X|R, \check{f}G) &= H^\bullet(\{P' \longrightarrow X\}, \check{f}G) \\ &= H^\bullet(\{P' \otimes_R P \longrightarrow X \otimes_R P\}, G) \\ &= \check{H}^\bullet(X \otimes_R P|R, G) \end{aligned}$$

for every $G \in \text{Ob}(\check{\underline{C}}_{A \otimes_R P|R})$. Thus $\check{f}$ sends acyclic sheaves to acyclic sheaves. On the other hand, f sends polynomial algebras over R to polynomial algebras over R , so

$$\tilde{f} : \tilde{\underline{C}}_{A \otimes_R P|R} \longrightarrow \tilde{\underline{C}}_{A|R}$$

is exact by 3.3(c). It follows that

$$H^\bullet(A \otimes_R P|R, F) \longrightarrow H^\bullet(A|R, \tilde{f}F)$$

is an isomorphism for every $F \in \text{Ob}(\tilde{\underline{C}}_{A \otimes_R P|R})$.

For each $X \in \text{Ob}(\underline{C}_{A|R})$ there is a canonical map

$$(\tilde{f}F)(X) = F(X \otimes_R P) \longrightarrow F(X) \oplus F(P),$$

and hence a canonical map $\tilde{f}F \longrightarrow F \oplus F(P)$, where $F(P)$ denotes the constant sheaf and F on the right denotes the restriction of F to $\underline{C}_{A|R}$. If F is additive, this map is an isomorphism. Therefore, for every $n > 0$,

$$H^n(A|R, \tilde{f}F) \xrightarrow{\sim} H^n(A|R, F \oplus F(P)) = H^n(A|R, F),$$

because the constant sheaf $F(P)$ is flasque. This proves (b).

(c) This argument was communicated by Rinehart. Consider the forgetful functor $\beta : \underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$. There is a spectral sequence

$$E^{p,q} = H^p(A|R, \beta^\# \underline{H}_{A|S}^q(F)) \implies H^{p+q}(A|S, F),$$

where

$$\beta^\# = G_{A|R} \circ \beta^\vee, \quad G_{A|R} : \check{\underline{C}}_{A|R} \longrightarrow \tilde{\underline{C}}_{A|R}$$

is the associated-sheaf functor.

Suppose now that F is additive. We claim that $\beta^\# \underline{H}_{A|S}^q(F)$ is a constant sheaf for every $q > 0$. Indeed, if P is a polynomial algebra over R , then $P = R \otimes_S P'$ for a polynomial algebra P' over S , and

$$[\beta^\# \underline{H}_{A|S}^q(F)](P) = H^q(P|S, F) = H^q(R \otimes_S P'|S, F) \longrightarrow H^q(R|S, F)$$

is an isomorphism for $q > 0$ by (b), and is surjective for $q = 0$. By 3.3(c), the canonical map

$$\beta^\# \underline{H}_{A|S}^q(F) \longrightarrow H^q(R|S, F)$$

to the corresponding constant sheaf is therefore an isomorphism for $q > 0$ and a surjection for $q = 0$. Hence $E^{p,q} = 0$ whenever $p, q > 0$, and the spectral sequence yields the exact sequence

$$(*) \quad 0 \longrightarrow E^{1,0} \longrightarrow H^1 \longrightarrow E^{0,1} \longrightarrow E^{2,0} \longrightarrow H^2 \longrightarrow E^{0,2} \longrightarrow E^{3,0} \longrightarrow H^3 \longrightarrow \dots$$

Here

$$E^{0,q} = [\beta^\# \underline{H}_{A|S}^q(F)](A) \xrightarrow{\sim} H^q(R|S, F) \quad (q > 0), \quad E^{p,0} = H^p(A|R, \tilde{\beta}F).$$

On the other hand, the exact sequence

$$0 \longrightarrow F_R \longrightarrow \tilde{\beta}F \longrightarrow F(R) \longrightarrow 0$$

gives the exact diagram

$$\begin{array}{ccccccccc} 0 & \rightarrow & H^0(A|R, F_R) & \rightarrow & H^0(A|R, \tilde{\beta}F) & \rightarrow & H^0(A|R, F(R)) & \rightarrow & H^1(A|R, F_R) & \rightarrow & H^1(A|R, \tilde{\beta}F) & \rightarrow & 0 \\ & & \parallel & & \parallel & & & & & & & & \\ & & H^0(A|S, F) & & H^0(R|S, F) & & & & & & & & \end{array}$$

labelled (**) in the source. Combining (*) and (**) gives the desired exact sequence. $\square$

Corollary 3.8.

(a) Let A and B be R -algebras such that

$$\mathrm{Tor}_i^R(A, B) = 0 \quad (0 < i \leq n).$$

Then, for every additive sheaf $F \in \mathrm{Ob}(\tilde{\mathcal{C}}_{A \otimes_R B|R})$, the canonical map

$$H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism for every $i \leq n$.

(b) Let A be an R -algebra such that $\mathrm{Tor}_i^R(A, A) = 0$ for every $i > 0$. If the map $A \otimes_R A \longrightarrow A$ is an isomorphism, then, for every additive sheaf $F \in \mathrm{Ob}(\tilde{\mathcal{C}}_{A|R})$, one has

$$H^i(A|R, F) = 0$$

for every i .

Proof.

(a) Let $F \in \mathrm{Ob}(\tilde{\mathcal{C}}_{A \otimes_R B|R})$ be additive. Applied to $R \longrightarrow A \longrightarrow A \otimes_R B$, Proposition 3.7(c) gives the exact sequence

$$(*) \quad \cdots \longrightarrow H^i(A \otimes_R B|A, F_A) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \longrightarrow \cdots.$$

Consider the commutative diagram

$$\begin{array}{ccc} H^i(A \otimes_R B|A, F_A) & \longrightarrow & H^i(A \otimes_R B|R, F) \\ \downarrow & & \downarrow \\ H^i(B|R, \tilde{A}F_A) & \xrightarrow{\approx} & H^i(B|R, F). \end{array}$$

Here

$$A : \underline{\mathcal{C}}_{B|R} \longrightarrow \underline{\mathcal{C}}_{A \otimes_R B|A}, \quad X \longmapsto A \otimes_R X,$$

and the bottom map is induced by $\tilde{A}F_A \longrightarrow F$, which is an isomorphism because F is additive. By functoriality, the commutativity of the diagram reduces to the case $i = 0$.

If $\mathrm{Tor}_i^R(A, B) = 0$ for $0 < i \leq n$, then Proposition 3.7(a) shows that the left vertical map is an isomorphism for $i \leq n$. Consequently, both composites

$$H^i(A \otimes_R B|A, F_A) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(B|R, F)$$

and, by symmetry,

$$H^i(A \otimes_R B|B, F_B) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F)$$

are isomorphisms for $i \leq n$. It follows that $(*)$ breaks into split short exact sequences

$$0 \longrightarrow H^i(A \otimes_R B|A, F_A) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \longrightarrow 0$$

for $i \leq n$, and hence that

$$H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism in that range.

(b) By (a), the map

$$H^i(A \otimes_R A|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(A|R, F)$$

is an isomorphism for every i . On the other hand, the assumed isomorphism $A \otimes_R A \longrightarrow A$ induces an isomorphism

$$H^i(A|R, F) \longrightarrow H^i(A \otimes_R A|R, F).$$

Their composite is the diagonal map

$$H^i(A|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(A|R, F).$$

Since it is an isomorphism, $H^i(A|R, F) = 0$ for every i . □

Corollary 3.9. Let S be a multiplicative subset of an R -algebra A . Suppose that

$$F \in \text{Ob}(\tilde{\mathcal{C}}_{S^{-1}A|R})$$

is additive and that F_A is also additive. Then the canonical map

$$H^i(S^{-1}A|R, F) \longrightarrow H^i(A|R, F)$$

is an isomorphism for every i . Here $F_A \in \text{Ob}(\tilde{\mathcal{C}}_{S^{-1}A|A})$ is defined by

$$F_A(X) = \ker(F(X) \longrightarrow F(A))$$

for every $X \in \text{Ob}(\mathcal{C}_{S^{-1}A|A})$.

Proof. Applied to $R \longrightarrow A \longrightarrow S^{-1}A$, Proposition 3.7(c) gives the exact sequence

$$\cdots \longrightarrow H^n(S^{-1}A|A, F_A) \longrightarrow H^n(S^{-1}A|R, F) \longrightarrow H^n(A|R, F) \longrightarrow \cdots$$

The map

$$S^{-1}A \otimes_A S^{-1}A \longrightarrow S^{-1}A$$

is an isomorphism, and F_A is additive by hypothesis. Corollary 3.8(b) therefore gives $H^n(S^{-1}A|A, F_A) = 0$ for every n . The middle map in the exact sequence is consequently an isomorphism for every n . □

Let A be an R -algebra. An A -module M defines an additive sheaf

$$\text{Der}_R(-, M) \in \text{Ob}(\tilde{\mathcal{C}}_{A|R}), \quad X \longmapsto \text{Der}_R(X, M).$$

We make the following definition.

Definition 3.10. Let A be an R -algebra. For every A -module M , set

$$D^i(A|R, M) = H^i(A|R, \text{Der}_R(-, M)).$$

We now compute the groups of low degree, namely $D^i(A|R, M)$ for $i = 1, 2$. Choose a surjective arrow $P \rightarrow A$ in $\underline{C}_{A|R}$ with P a polynomial algebra over R , and write

$$0 \rightarrow I \rightarrow P \rightarrow A \rightarrow 0.$$

Lemma 3.10.

$$\begin{aligned} D^1(A|R, M) &= \text{Coker}(\text{Der}_R(P, M) \rightarrow \text{Hom}_P(I, M)), \\ D^2(A|R, M) &= H^1(A|R, \underline{H}_{A|R}^1(\text{Der}_R(-, M))). \end{aligned}$$

Proof. Consider the simplicial algebra

$$(*) \quad \cdots P_A^{(2)} \rightrightarrows P_A^{(1)} \rightrightarrows P_A^{(0)},$$

where

$$P_A^{(n)} = \overbrace{P \times_A \cdots \times_A P}^{n+1}.$$

For every n there is an exact sequence

$$0 \rightarrow I^{(n)} \rightarrow P_A^{(n)} \rightarrow A \rightarrow 0, \quad I^{(n)} = \overbrace{I \times \cdots \times I}^{n+1}.$$

Put $\bar{I} = I/I^2$, and let Δ denote the diagonal map. It is straightforward to verify that, for every n , the sequence

$$(a) \quad 0 \rightarrow \text{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \rightarrow \text{Der}_R(P_A^{(n)}, M) \xrightarrow{\text{Der}(\Delta, M)} \text{Der}_R(P, M) \rightarrow 0$$

is exact. We also have the exact sequence

$$(b) \quad 0 \rightarrow \text{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \rightarrow \text{Hom}_A(\bar{I}^{(n)}, M) \xrightarrow{\text{Hom}(\Delta, M)} \text{Hom}_A(\bar{I}, M) \rightarrow 0.$$

The sequences (a) and (b) yield the corresponding exact sequences of simplicial A -modules. Since $\{I \rightarrow 0\}$ and $\{I \rightarrow I\}$ admit sections,

$$H^i(\{I \rightarrow 0\}, \text{Hom}_A(-, M)) = H^i(\{I \rightarrow I\}, \text{Hom}_A(-, M)) = 0 \quad (i > 0).$$

It follows that

$$\left\{ \begin{array}{l} 0 \rightarrow \text{Der}_R(A, M) \rightarrow \text{Der}_R(P, M) \rightarrow \text{Hom}_A(\bar{I}, M) \\ \quad \rightarrow H^1(\{P \rightarrow A\}, \text{Der}_R(-, M)) \rightarrow 0 \quad \text{is exact,} \\ H^i(\{P \rightarrow A\}, \text{Der}_R(-, M)) = 0 \quad (i > 1). \end{array} \right.$$

By 3.1(b), therefore,

$$\begin{aligned} D^1(A|R, M) &= \check{H}^1(A|R, \text{Der}_R(-, M)) \\ &= H^1(\{P \rightarrow A\}, \text{Der}_R(-, M)) \\ &= \text{Coker}(\text{Der}_R(P, M) \rightarrow \text{Hom}_A(\bar{I}, M)) \\ &= \text{Coker}(\text{Der}_R(P, M) \rightarrow \text{Hom}_P(I, M)), \\ D^2(A|R, M) &= H^1(A|R, \underline{H}_{A|R}^1(\text{Der}_R(-, M))). \end{aligned}$$

□

Choose a free P -module F and a P -linear map $\gamma : F \rightarrow P$ such that $\gamma(F) = I$. Associated with γ is a Koszul complex, denoted by K_I by abuse of notation. Lemma 3.11. There is an isomorphism

$$D^2(A|R, M) \simeq \text{Coker}(H^1(K_I, M) \rightarrow \text{Hom}_A(H_1(K_I), M)).$$

Proof. We have

$$\begin{aligned} D^2(A|R, M) &= H^1(A|R, \underline{H}_{A|R}^1(\text{Der}_R(-, M))) \\ &= \ker \left(H^1(P_A^{(1)}|R, M) \rightrightarrows H^1(P_A^{(2)}|R, M) \right), \end{aligned}$$

because

$$[\underline{H}_{A|R}^1(\text{Der}_R(-, M))](P_A^{(0)}) = [\underline{H}_{A|R}^1(\text{Der}_R(-, M))](P) = 0.$$

Let

$$0 \longrightarrow K \longrightarrow F \xrightarrow{\gamma} I \longrightarrow 0$$

be exact. Write $P[F]$ for the symmetric algebra over P generated by F . Then $P[F]$ and $P[F] \otimes_P P[F]$ are polynomial algebras over R , and there are surjective maps

$$\begin{aligned} P[F] &\longrightarrow P_A^{(1)} = P \times_A P, & X &\longmapsto (0, \gamma(X)), \\ P[F] \otimes_P P[F] &\longrightarrow P_A^{(2)} = P \times_A P \times_A P, & X \otimes 1 &\longmapsto (0, \gamma(X), 0), \\ & & 1 \otimes X &\longmapsto (0, 0, \gamma(X)), \end{aligned}$$

for $X \in F$. Put

$$J = \ker(P[F] \longrightarrow P_A^{(1)}), \quad J' = \ker(P[F] \otimes_P P[F] \longrightarrow P_A^{(2)}).$$

If F_γ denotes the ideal generated by $\{X - \gamma(X) \mid X \in F\}$, then

$$\begin{aligned} J &= (F) \cap (F_\gamma) = (K) + (F \cdot F_\gamma), \\ J' &= (F \otimes 1 + 1 \otimes F) \cap (F_\gamma \otimes 1 + 1 \otimes F) \cap (F \otimes 1 + 1 \otimes F_\gamma) \\ &= (K \otimes 1 + 1 \otimes K) + (F \cdot F_\gamma \otimes 1 + 1 \otimes F \cdot F_\gamma) + (F \otimes F). \end{aligned}$$

The three maps $\pi_i : P_A^{(2)} \longrightarrow P_A^{(1)}$ admit lifts $\widehat{\pi}_i : P[F] \otimes_P P[F] \longrightarrow P[F]$ given, for $X \in F$, by

$$\widehat{\pi}_0 : \begin{cases} X \otimes 1 \longmapsto \gamma(X) - X, \\ 1 \otimes X \longmapsto X, \end{cases} \quad \widehat{\pi}_1 : \begin{cases} X \otimes 1 \longmapsto 0, \\ 1 \otimes X \longmapsto X, \end{cases} \quad \widehat{\pi}_2 : \begin{cases} X \otimes 1 \longmapsto X, \\ 1 \otimes X \longmapsto 0. \end{cases}$$

A direct computation shows that $\widehat{\pi}_0 - \widehat{\pi}_1 + \widehat{\pi}_2$ annihilates

$$K \otimes 1 + 1 \otimes K + 1 \otimes F \cdot F_\gamma$$

and maps $F \cdot F_\gamma \otimes 1 + F \otimes F$ onto $F \cdot F_\gamma$. For every

$$D \in \text{Der}_R(P[F] \otimes_P P[F], M)$$

one has $D(F \cdot F_\gamma \otimes 1 + F \otimes F) = 0$, because $I \cdot M = 0$. Consequently,

$$\begin{aligned} D^2(A|R, M) &= \ker \left(D^1(P_A^{(1)}|R, M) \rightrightarrows D^1(P_A^{(2)}|R, M) \right) \\ &= \{[f] \in D^1(P_A^{(1)}|R, M) \mid f(F \cdot F_\gamma) = 0\} \\ &= \text{Coker} \left(\text{Der}_R(P[F], M) \longrightarrow \text{Hom}_{P[F]} \left(\frac{(F) \cap (F_\gamma)}{F \cdot F_\gamma}, M \right) \right) \\ &= \text{Coker} \left(\text{Der}_P(P[F], M) \longrightarrow \text{Hom}_{P[F]} \left(\frac{(F) \cap (F_\gamma)}{(F)(F_\gamma)}, M \right) \right). \end{aligned}$$

The sequences

$$0 \longrightarrow (F) \longrightarrow P[F] \xrightarrow{P[0]} P \longrightarrow 0, \quad 0 \longrightarrow (F_\gamma) \longrightarrow P[F] \xrightarrow{P[\gamma]} P_\gamma \longrightarrow 0$$

are exact. Hence

$$\frac{(F) \cap (F_\gamma)}{(F)(F_\gamma)} = \text{Tor}_1^{P[F]}(P, P_\gamma) = H_1(K_I),$$

where K_I is the Koszul complex associated with $F \xrightarrow{\gamma} I \subset P$. We therefore obtain

$$\begin{aligned} D^2(A|R, M) &= \text{Coker}(\text{Hom}_P(F, M) \longrightarrow \text{Hom}_P(H_1(K_I), M)) \\ &= \text{Coker}(H^1(K_I, M) \longrightarrow \text{Hom}_A(H_1(K_I), M)). \end{aligned}$$

□

We can now state the basic facts needed below, summarizing the results already established.
Theorem 3.12.

(1) If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A -modules, then the sequence

$$\begin{aligned} 0 \longrightarrow D^0(A|R, M') \longrightarrow D^0(A|R, M) \longrightarrow D^0(A|R, M'') \longrightarrow D^1(A|R, M') \\ \longrightarrow D^1(A|R, M) \longrightarrow D^1(A|R, M'') \longrightarrow \cdots \longrightarrow D^n(A|R, M') \\ \longrightarrow D^n(A|R, M) \longrightarrow D^n(A|R, M'') \longrightarrow \cdots \end{aligned}$$

is exact.

(2) Let $S \longrightarrow R \longrightarrow A$ be ring homomorphisms. Then, for every A -module M , the sequence

$$\begin{aligned} 0 \longrightarrow D^0(A|R, M) \longrightarrow D^0(A|S, M) \longrightarrow D^0(R|S, M) \longrightarrow D^1(A|R, M) \\ \longrightarrow D^1(A|S, M) \longrightarrow D^1(R|S, M) \longrightarrow \cdots \longrightarrow D^n(A|R, M) \\ \longrightarrow D^n(A|S, M) \longrightarrow D^n(R|S, M) \longrightarrow \cdots \end{aligned}$$

is exact.

(3) Let A and S be R -algebras such that $\text{Tor}_i^R(A, S) = 0$ for every $i > 0$. Then, for every $A \otimes_R S$ -module M , the canonical map

$$D^i(A \otimes_R S|S, M) \longrightarrow D^i(A|R, M)$$

is an isomorphism for every i . In particular, if A is flat over R , then

$$D^i(A \otimes_R S|S, M) \xrightarrow{\sim} D^i(A|R, M)$$

for every i , every R -algebra S , and every $A \otimes_R S$ -module M .

(4) Let S be a multiplicative subset of A . For every $S^{-1}A$ -module N , the canonical map

$$D^i(S^{-1}A|R, N) \longrightarrow D^i(A|R, N)$$

is an isomorphism for every i . Consequently, for every A -module M , there is a canonical isomorphism

$$D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M)$$

for every i .

(5) Let A and B be R -algebras such that $\text{Tor}_i^R(A, B) = 0$ for $0 < i \leq n$. Then, for every $A \otimes_R B$ -module M , the canonical map

$$D^i(A \otimes_R B|R, M) \longrightarrow D^i(A|R, M) \oplus D^i(B|R, M)$$

is an isomorphism for $i \leq n$. In particular, suppose that $\mathrm{Tor}_i^R(A, A) = 0$ for every $i > 0$ and that the diagonal

$$\Delta : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(A)$$

is an open immersion—for example, suppose that A is *étale* over R . Then

$$D^i(A|R, M) = 0$$

for every $i > 0$ and every A -module M .

- (6) Let R be noetherian, let A be an R -algebra essentially of finite type, and let E be a flat A -module. Then the canonical map

$$E \otimes_A D^i(A|R, M) \xrightarrow{\sim} D^i(A|R, E \otimes_A M)$$

is an isomorphism for every $i \geq 0$ and every A -module M .

- (7) If R is noetherian, A is an R -algebra essentially of finite type, and M is an A -module of finite type, then $D^i(A|R, M)$ is an A -module of finite type for every i .
(8) Let P be a polynomial algebra over R , and let

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0$$

be exact. Then

$$D^0(A|R, M) = \mathrm{Der}_R(A, M),$$

$$\begin{aligned} D^1(A|R, M) &= \mathrm{Coker}(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)) \\ &= \{ \text{isomorphism classes of commutative } R\text{-algebra extensions of } A \text{ by } M \}, \\ D^2(A|R, M) &= \mathrm{Coker}(H^1(K_I, M) \longrightarrow \mathrm{Hom}_A(H_1(K_I), M)). \end{aligned}$$

Proof.

- (1) If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A -modules, then, for every polynomial algebra P over R , the sequence

$$0 \longrightarrow \mathrm{Der}_R(P, M') \longrightarrow \mathrm{Der}_R(P, M) \longrightarrow \mathrm{Der}_R(P, M'') \longrightarrow 0$$

is exact. Hence, by 3.3(c),

$$0 \longrightarrow \mathrm{Der}_R(-, M') \longrightarrow \mathrm{Der}_R(-, M) \longrightarrow \mathrm{Der}_R(-, M'') \longrightarrow 0$$

is an exact sequence of sheaves.

- (2) The functor $\mathrm{Der}_S(-, M)$ is an additive sheaf on $\underline{\mathcal{C}}_{A|S}$, and

$$\mathrm{Der}_R(B, M) = \ker(\mathrm{Der}_S(B, M) \longrightarrow \mathrm{Der}_S(R, M))$$

for every $B \in \mathrm{Ob}(\underline{\mathcal{C}}_{A|R})$. The assertion therefore follows from 3.7(c).

- (3) Let

$$S : \underline{\mathcal{C}}_{A|R} \longrightarrow \underline{\mathcal{C}}_{A \otimes_R S|S}, \quad X \longmapsto X \otimes_R S.$$

Then $\tilde{S} \mathrm{Der}_S(-, M) = \mathrm{Der}_R(-, M)$, so the assertion follows from 3.7(a).

- (4) This follows from 3.9. Indeed, $F = \text{Der}_R(-, N)$ is an additive sheaf on $\underline{C}_{S^{-1}A|R}$, and $F_A \in \text{Ob}(\tilde{\underline{C}}_{S^{-1}A|A})$ is also additive, since

$$\begin{aligned} F_A(X) &= \ker(F(X) \longrightarrow F(A)) \\ &= \ker(\text{Der}_R(X, N) \longrightarrow \text{Der}_R(A, N)) \\ &= \text{Der}_A(X, N). \end{aligned}$$

Thus $F_A = \text{Der}_A(-, N)$.

- (5) The first assertion follows from 3.8(a). Now suppose that $\text{Tor}_i^R(A, A) = 0$ for every $i > 0$ and that

$$\Delta : \text{Spec}(A) \longrightarrow \text{Spec}(A) \times_{\text{Spec}(R)} \text{Spec}(A)$$

is an open immersion. Let $\varphi : A \otimes_R A \longrightarrow A$ be the multiplication map. For every maximal ideal $\mathfrak{m}$ of A , the induced map

$$(A \otimes_R A)_{\varphi^{-1}(\mathfrak{m})} \longrightarrow A_{\mathfrak{m}}$$

is an isomorphism. The first assertion, together with 3.12(4), therefore implies that the diagonal map

$$\Delta_{\mathfrak{m}} : D^i(A|R, M)_{\mathfrak{m}} \longrightarrow D^i(A|R, M)_{\mathfrak{m}} \oplus D^i(A|R, M)_{\mathfrak{m}}$$

is an isomorphism for every maximal ideal $\mathfrak{m}$ of A . Consequently,

$$\Delta : D^i(A|R, M) \longrightarrow D^i(A|R, M) \oplus D^i(A|R, M)$$

is an isomorphism, and hence $D^i(A|R, M) = 0$ for every $i > 0$.

- (6) In view of 3.12(4), we may assume that A is an R -algebra of finite type. Consider the morphism of sheaves

$$E \otimes_A \text{Der}_R(-, M) \longrightarrow \text{Der}_R(-, E \otimes_A M)$$

on $\tilde{\underline{C}}_{A|R}$. Since E is flat over A , the map

$$E \otimes_A \text{Der}_R(B, M) \xrightarrow{\sim} \text{Der}_R(B, E \otimes_A M)$$

is an isomorphism whenever $B \in \text{Ob}(\underline{C}_{A|R})$ is of finite type over R . By 3.3(b), it follows that

$$D^i(A|R, E \otimes_A \text{Der}_R(-, M)) \longrightarrow D^i(A|R, \text{Der}_R(-, E \otimes_A M))$$

is an isomorphism. On the other hand, $E \otimes_A -$ is exact, and therefore

$$E \otimes_A D^i(A|R, M) = D^i(A|R, E \otimes_A \text{Der}_R(-, M)).$$

This proves the assertion.

- (7) This follows from 3.3(a).
 (8) In view of 3.10 and 3.11, it suffices to show that

$$\text{Coker}(\text{Der}_R(P, M) \longrightarrow \text{Hom}_A(I/I^2, M)) = \text{Exalcomm}(A|R, M),$$

where $\text{Exalcomm}(A|R, M)$ denotes the set of isomorphism classes of R -algebra extensions of A by M . Let

$$0 \longrightarrow M \longrightarrow A' \longrightarrow A \longrightarrow 0$$

be such an extension. Since P is a polynomial algebra over R , we obtain a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A \longrightarrow 0. \end{array}$$

The map $I \longrightarrow M$ is determined by the extension A' , up to the action of $\text{Der}_R(P, M)$. We thus obtain a map

$$\Phi : \text{Exalcomm}(A|R, M) \longrightarrow \text{Coker}(\text{Der}_R(P, M) \longrightarrow \text{Hom}_A(I/I^2, M)).$$

Conversely, a P -linear map $I \longrightarrow M$ yields, by pushout, an extension fitting into a diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \parallel & & \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A & \longrightarrow & 0. \end{array}$$

Two P -linear maps $I \rightrightarrows M$ that differ by an element of $\text{Der}_R(P, M)$ yield isomorphic extensions. Hence there is a map

$$\Psi : \text{Coker}(\text{Der}_R(P, M) \longrightarrow \text{Hom}_A(I/I^2, M)) \longrightarrow \text{Exalcomm}(A|R, M).$$

It is straightforward to verify that Φ and Ψ are inverse to each other. □

Corollary 3.13.

- (a) The R -algebra A is formally smooth over R if and only if $D^1(A|R, M) = 0$ for every A -module M .
- (b) Let R be noetherian and let A be an R -algebra of finite type. Then A is a relative complete intersection over R if and only if $D^2(A|R, M) = 0$ for every A -module M .
- (c) Let R be noetherian and let A be an R -algebra of finite type. For every multiplicative subset S of A , there is a canonical isomorphism

$$D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M) \quad (i \geq 0).$$

- (d) Let R be noetherian and let A be a local R -algebra essentially of finite type. Write $\hat{A}$ for the $\mathfrak{m}$ -adic completion of A , where $\mathfrak{m}$ is the maximal ideal of A . Then, for every A -module E of finite type, there is a canonical isomorphism

$$D^1(\hat{A}|R, \hat{A} \otimes_A E) \xrightarrow{\sim} \hat{A} \otimes_A D^1(A|R, E).$$

Proof.

Part (a) follows immediately from $D^1(A|R, M) = \text{Exalcomm}(A|R, M)$.

For (b), let P be a polynomial algebra over R in finitely many variables, and let

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0$$

be exact. After localization, we may suppose that I is generated by a P -regular sequence. Then $H_1(K_I) = 0$, and hence

$$D^2(A|R, M) = \text{Coker}(H^1(K_I, M) \longrightarrow \text{Hom}_A(H_1(K_I), M)) = 0.$$

Conversely, suppose that $D^2(A|R, M) = 0$ for every A -module M . Choose an exact sequence

$$0 \longrightarrow K \longrightarrow F \longrightarrow I \longrightarrow 0,$$

where F is a free P -module of finite type. If

$$K_0 = \text{Im}(\bigwedge^2 F \longrightarrow \bigwedge^1 F),$$

then $H_1(K_I) = K/K_0$. Since $D^2(A|R, H_1(K_I)) = 0$, the map

$$H^1(K_I, H_1(K_I)) \longrightarrow \text{Hom}_A(H_1(K_I), H_1(K_I))$$

is surjective. Consequently, the exact sequence

$$0 \longrightarrow K/K_0 \longrightarrow F/IF \longrightarrow I/I^2 \longrightarrow 0$$

splits as a sequence of A -modules. In particular, I/I^2 is projective. After localizing A once more, we may suppose that I/I^2 is free and that $F/IF \longrightarrow I/I^2$ is an isomorphism. The split exact sequence then gives $H_1(K_I) = K/K_0 = 0$. Thus I is generated by a P -regular sequence, so A is a relative complete intersection over R .

Part (c) follows immediately from 3.12(4) and (6).

For (d), 3.12(6) first gives canonical isomorphisms

$$D^i(A|R, \widehat{A} \otimes_A E) \xrightarrow{\sim} \widehat{A} \otimes_A D^i(A|R, E)$$

for every i . It therefore remains to prove that

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \longrightarrow D^1(A|R, \widehat{A} \otimes_A E)$$

is an isomorphism. We use the description of D^1 in terms of algebra extensions.

Let

$$0 \longrightarrow \widehat{A} \otimes_A E \longrightarrow A' \xrightarrow{\varphi} \widehat{A} \longrightarrow 0$$

be an R -algebra extension whose restriction to A splits. Since $\widehat{A} \otimes_A E$ is an $\widehat{A}$ -module of finite type, A' is a local R -algebra complete for its $\mathfrak{m}'$ -adic topology, where $\mathfrak{m}'$ is the maximal ideal of A' . If $\psi : A \longrightarrow A'$ is an R -algebra map lifting the canonical map $A \longrightarrow \widehat{A}$, then ψ extends uniquely to a map $\widehat{\psi} : \widehat{A} \longrightarrow A'$ satisfying $\varphi \circ \widehat{\psi} = \text{id}_{\widehat{A}}$. This proves that

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \longrightarrow D^1(A|R, \widehat{A} \otimes_A E)$$

is injective.

Now let

$$0 \longrightarrow \widehat{A} \otimes_A E \longrightarrow B \xrightarrow{\varphi} A \longrightarrow 0$$

be an R -algebra extension, and put $\widehat{E} = \widehat{A} \otimes_A E$. We first claim that the topology on $\widehat{E}$ induced by the $\mathfrak{m}_B$ -adic topology on B coincides with its topology as an $\widehat{A}$ -module. Indeed, since A is essentially of finite type over R , there is a subalgebra $B_0 \subset B$ such that B_0 is a local R -algebra essentially of finite type and $\varphi(B_0) = A$. Put $E_0 = \widehat{E} \cap B_0$, and denote the maximal ideals of B , B_0 , and A by $\mathfrak{m}_B$, $\mathfrak{m}_0$, and $\mathfrak{m}$, respectively. Then

$$\mathfrak{m}_B = \mathfrak{m}_0 + \widehat{E}, \quad \mathfrak{m}_B^{n+1} = \mathfrak{m}_0^{n+1} + \mathfrak{m}^n \widehat{E}$$

for every n . Since B_0 is noetherian, the Artin–Rees theorem gives an integer $r > 0$ such that, for every $n \geq r$,

$$\begin{aligned} \widehat{E} \cap \mathfrak{m}_B^{n+1} &= \mathfrak{m}^n \widehat{E} + \widehat{E} \cap \mathfrak{m}_0^{n+1} \\ &= \mathfrak{m}^n \widehat{E} + E_0 \cap \mathfrak{m}_0^{n+1} \\ &= \mathfrak{m}^n \widehat{E} + \mathfrak{m}^{n+1-r} (E_0 \cap \mathfrak{m}_0^r) \subset \mathfrak{m}^{n+1-r} \widehat{E}. \end{aligned}$$

This proves the assertion about the two topologies. In particular, $\widehat{E}$ is complete for the topology induced by the $\mathfrak{m}_B$ -adic topology on B , and hence completion gives an exact commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \widehat{E} & \longrightarrow & B & \longrightarrow & A \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \widehat{E} & \longrightarrow & \widehat{B} & \longrightarrow & \widehat{A} \longrightarrow 0. \end{array}$$

Here $\widehat{B}$ denotes the $\mathfrak{m}_B$ -adic completion of B . Thus

$$D^1(\widehat{A}|R, \widehat{E}) \longrightarrow D^1(A|R, \widehat{E})$$

is surjective, and the proof is complete. $\square$

Corollary 3.14. Let R be noetherian, and let A be an R -algebra of finite type that is smooth everywhere except at one closed point $x \in \operatorname{Spec}(A)$. Then there are isomorphisms

$$D^1(A|R, A) \simeq D^1(A_x|R, A_x) \simeq D^1(\widehat{A}_x|R, \widehat{A}_x).$$

Proof. Since A is smooth over R away from x , $D^1(A|R, A)$ is an A -module of finite type supported on $\{x\}$. Hence

$$D^1(A|R, A) = D^1(A|R, A)_x = D^1(\widehat{A}|R, \widehat{A})_x.$$

The assertion now follows from 3.13. $\square$

Remark 3.15. Let R be a topological ring and A a topological R -algebra. For every A -module E , define

$$D_{\text{top}}^i(A|R, E) = \varinjlim_{\alpha} D^i(A_{\alpha}|R_{\alpha}, A_{\alpha} \otimes_A E),$$

where $A_{\alpha} = A/J_{\alpha}$, $R_{\alpha} = R/I_{\alpha}$, and J_{α} and I_{α} are open in A and R , respectively. Thus A is formally smooth as a topological R -algebra precisely when $D_{\text{top}}^1(A|R, E) = 0$ for every A -module E .

Now let X be a scheme over a ring R , and let $\underline{E}$ be a quasi-coherent $\mathcal{O}_X$ -module. We define sheaves

$$\underline{D}_{X|R}^i(\underline{E}) = \underline{D}^i(X|R, \underline{E})$$

on X from the presheaves on affine open subsets $U \subset X$ given by

$$[\underline{D}_{X|R}^i(\underline{E})](U) = D^i(\Gamma(U, \mathcal{O}_X)|R, \Gamma(U, \underline{E})).$$

If R is noetherian and X is of finite type over R , the localization property 3.13(c) shows that these presheaves canonically form quasi-coherent $\mathcal{O}_X$ -modules.

When no confusion is possible, we abbreviate

$$\underline{D}_{X|R}^i = \underline{D}_{X|R}^i(\mathcal{O}_X).$$

All the assertions of 3.13 have corresponding formulations for the quasi-coherent sheaves $\underline{D}_{X|R}^i(\underline{E})$. We leave those formulations to the reader and record below the properties most relevant to our purpose.

Corollary 3.15. Let R be noetherian and let X be an R -scheme of finite type.

(a) For every coherent $\mathcal{O}_X$ -module $\underline{E}$, $\underline{D}_{X|R}^i(\underline{E})$ is again coherent for every i . Moreover,

$$\operatorname{Supp} \underline{D}_{X|R}^i(\underline{E}) \subset X^{\text{Sing}} \quad (i > 0),$$

where

$$X^{\text{Sing}} = \{x \in X \mid X \text{ is not smooth over } R \text{ at } x\}.$$

(b) If X is flat over R , then for every R -algebra S there is a canonical isomorphism

$$\underline{D}_{X \otimes_R S | S}^i(\underline{E}) \xrightarrow{\sim} \underline{D}_{X|R}^i(\underline{E})$$

for every i .

- (c) Suppose that X is a relative complete intersection over R . For every surjective ring homomorphism $S \rightarrow R$, there is an exact sequence

$$0 \rightarrow \underline{D}_{X|R}^1(\underline{E}) \rightarrow \underline{D}_{X|S}^1(\underline{E}) \rightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_X}((I/I^2) \otimes_R \mathcal{O}_X, \underline{E}) \rightarrow 0,$$

where $I = \ker(S \rightarrow R)$. If, moreover, R is local with residue field k and X is flat over R , then

$$0 \rightarrow \underline{D}_{X_0|k}^1(\mathcal{O}_{X_0}) \rightarrow \underline{D}_{X|S}^1(\mathcal{O}_{X_0}) \rightarrow D^1(R|S, k) \otimes_k \mathcal{O}_{X_0} \rightarrow 0,$$

where $X_0 = X \otimes_R k$.

Proof. Part (a) follows from the finiteness and low-degree descriptions in 3.12, while (b) follows from 3.12(3).

For (c), because X is a relative complete intersection over R , we have $\underline{D}_{X|R}^i(\underline{E}) = 0$ for every $i \geq 2$. The transitivity sequence 3.12(2) therefore gives an exact sequence

$$0 \rightarrow \underline{D}_{X|R}^0(\underline{E}) \rightarrow \underline{D}_{X|S}^0(\underline{E}) \rightarrow \underline{D}_{R|S}^0(\underline{E}) \rightarrow \underline{D}_{X|R}^1(\underline{E}) \rightarrow \underline{D}_{X|S}^1(\underline{E}) \rightarrow \underline{D}_{R|S}^1(\underline{E}) \rightarrow 0,$$

where the quasi-coherent sheaves $\underline{D}_{R|S}^i(\underline{E})$ on X are defined on every affine open subset $U \subset X$ by

$$[\underline{D}_{R|S}^i(\underline{E})](U) = D^i(R|S, \underline{E}(U)).$$

Since $S \rightarrow R$ is surjective, $\underline{D}_{R|S}^0(\underline{E}) = 0$; and the low-degree description in 3.12(8) gives

$$\underline{D}_{R|S}^1(\underline{E}) = \underline{\mathrm{Hom}}_{\mathcal{O}_X}((I/I^2) \otimes_R \mathcal{O}_X, \underline{E}).$$

This proves the first exact sequence.

If X is flat over R , then, for every quasi-coherent $\mathcal{O}_{X_0}$ -module $\underline{E}$, there are canonical isomorphisms

$$\underline{D}_{X_0|k}^i(\underline{E}) \xrightarrow{\sim} \underline{D}_{X|R}^i(\underline{E})$$

for every i . In particular,

$$\underline{D}_{X_0|k}^1(\mathcal{O}_{X_0}) \xrightarrow{\sim} \underline{D}_{X|R}^1(\mathcal{O}_{X_0}),$$

from which the second exact sequence follows. $\square$

Remark 3.16. The quasi-coherent sheaves $\underline{D}_{X|R}^i(\underline{E})$ defined above should occur as the terms $E_2^{0,i}$ of a spectral sequence relating the local theory to the global theory. L. Illusie has obtained such a theory in *Lecture Notes in Mathematics* 239. His method uses “homotopical algebra,” generalizing the methods of M. André and D. Quillen, rather than “cohomological algebra.” It seems that the approach taken here can be globalized by means of Grothendieck cohomology on the category of schemes, equipped with an appropriate topology generalizing the one on the affine category.

4. Formal moduli of deformations

Among the natural cofibered categories whose isomorphism classes of objects are generally not prorepresentable but nevertheless admit a formal versal object is the cofibered category of deformations. We return to the situation and notation of section 1. Thus, throughout this section, Λ is a fixed complete noetherian local ring with residue field k , and $\underline{\mathcal{C}}_\Lambda$ is the category of artinian local Λ -algebras with residue field k .

Let X_0 be a fixed scheme over k . By an (infinitesimal) deformation of X_0 we mean a pair (R, X) , where $R \in \text{Ob}(\underline{\mathcal{C}}_\Lambda)$ and X is a flat R -scheme, together with a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \text{Spec}(k) & \longrightarrow & \text{Spec}(R) \end{array}$$

such that the induced morphism

$$X_0 \longrightarrow X \times_{\text{Spec}(R)} \text{Spec}(k)$$

is an isomorphism. Since R is artinian local, $i_X : X_0 \longrightarrow X$ is necessarily a closed immersion and a homeomorphism on the underlying topological spaces.

If (R, X) and (R', X') are deformations of X_0 over R and R' , respectively, an arrow

$$(R', X') \longrightarrow (R, X)$$

consists of a map $\varphi : R' \longrightarrow R$ in $\underline{\mathcal{C}}_\Lambda$ and a morphism of schemes $\Phi : X \longrightarrow X'$ fitting into a commutative diagram

$$\begin{array}{ccccc} X & & \xleftarrow{\Phi} & & X' \\ & \swarrow i_X & & \searrow i_{X'} & \\ & X_0 & & & \\ & \downarrow & & & \\ & \text{Spec}(k) & & & \\ & \swarrow & & \searrow & \\ \text{Spec}(R) & & \xrightarrow{\text{Spec}(\varphi)} & & \text{Spec}(R'). \end{array}$$

This defines the category $\underline{M}_{X_0}$ of infinitesimal deformations of X_0 . It is equipped with the functor

$$p : \underline{M}_{X_0} \longrightarrow \underline{\mathcal{C}}_\Lambda, \quad (R, X) \longmapsto R.$$

Let $(R, X) \in \text{Ob}(\underline{M}_{X_0})$ and let $\varphi : R \longrightarrow S$ be a map in $\underline{\mathcal{C}}_\Lambda$. There is a commutative diagram

$$\begin{array}{ccc} X \times_{\text{Spec}(R)} \text{Spec}(S) & \xrightarrow{p_1} & X \\ \downarrow \text{flat} & & \downarrow \text{flat} \\ \text{Spec}(S) & \xrightarrow{\text{Spec}(\varphi)} & \text{Spec}(R), \end{array}$$

and the φ -arrow

$$(\varphi, p_1) : (R, X) \longrightarrow (S, X \times_{\text{Spec}(R)} \text{Spec}(S))$$

is cocartesian in $\underline{M}_{X_0}$. Hence $\underline{M}_{X_0}$ is a cofibered category over $\underline{\mathcal{C}}_\Lambda$. We record the following facts.

4.1(a). Let

$$(1_R, \Phi) : (R, X_1) \longrightarrow (R, X_2)$$

be an arrow in $\underline{M}_{X_0}$. Thus $\Phi : X_2 \rightarrow X_1$ is an R -morphism whose reduction

$$X_2 \otimes_R k \rightarrow X_1 \otimes_R k$$

is an isomorphism. Since R is artinian and the X_i are flat over R , Φ is an isomorphism. Consequently, $\underline{M}_{X_0}$ is a cofibered category in groupoids over $\underline{C}_\Lambda$.

4.1(b). Suppose that X_0 is affine, say $X_0 = \text{Spec}(A_0)$. Then every $(R, X) \in \text{Ob}(\underline{M}_{X_0})$ has X affine as well. Consequently, for every $R \in \text{Ob}(\underline{C}_\Lambda)$, taking global sections identifies $\underline{M}_{X_0}(R)$ with the following category. An object is a flat R -algebra A together with an R -algebra map $j_A : A \rightarrow A_0$ inducing an isomorphism

$$A \otimes_R k \xrightarrow{\sim} A_0.$$

A morphism $\Phi : A_1 \rightarrow A_2$ is an R -algebra map for which the diagram

$$\begin{array}{ccc} A_1 & \xrightarrow{\Phi} & A_2 \\ j_{A_1} \downarrow & & \downarrow j_{A_2} \\ A_0 & \xlongequal{\quad} & A_0 \end{array}$$

commutes.

4.1(c). Consider the cofibered category $\widehat{\underline{M}}_{X_0}$ in groupoids over $\widehat{\underline{C}}_\Lambda$. By definition, for every $R \in \text{Ob}(\widehat{\underline{C}}_\Lambda)$, the category $\widehat{\underline{M}}_{X_0}(R)$ is canonically equivalent to the category of formal schemes $\mathfrak{X}$, adic over $\text{Spf}(R)$, such that

$$\mathfrak{X}_n = \mathfrak{X} \otimes_R R/\mathfrak{m}^{n+1}$$

is flat over $R/\mathfrak{m}^{n+1}$ for every $n \geq 0$. If X_0 is noetherian, this is equivalent to requiring $\mathfrak{X}$ to be flat over $\text{Spf}(R)$. Thus, in that case, $\widehat{\underline{M}}_{X_0}$ is canonically equivalent over $\widehat{\underline{C}}_\Lambda$ to the category of deformations of X_0 over the objects of $\widehat{\underline{C}}_\Lambda$, in the sense of formal schemes.

4.1(d). Let $(R, X) \in \text{Ob}(\underline{M}_{X_0})$, so that there is a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \text{Spec}(k) & \longrightarrow & \text{Spec}(R) \end{array}$$

whose induced map $X_0 \rightarrow X \times_{\text{Spec}(R)} \text{Spec}(k)$ is an isomorphism.

Let $U_0 \subset X_0$ be an open subscheme. Since $i_X : X_0 \rightarrow X$ is a closed immersion and a homeomorphism on underlying spaces, restriction gives a commutative diagram

$$\begin{array}{ccc} U_0 & \xrightarrow{i_X|_{U_0}} & X|_{U_0} \\ \downarrow & & \downarrow \text{flat} \\ \text{Spec}(k) & \longrightarrow & \text{Spec}(R). \end{array}$$

The induced map

$$U_0 \rightarrow (X|_{U_0}) \times_{\text{Spec}(R)} \text{Spec}(k) = (X \otimes_R k)|_{U_0}$$

is an isomorphism, so $(R, X|_{U_0})$ is a deformation of U_0 over R . Moreover, every arrow $(\varphi, \Phi) : (R, X) \longrightarrow (R', X')$ in $\underline{M}_{X_0}$ restricts to an arrow

$$(\varphi, \Phi|_{U_0}) : (R, X|_{U_0}) \longrightarrow (R', X'|_{U_0})$$

in $\underline{M}_{U_0}$. We therefore have a restriction functor

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{U_0}$$

over $\underline{C}_\Lambda$.

One may likewise localize and complete at a point. Let $x_0 \in X_0$. If $(R, X) \in \text{Ob}(\underline{M}_{X_0})$, then $\text{Spec}(\mathcal{O}_{X, x_0})$ is a deformation of $\text{Spec}(\mathcal{O}_{X_0, x_0})$ over R , while $\text{Spec}(\widehat{\mathcal{O}}_{X, x_0})$ is a deformation of $\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_0})$ over R ; the hats denote completion for the topology defined by powers of the maximal ideal. Thus there are functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\text{Spec}(\mathcal{O}_{X_0, x_0})} \longrightarrow \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_0})}.$$

We first prove that $\underline{M}_{X_0}$ is a homogeneous cofibered category over $\underline{C}_\Lambda$.

Lemma 4.2. Consider a diagram

$$\begin{array}{ccc} & & R' \\ & & \downarrow \\ R & \xrightarrow{\text{surjective}} & \bar{R} \end{array}$$

in $\underline{C}_\Lambda$, and let E and E' be flat modules over R and R' , respectively. Then, for every R' -homomorphism

$$E' \longrightarrow \bar{E} = E \otimes_R \bar{R},$$

the fiber product $E \times_{\bar{E}} E'$ is flat as a module over $R \times_{\bar{R}} R'$.

Proof. Let

$$0 \longrightarrow I \longrightarrow R \longrightarrow \bar{R} \longrightarrow 0$$

be exact, so that

$$0 \longrightarrow I \longrightarrow R \times_{\bar{R}} R' \longrightarrow R' \longrightarrow 0$$

is exact. Since E is flat over R ,

$$0 \longrightarrow I \otimes_R E \longrightarrow E \longrightarrow \bar{E} \longrightarrow 0$$

is exact, and hence we obtain the exact sequence

$$(*) \quad 0 \longrightarrow I \otimes_R E \longrightarrow E \times_{\bar{E}} E' \longrightarrow E' \longrightarrow 0.$$

Set $S = R \times_{\bar{R}} R'$. Since I is nilpotent, $E \times_{\bar{E}} E'$ is flat as an S -module if and only if $(E \times_{\bar{E}} E') \otimes_S R'$ is flat as an R' -module and

$$\text{Tor}_1^S(E \times_{\bar{E}} E', R') = 0.$$

But

$$I \otimes_R E = I \otimes_S (E \times_{\bar{E}} E'), \quad E' = R' \otimes_S (E \times_{\bar{E}} E'),$$

and therefore the exact sequence $(*)$ may be rewritten as

$$0 \longrightarrow I \otimes_S (E \times_{\bar{E}} E') \longrightarrow E \times_{\bar{E}} E' \longrightarrow R' \otimes_S (E \times_{\bar{E}} E') \longrightarrow 0.$$

Thus

$$\text{Tor}_1^S(R', E \times_{\bar{E}} E') = 0, \quad R' \otimes_S (E \times_{\bar{E}} E') = E',$$

and E' is flat over R' . It follows that $E \times_{\bar{E}} E'$ is flat over S . $\square$

Corollary 4.3. Given a diagram

$$\begin{array}{ccc} (R, X) & & (R', X') \\ & \searrow (\varphi, \Phi) & \swarrow (\varphi', \Phi') \\ & (\bar{R}, \bar{X}) & \end{array}$$

in $\underline{M}_{X_0}$, the amalgamated sum $X \amalg_{\bar{X}} X'$ (in the category of schemes) is a flat scheme over $R \times_{\bar{R}} R'$, provided that $\varphi : R \rightarrow \bar{R}$ is surjective. Consequently, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Proof. For every affine open $U_0 \subset X_0$, by definition we have

$$(X \amalg_{\bar{X}} X')|_{U_0} \simeq \text{Spec} \left(\Gamma(U_0, X) \times_{\Gamma(U_0, \bar{X})} \Gamma(U_0, X') \right).$$

Consequently, if $\varphi : R \rightarrow \bar{R}$ is surjective, $X \amalg_{\bar{X}} X'$ is flat over $R \times_{\bar{R}} R'$ by Lemma 4.2. Moreover, the morphism

$$i_X \amalg_{i_{\bar{X}}} i_{X'} : X_0 \rightarrow X \amalg_{\bar{X}} X'$$

induces an isomorphism

$$X_0 \rightarrow (X \amalg_{\bar{X}} X') \otimes_S k, \quad S = R \times_{\bar{R}} R'.$$

Therefore

$$(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X') \in \text{Ob}(\underline{M}_{X_0}),$$

and, in $\underline{M}_{X_0}$, we have

$$(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X') = (R, X) \times_{(\bar{R}, \bar{X})} (R', X').$$

Thus $\underline{M}_{X_0}$ is homogeneous over $\underline{C}_\Lambda$. $\square$

Lemma 4.4. For every scheme X_0 over k , there is a canonical exact sequence

$$0 \rightarrow H^1(X_0, \underline{D}_{X_0}^0) \rightarrow [\underline{M}_{X_0}(k[\varepsilon])] \rightarrow H^0(X_0, \underline{D}_{X_0}^1),$$

where the $\underline{D}_{X_0}^i$ are as defined in section 3.

Proof. First suppose that X_0 is affine, and put $\Gamma(X_0, \mathcal{O}_{X_0}) = A_0$. As noted in 4.1(b), the category $\underline{M}_{X_0}(k[\varepsilon])$ is canonically equivalent to a category $\underline{E}$ whose objects are flat $k[\varepsilon]$ -algebras A equipped with a $k[\varepsilon]$ -algebra map $j_A : A \rightarrow A_0$ inducing an isomorphism

$$k \otimes_{k[\varepsilon]} A \xrightarrow{\sim} A_0.$$

Equivalently, an object of $\underline{E}$ is a k -algebra A equipped with an exact sequence

$$0 \rightarrow A_0 \xrightarrow{\varepsilon} A \xrightarrow{j_A} A_0 \rightarrow 0$$

in which $\varepsilon(A_0)^2 = 0$ as an ideal of A .

An arrow $A_1 \rightarrow A_2$ in $\underline{E}$, in the first description, is a $k[\varepsilon]$ -algebra map $\Phi : A_1 \rightarrow A_2$ such that $j_{A_1} = j_{A_2} \circ \Phi$. In the second description it is a k -algebra map $\Phi : A_1 \rightarrow A_2$ yielding a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A_0 & \xrightarrow{\varepsilon} & A_1 & \xrightarrow{j_{A_1}} & A_0 \longrightarrow 0 \\ & & \parallel & & \downarrow \Phi & & \parallel \\ 0 & \longrightarrow & A_0 & \longrightarrow & A_2 & \xrightarrow{j_{A_2}} & A_0 \longrightarrow 0. \end{array}$$

It follows that, via $\Gamma(X_0, -)$, $[\underline{M}_{X_0}(k[\varepsilon])]$ is canonically isomorphic to the set of isomorphism classes of k -algebra extensions of A_0 by A_0 . In other words, there is a canonical bijection

$$[\underline{M}_{X_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(X_0, \underline{D}_{X_0}^1)$$

which plainly respects the vector-space structures on both sides.

In general, for every affine open $U_0 \subset X_0$, restriction and the affine case give the composite map

$$[\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{U_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(U_0, \underline{D}_{X_0}^1).$$

These maps give a canonical map

$$[\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}_{X_0}^1).$$

Its kernel consists of the isomorphism classes of locally trivial deformations of X_0 over $k[\varepsilon]$, that is, of deformations locally isomorphic to $X_0 \otimes_k k[\varepsilon]$. The sheaf of automorphisms of $X_0 \otimes_k k[\varepsilon]$ inducing the identity on X_0 is

$$\underline{D}_{X_0}^0 = \underline{\mathrm{Der}}_k(\mathcal{O}_{X_0}, \mathcal{O}_{X_0}).$$

The isomorphism classes of such locally trivial deformations are therefore canonically identified with $H^1(X_0, \underline{D}_{X_0}^0)$, which gives the asserted exact sequence. $\square$

Theorem 4.5. Let X_0 be a scheme of finite type over k for which either

- (i) X_0 is proper over k , or
- (ii) X_0 is affine and

$$X_0^{\mathrm{sing}} = \{x \in X_0 \mid X_0 \text{ is not smooth over } k \text{ at } x\}$$

is finite.

(a) The category $\underline{M}_{X_0}$ admits a formal versal object over $\underline{C}_\Lambda$, called a formal versal deformation of X_0 . If $(R, \mathcal{X})$ is a formal versal deformation of X_0 , then for every $(S, \mathcal{Y}) \in \mathrm{Ob}(\widehat{\underline{M}}_{X_0})$ there is a map

$$(\varphi, \Phi) : (R, \mathcal{X}) \longrightarrow (S, \mathcal{Y})$$

in $\widehat{\underline{M}}_{X_0}$ such that the following diagram is cartesian:

$$\begin{array}{ccc} \mathcal{Y} & \xrightarrow{\Phi} & \mathcal{X} \\ \downarrow & & \downarrow \\ \mathrm{Spf}(S) & \xrightarrow{\mathrm{Spf}(\varphi)} & \mathrm{Spf}(R). \end{array}$$

Furthermore, a formal versal deformation $(R, \mathcal{X})$ of X_0 prorepresents the functor $[\underline{M}_{X_0}]$ if and only if, for every surjective map $R' \rightarrow R$ in $\underline{C}_\Lambda$ and every deformation (R', X') of X_0 over R' , the homomorphism

$$\mathrm{Aut}_{R'}(X' \mid X_0) \longrightarrow \mathrm{Aut}_R(X' \otimes_{R'} R \mid X_0)$$

is surjective.

(b) If X_0 is proper over k , then $\underline{M}_{X_0}$ is smoothly pseudo-representable over $\underline{C}_\Lambda$.

Proof. (a) By Corollary 4.3, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$. By 1.11 and 2.9(b), it therefore suffices to prove that $[\underline{M}_{X_0}(k[\varepsilon])]$ is finite-dimensional over k .

Consider the exact sequence

$$(*) \quad 0 \longrightarrow H^1(X_0, \underline{D}_{X_0}^0) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}_{X_0}^1).$$

If X_0 is proper over k , both $H^1(X_0, \underline{D}_{X_0}^0)$ and $H^0(X_0, \underline{D}_{X_0}^1)$ are finite-dimensional, since the $\underline{D}_{X_0}^i$ are coherent for all $i \geq 0$ by 2.13(a).

Now suppose that X_0 is affine and X_0^{sing} is finite. The sheaf $\underline{D}_{X_0}^1$ is coherent and

$$\text{Supp } \underline{D}_{X_0}^1 \subset X_0^{\text{sing}}$$

by 2.13(a). Hence $H^0(X_0, \underline{D}_{X_0}^1)$ is finite-dimensional over k , whereas $H^1(X_0, \underline{D}_{X_0}^0) = 0$ because X_0 is affine. In either case, the exact sequence (*) shows that $[\underline{M}_{X_0}(k[\varepsilon])]$ is finite-dimensional over k . The second assertion of part (a) follows from 2.7.

(b) If X_0 is proper, then

$$\text{Aut}(X_0 \otimes_k k[\varepsilon] \mid X_0) = H^0(X_0, \underline{D}_{X_0}^0)$$

is finite-dimensional, and the assertion follows from 2.13. $\square$

Remark 4.6(a). Let X_0 be affine. A formal versal deformation of X_0 exists if and only if the support of $\underline{D}_{X_0}^1$ is a finite set. This does not imply, however, that all nonsmooth points of X_0 are isolated: there are affine schemes X_0 over k having a nonisolated nonsmooth point for which $\text{Supp } \underline{D}_{X_0}^1$ is finite.

Remark 4.6(b). Henceforth, a deformation of X_0 means an object of $\underline{M}_{X_0}$. Where confusion is possible, such an object will be called an infinitesimal deformation of X_0 .

Remark 4.6(c). Let X_0 be a scheme over k . We have fixed a complete noetherian local ring Λ with residue field k and considered deformations of X_0 over $\underline{C}_\Lambda$.

In practice one usually takes $\Lambda = k$, or $\Lambda = W(k)$ when k is a perfect field of positive characteristic. The inclusion $\underline{C}_k \subset \underline{C}_\Lambda$ is fully faithful, and $\underline{M}_{X_0}|_{\underline{C}_k}$ is simply the category of deformations of X_0 over the objects of $\underline{C}_k$. Consequently, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0 over $\underline{C}_\Lambda$, then

$$(k \otimes_\Lambda R, k \otimes_\Lambda \mathfrak{X})$$

is a formal versal deformation of X_0 over $\underline{C}_k$. It follows, for example, that

$$\dim_{\underline{C}_\Lambda} \underline{M}_{X_0} = \dim(k \otimes_\Lambda R)$$

does not depend on the choice of Λ , but only on X_0 .

The preceding theorem admits many generalizations. For example, consider a pair $Y_0 \rightarrow X_0$, where X_0 is a scheme of finite type over k and Y_0 is a fixed closed subscheme of X_0 . For $R \in \text{Ob}(\underline{C}_\Lambda)$, a deformation of (X_0, Y_0) over R is a commutative diagram

$$\begin{array}{ccccc} Y_0 & \xrightarrow{\quad\quad\quad} & & & Y \\ & \searrow & & \swarrow & \\ & & X_0 & \xrightarrow{\quad\quad\quad} & X \\ & \searrow & \downarrow & \downarrow & \swarrow \\ & & \text{Spec}(k) & \xrightarrow{\quad\quad\quad} & \text{Spec}(R) \end{array} \quad \begin{array}{l} \\ \\ \text{flat} \\ \text{flat} \\ \text{flat} \end{array}$$

such that the induced map

$$(X_0, Y_0) \longrightarrow (X \times_{\text{Spec}(R)} \text{Spec}(k), Y \times_{\text{Spec}(R)} \text{Spec}(k))$$

is an isomorphism. Defining morphisms

$$(R'; X', Y') \longrightarrow (R; X, Y)$$

of deformations of (X_0, Y_0) in the evident way gives a cofibered category $\underline{M}_{X_0, Y_0}$ in groupoids over $\underline{C}_\Lambda$. This construction is frequently used to rigidify automorphisms in deformation-moduli

problems. The argument of Corollary 4.3, based on Lemma 4.2, immediately shows that $\underline{M}_{X_0, Y_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Lemma 4.7. The sequence

$$0 \longrightarrow H^0\left(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\right) \longrightarrow [\underline{M}_{X_0, Y_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])]$$

is exact, where $\underline{I}$ is the ideal sheaf defining Y_0 in X_0 ; thus

$$0 \longrightarrow \underline{I} \longrightarrow \mathcal{O}_{X_0} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow 0$$

is exact.

Proof. Let $\underline{B}$ be the full subcategory of $\underline{M}_{X_0, Y_0}(k[\varepsilon])$ whose objects have the form

$$(k[\varepsilon]; X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]), Y).$$

Then

$$[\underline{B}] = \ker([\underline{M}_{X_0, Y_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])]).$$

The category $\underline{B}$ is the groupoid of deformations of Y_0 over $k[\varepsilon]$ inside $X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon])$. Thus an object is a commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{i_Y} & X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]) \\ & \searrow \text{flat} & \swarrow \\ & \mathrm{Spec}(k[\varepsilon]) & \end{array}$$

whose special fiber is the fixed closed immersion i_{Y_0} . An arrow $(Y_1, i_{Y_1}) \longrightarrow (Y_2, i_{Y_2})$ is a $k[\varepsilon]$ -morphism $\Phi : Y_1 \longrightarrow Y_2$ (necessarily an isomorphism) whose special fiber is id_{Y_0} and for which $i_{Y_2} \circ \Phi = i_{Y_1}$.

Equivalently, an object of $\underline{B}$ is a pair (Y, j_Y) , where Y is a deformation of Y_0 over $k[\varepsilon]$ and $j_Y : Y \longrightarrow X_0$ is a k -morphism whose composite $Y_0 \longrightarrow Y \xrightarrow{j_Y} X_0$ is i_{Y_0} . An arrow $(Y_1, j_{Y_1}) \longrightarrow (Y_2, j_{Y_2})$ is a morphism $\Phi : Y_1 \longrightarrow Y_2$ whose special fiber is id_{Y_0} and for which $j_{Y_2} \circ \Phi = j_{Y_1}$.

The assignment sending (Y, j_Y) to the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \underline{I} & \longrightarrow & \mathcal{O}_{X_0} & \longrightarrow & \mathcal{O}_{Y_0} \longrightarrow 0 \\ & & \downarrow & & \downarrow j_Y & & \parallel \\ 0 & \longrightarrow & \mathcal{O}_{Y_0} & \xrightarrow{\varepsilon} & \mathcal{O}_Y & \longrightarrow & \mathcal{O}_{Y_0} \longrightarrow 0 \end{array}$$

gives an isomorphism

$$\begin{aligned} [\underline{B}] &\xrightarrow{j_Y|_{\underline{I}}} \mathrm{Hom}_{\mathcal{O}_{X_0}}(\underline{I}, \mathcal{O}_{Y_0}) \\ &= \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0}) \\ &= H^0\left(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\right). \end{aligned}$$

This proves exactness. $\square$

Remark 4.8. For a more general formula implying Lemma 4.7, see SGA 3, III, 4.4. One may also note the following. For every S -scheme X and every quasi-coherent $\mathcal{O}_X$ -module E , one can define quasi-coherent $\mathcal{O}_X$ -modules $\underline{D}_{X|S}^i(E)$ for all $i \geq 0$, generalizing the case in which S is affine. As usual, we write

$$\underline{D}_{X|S}^i(\mathcal{O}_X) = \underline{D}_{X|S}^i.$$

For the closed immersion $Y_0 \hookrightarrow X_0$, we then have

$$H^0\left(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\right) = H^0(Y_0, \underline{D}_{Y_0|X_0}^1),$$

where $0 \longrightarrow \underline{I} \longrightarrow \mathcal{O}_{X_0} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow 0$ is exact.

Theorem 4.9. Let X_0 be a k -scheme of finite type and Y_0 a closed subscheme of X_0 . Suppose either

- (i) X_0 is proper over k , or
 - (ii) X_0 is affine and smooth over k outside finitely many points, and Y_0 is proper over k .
- (a) The category $\underline{M}_{X_0, Y_0}$ admits a formal versal deformation over $\underline{C}_\Lambda$. It prorepresents the functor $[\underline{M}_{X_0, Y_0}]$ if and only if, for every surjective map $R' \longrightarrow R$ in $\underline{C}_\Lambda$ and every deformation $(R'; X', Y')$, the map

$$\mathrm{Aut}_{R'}((X', Y') \mid X_0) \longrightarrow \mathrm{Aut}_R((X' \otimes_{R'} R, Y' \otimes_{R'} R) \mid X_0)$$

is surjective.

- (b) If, in addition,

$$H^0\left(X_0, \underline{D}_{Y_0|k}^0 \times_{\underline{D}_{X_0|k}^0(\mathcal{O}_{Y_0})} \underline{D}_{X_0|k}^0\right)$$

is finite-dimensional, then $\underline{M}_{X_0, Y_0}$ is smoothly pseudo-representable.

Proof. This follows immediately from Lemmas 4.4 and 4.7 together with 1.11, 2.7, and 2.13. $\square$

We now consider localization and completion in the deformation theory of schemes. We first fix notation. Let X_0 be a k -scheme of finite type and let $x_0 \in X_0$ be a fixed point. There are natural functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0, x_0})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0, x_0})},$$

and hence functors

$$\widehat{\underline{M}}_{X_0} \longrightarrow \widehat{\underline{M}}_{\mathrm{Spec}(\mathcal{O}_{X_0, x_0})} \longrightarrow \widehat{\underline{M}}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0, x_0})}.$$

As observed in 4.1(c), $\widehat{\underline{M}}_{X_0}$ is canonically equivalent to the category whose objects are pairs $(R, \mathfrak{X})$, where $R \in \mathrm{Ob}(\widehat{\underline{C}}_\Lambda)$ and $\mathfrak{X}$ is a flat R -adic formal scheme equipped with a fixed isomorphism $\mathfrak{X}_0 \xrightarrow{\sim} X_0$.

When there is no danger of confusion, we denote the images of $(R, \mathfrak{X})$ under localization and completion by $(R, \mathfrak{X}_{(x_0)})$ and $(R, \mathfrak{X}_{(\widehat{x}_0)})$, respectively. Thus, if

$$\mathfrak{X} = \varinjlim_{\nu} \mathfrak{X}_{\nu}, \quad \mathfrak{X}_{\nu} = \mathfrak{X} \times_{\mathrm{Spf}(R)} \mathrm{Spec}(R/\mathfrak{m}^{\nu+1}),$$

then

$$\mathfrak{X}_{(x_0)} = \varinjlim_{\nu} \mathrm{Spec}(\mathcal{O}_{\mathfrak{X}_{\nu}, x_0})$$

and

$$\mathfrak{X}_{(\widehat{x}_0)} = \varinjlim_{\nu} \mathrm{Spec}(\widehat{\mathcal{O}}_{\mathfrak{X}_{\nu}, x_0}).$$

Theorem 4.10. Let X_0 be an affine scheme of finite type over k .

- (a) If X_0 is locally a complete intersection, then $\underline{M}_{X_0}$ is smooth over $\underline{C}_\Lambda$.
- (b) Suppose that, for a fixed closed point $x \in X_0$, the open $X_0 - \{x\}$ is smooth over k . Then both functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0, x})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0, x})}$$

are minimally smooth. In particular, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0 , then $(R, \mathfrak{X}_{(x)})$ and $(R, \mathfrak{X}_{(\widehat{x})})$ are formal versal deformations of $\mathrm{Spec}(\mathcal{O}_{X_0, x})$ and $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0, x})$, respectively.

Proof. We use the notation for the cohomology of commutative algebras from section 3, and write $X_0 = \text{Spec}(A_0)$.

(a) It suffices to show that $P : [\underline{M}_{X_0}] \rightarrow \underline{C}_\Lambda$ is smooth. Let $(R, \text{Spec}(A))$ be a deformation of $\text{Spec}(A_0)$ over $R \in \text{Ob}(\underline{C}_\Lambda)$. Given a small extension $R' \rightarrow R$ in $\underline{C}_\Lambda$, so that

$$0 \rightarrow k \rightarrow R' \rightarrow R \rightarrow 0$$

is exact, we must lift $(R, \text{Spec}(A))$ to R' . The transitivity sequence for D^i with coefficients in A_0 , applied to $R' \rightarrow R \rightarrow A$, gives

$$\cdots \rightarrow D^1(A|R, A_0) \rightarrow D^1(A|R', A_0) \rightarrow D^1(R|R', k) \otimes_k A_0 \rightarrow D^2(A|R, A_0) \rightarrow \cdots$$

Because $R' \rightarrow R$ is a small extension, $D^1(R|R', k)$ is one-dimensional over k , generated by the class $[R']$ represented by $0 \rightarrow k \rightarrow R' \rightarrow R \rightarrow 0$. Since A is flat over R ,

$$D^i(A|R, A_0) = D^i(A_0|k, A_0) \quad (i > 0).$$

In particular, $D^2(A|R, A_0) = 0$ because A_0 is locally a complete intersection, by the low-degree description in 3.12(8). We therefore obtain an exact sequence

$$\rightarrow D^1(A_0|k, A_0) \rightarrow D^1(A|R', A_0) \rightarrow D^1(R|R', k) \otimes_k A_0 \rightarrow 0.$$

Hence there is an R' -algebra extension A' of A by A_0 whose class maps to $[R'] \otimes 1$. Equivalently, $(R, \text{Spec}(A))$ lifts to the deformation $(R', \text{Spec}(A'))$.

(b) By 3.14 there are isomorphisms

$$D^1(A_0|k, A_0) \xrightarrow{\sim} D^1((A_0)_x|k, (A_0)_x) \xrightarrow{\sim} D^1(\widehat{(A_0)_x}|k, \widehat{(A_0)_x}),$$

and hence

$$[\underline{M}_{X_0}(k[\varepsilon])] \xrightarrow{\sim} [\underline{M}_{\text{Spec}(\mathcal{O}_{X_0, x})}(k[\varepsilon])] \xrightarrow{\sim} [\underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x})}(k[\varepsilon])].$$

By 2.7(a), it remains to show that

$$[\underline{M}_{X_0}] \rightarrow [\underline{M}_{\text{Spec}(\mathcal{O}_{X_0, x})}] \rightarrow [\underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x})}]$$

are smooth functors.

Let $(R, \text{Spec}(A))$ be a deformation of $X_0 = \text{Spec}(A_0)$, and let $0 \rightarrow k \rightarrow R' \rightarrow R \rightarrow 0$ be a small extension. As above, the homomorphisms $R' \rightarrow R \rightarrow A$ give an exact commutative diagram

$$\begin{array}{ccccccc} \rightarrow D^1(A_0|k, A_0) & \longrightarrow & D^1(A|R', A_0) & \xrightarrow{p} & D^1(R|R', k) \otimes_k A_0 & \longrightarrow & D^2(A_0|k, A_0) \rightarrow \cdots \\ & \sim \downarrow & \downarrow & & \downarrow & & \downarrow \sim \\ \rightarrow D^1(A_0|k, (A_0)_x) & \longrightarrow & D^1(A|R', (A_0)_x) & \rightarrow & D^1(R|R', k) \otimes_k (A_0)_x & \rightarrow & D^2(A_0|k, (A_0)_x) \rightarrow \cdots \\ & \uparrow & \uparrow & & \uparrow & & \uparrow \\ \rightarrow D^1((A_0)_x|k, (A_0)_x) & \rightarrow & D^1(A_x|R', (A_0)_x) & \xrightarrow{p_x} & D^1(R|R', k) \otimes_k (A_0)_x & \rightarrow & D^2((A_0)_x|k, (A_0)_x) \rightarrow \cdots \end{array}$$

The indicated isomorphisms follow from the fact that A_0 is smooth over k away from the single closed point x , so that

$$D^i(A_0|k, (A_0)_x) = D^i(A_0|k, A_0)_x = D^i(A_0|k, A_0) \quad (i > 0).$$

Suppose we are given an arrow

$$(R', \text{Spec}(B)) \rightarrow (R, \text{Spec}(A_x))$$

in $[\underline{M}_{\text{Spec}((A_0)_x)}]$. This amounts to a class $[B] \in D^1(A_x|R', (A_0)_x)$ satisfying $p_x([B]) = [R']$. A diagram chase gives a class $[A'] \in D^1(A|R', A_0)$ such that $p([A']) = [R']$ and $A'_x = B$. Thus $(R', \text{Spec}(A'))$ is a deformation of $\text{Spec}(A_0)$ over R' whose localization at x is $\text{Spec}(B)$. This proves that

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\text{Spec}(\mathcal{O}_{X_0, x})}$$

is smooth.

The smoothness of

$$\underline{M}_{\text{Spec}((A_0)_x)} \longrightarrow \underline{M}_{\text{Spec}(\widehat{(A_0)_x})}$$

follows from the analogous diagram and the identities

$$D^i((A_0)_x|k, \widehat{(A_0)_x}) = D^i(\widehat{(A_0)_x|k}, (A_0)_x) = D^i((A_0)_x|k, (A_0)_x) \quad (i > 0),$$

since x is the only possible nonsmooth point. $\square$

Theorem 4.11. Let X_0 be a separated scheme of finite type over k , smooth away from a finite closed subset

$$X_0^{\text{Sing}} = \{x_1, \dots, x_r\}.$$

Let $U_0^1, \dots, U_0^r$ be affine open neighbourhoods of $x_1, \dots, x_r$ such that $U_0^i \cap X_0^{\text{Sing}} = \{x_i\}$. If $H^2(X_0, \underline{D}_{X_0}^0) = 0$, then the natural functor

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{U_0^1} \times \dots \times \underline{M}_{U_0^r}$$

is smooth.

Proof. Complete $U_0^1, \dots, U_0^r$ to an affine open covering of X_0 by choosing $U_0^{r+1}, \dots, U_0^N$ such that $U_0^j \cap X_0^{\text{Sing}} = \emptyset$ for every $j > r$. Let $R' \rightarrow R$ be a small extension in $\underline{C}_\Lambda$, and let $(R', U^{j1}), \dots, (R', U^{jr})$ be deformations of $U_0^1, \dots, U_0^r$ over R' such that

$$U^{ji} \otimes_{R'} R = X|_{U_0^i}, \quad 1 \leq i \leq r,$$

for some deformation (R, X) of X_0 over R . We must find a deformation (R', X') of X_0 over R' such that

$$X' \otimes_{R'} R = X, \quad X'|_{U_0^i} = U^{ji} \quad (1 \leq i \leq r).$$

For $j > r$, the scheme U_0^j is smooth over k ; hence, by 4.9(a), $X|_{U_0^j}$ lifts to a deformation (R', U^{jj}) of U_0^j over R' . Thus

$$U^{jj} \otimes_{R'} R = X|_{U_0^j} \quad (j > r).$$

For every pair $i \neq j$, the intersection $U_0^i \cap U_0^j$ is affine and smooth over k . We therefore obtain an isomorphism

$$\tau_{ij} : U^{ji}|_{U_0^i \cap U_0^j} \xrightarrow{\sim} U^{jj}|_{U_0^i \cap U_0^j}$$

such that $\tau_{ij} \otimes_{R'} R = \text{id}_{X|_{U_0^i \cap U_0^j}}$. On triple intersections, the automorphisms

$$\tau_{ij}\tau_{jk}\tau_{ik}^{-1}$$

define a cohomology class in $H^2(X_0, \underline{D}_{X_0}^0)$. This class is the obstruction to gluing the U^{ji} to a global deformation. It vanishes by hypothesis, so the U^{ji} glue to a deformation (R', X') with $X' \otimes_{R'} R = X$ and $X'|_{U_0^i} \simeq U^{ji}$. $\square$

Remark 4.12. The preceding proof assumes that X_0 is separated, but this assumption is unnecessary. Indeed, fix a deformation (R, X) of X_0 and deformations (R', U^{ji}) of the members U_0^i of

an open covering. The deformations (R', V') of open subsets V of X over R' that are compatible with the already-given deformations (R', U^i) form a “gerbe” (in Giraud’s terminology) whose structure sheaf is $\underline{D}_{X_0}^0$. The obstruction to the existence of a section of this gerbe—that is, to a global deformation (R', X') of X compatible with the given data (R', U^i) —lies in $H^2(X_0, \underline{D}_{X_0}^0)$, and hence vanishes under the hypothesis of the theorem.

Corollary 4.13. Let X_0 be a scheme of finite type over k , smooth away from a finite subset

$$X_0^{\text{Sing}} = \{x_1, \dots, x_r\},$$

and suppose that $H^2(X_0, \underline{D}_{X_0}^0) = 0$. Then:

(a) The functor

$$\underline{M}_{X_0} \longrightarrow \prod_{i=1}^r \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}$$

is smooth. In particular,

$$\dim \underline{M}_{X_0} = \dim_k H^1(X_0, \underline{D}_{X_0}^0) + \sum_{i=1}^r \dim \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}.$$

(b) If, in addition, X_0 is locally a complete intersection, then $\underline{M}_{X_0}$ is smooth.

Proof. (a) The first assertion follows from 4.10 and 4.9(b). Let $(R, \mathfrak{X})$ and $(R_1, \mathfrak{X}_1), \dots, (R_r, \mathfrak{X}_r)$ be formal versal deformations of X_0 and of

$$\text{Spec}(\mathcal{O}_{X_0, x_1}), \dots, \text{Spec}(\mathcal{O}_{X_0, x_r}),$$

respectively. We obtain a commutative diagram

$$\begin{array}{ccc} \underline{h}_R & \longrightarrow & \underline{h}_{R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r} \\ \downarrow & & \downarrow \\ \underline{M}_{X_0} & \longrightarrow & \prod_{i=1}^r \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}. \end{array}$$

The composite functor

$$\underline{h}_R \longrightarrow \underline{M}_{X_0} \longrightarrow \prod_{i=1}^r \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}$$

is smooth, whereas

$$\underline{h}_{R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r} \longrightarrow \prod_{i=1}^r \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}$$

is minimally smooth. It follows from 2.7(a) that

$$\underline{h}_R \longrightarrow \underline{h}_{R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r}$$

is smooth; equivalently, R is a formal power-series ring over $R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r$.

By 4.4, the sequence

$$0 \longrightarrow H^1(X_0, \underline{D}_{X_0}^0) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow \prod_{i=1}^r [\underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}(k[\varepsilon])]$$

is exact. Equivalently,

$$0 \longrightarrow H^1(X_0, \underline{D}_{X_0}^0) \longrightarrow \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow \text{Hom}_{\Lambda\text{-alg}}(R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r, k[\varepsilon])$$

is exact. Consequently, R is a formal power-series ring over $R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r$ in d variables, where

$$d = \dim_k H^1(X_0, \underline{D}_{X_0}^0).$$

Therefore

$$\begin{aligned} \dim \underline{M}_{X_0} &= \dim(k \otimes_\Lambda R) \\ &= d + \dim(k \otimes_\Lambda (R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r)) \\ &= d + \dim((k \otimes_\Lambda R_1) \widehat{\otimes}_k \cdots \widehat{\otimes}_k (k \otimes_\Lambda R_r)) \\ &= d + \sum_{i=1}^r \dim(k \otimes_\Lambda R_i) \\ &= d + \sum_{i=1}^r \dim \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}. \end{aligned}$$

(b) This follows from 4.10 and 4.9(a). $\square$

4.14. We close this section by constructing explicitly a formal versal deformation in a simple case. When X_0 is affine and a complete intersection, such a deformation is immediate to write down. Let

$$A_0 = k[x_1, \dots, x_n]/(f_1, \dots, f_r) = S/J,$$

where $S = k[x_1, \dots, x_n]$, $J = (f_1, \dots, f_r)$, and $f_1, \dots, f_r$ is an S -regular sequence. We have an exact sequence

$$\cdots \longrightarrow \text{Der}_k(S, A_0) \longrightarrow \text{Hom}_{A_0}(J/J^2, A_0) \longrightarrow D^1(A_0|k, A_0) \longrightarrow 0.$$

The A_0 -module J/J^2 is free of rank r , generated by the classes of $f_1, \dots, f_r$, because these elements form an S -regular sequence.

Assume now that $D^1(A_0|k, A_0)$ is finite-dimensional over k ; for example, assume that A_0 has only isolated nonsmooth points. Put

$$d = \dim_k D^1(A_0|k, A_0),$$

and choose maps

$$p_i : J/J^2 \longrightarrow A_0 \quad (1 \leq i \leq d)$$

whose classes form a k -basis of $D^1(A_0|k, A_0)$. For $1 \leq i \leq d$ and $1 \leq j \leq r$, choose $P_{ij} \in \Lambda[x_1, \dots, x_n]$ such that

$$P_{ij} \equiv p_i(f_j) \pmod{\mathfrak{m}_\Lambda + J},$$

where $\mathfrak{m}_\Lambda$ is the maximal ideal of Λ . Set

$$R = \Lambda[[t_1, \dots, t_d]], \quad B = R[x_1, \dots, x_n]/(G_1, \dots, G_r),$$

where

$$G_j = F_j - \sum_{i=1}^d t_i P_{ij},$$

and $F_j \in \Lambda[x_1, \dots, x_n]$ is obtained from f_j by lifting its coefficients from k to Λ .

The algebra $R[x_1, \dots, x_n]$ is flat over R , and the images of $G_1, \dots, G_r$ modulo $\mathfrak{m}_R$ form the regular sequence $f_1, \dots, f_r$ in $k \otimes_R R[x_1, \dots, x_n]$. Hence B is R -flat and $k \otimes_R B = A_0$. By construction, the map

$$\text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow D^1(A_0|k, A_0) = [\underline{M}_{\text{Spec}(A_0)}(k[\varepsilon])]$$

is bijective. Thus, if

$$\mathrm{Spf}(B) = \varinjlim_{\nu} \mathrm{Spec}((R/\mathfrak{m}_R^{\nu+1}) \otimes_R B),$$

then $(R, \mathrm{Spf}(B))$ is an object of $\widehat{\underline{M}}_{\mathrm{Spec}(A_0)}$. The functor

$$\underline{h}_R \longrightarrow \underline{M}_{\mathrm{Spec}(A_0)}$$

that it induces is smooth by 1.6, since R is a formal power-series ring over Λ . In other words, $(R, \mathrm{Spf}(B))$ is a formal versal object of $\underline{M}_{\mathrm{Spec}(A_0)}$.

By 4.10(b), if instead we put

$$B = R[[x_1, \dots, x_n]]/(G_1, \dots, G_r),$$

then $(R, \mathrm{Spf}(B))$ is a formal versal deformation of $\mathrm{Spec}(A_0)$ for

$$A_0 = k[[x_1, \dots, x_n]]/(f_1, \dots, f_r).$$

Remark 4.15. Let X_0 be a scheme of finite type over k with only isolated nonsmooth points, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . The preceding construction shows that, when X_0 is affine and locally a complete intersection, $\mathfrak{X}$ is the formalization of an R -scheme of finite type. The same fact extends readily to the nonaffine case provided that $H^2(X_0, \underline{D}_{X_0}^0) = 0$.

Remark 4.16. The global theory of the relative cotangent complex—here reduced to the truncated complex of length one, which nevertheless suffices for many purposes in deformation theory—generalizes and simplifies results such as 4.10. Let S be a scheme, and let $S_0 \subset S' \subset S$ be subschemes defined by ideals $J \supset I$ with $JI = 0$. Let X' be flat over S' , and write

$$L_{\bullet} = L^{X_0/S_0}$$

for the relative cotangent complex of X_0 over S_0 , regarded as an object of $D(X_0, \mathcal{O}_{X_0})$. Let $\underline{F}$ be the category of flat S -schemes X extending X'/S' ; thus an object is equipped with an S' -isomorphism

$$X \times_S S' \xrightarrow{\sim} X'.$$

Then:

(a) For an object X of $\underline{F}$, its group of automorphisms inducing the identity on X' is canonically isomorphic to

$$\mathrm{Ext}^0(X_0; L_{\bullet}, I_{X_0}),$$

where I_{X_0} is the inverse image on X_0 of the $\mathcal{O}_{S_0}$ -module I .

(b) The set of isomorphism classes of objects of $\underline{F}$ is either empty or a torsor (a principal homogeneous space) under $\mathrm{Ext}^1(X_0; L_{\bullet}, I_{X_0})$.

(c) There is a canonical element

$$c \in \mathrm{Ext}^2(X_0; L_{\bullet}, I_{X_0})$$

whose vanishing is necessary and sufficient for $\underline{F}$ to be nonempty.

To study these groups, it is often convenient to write

$$\mathrm{Ext}^i(X_0; L_{\bullet}, I_{X_0}) = H^i(X_0, R\mathrm{Hom}(L_{\bullet}, I_{X_0})).$$

The cohomology sheaves of $R\mathrm{Hom}(L_{\bullet}, I_{X_0})$ are precisely the sheaves $\underline{D}_{X_0}^i(I_{X_0})$ introduced in section 3. For $i = 0, 1, 2$, they may be viewed respectively as the sheaf of automorphisms of an object X of $\underline{F}$ (automorphisms that induce the identity on X'), the sheaf measuring the indeterminacy of isomorphism classes of local solutions to the flat-deformation problem, and the sheaf of obstructions to the existence of local solutions.

The standard spectral sequence is

$$E_2^{p,q} = H^p(X_0, \underline{D}_{X_0}^q(I_{X_0})) \implies \text{Ext}^{p+q}(X_0; L_\bullet, I_{X_0}).$$

It gives the familiar isomorphisms

$$\begin{aligned} \text{Aut}(X/X') &\simeq H^0(X_0, \underline{D}_{X_0}^0(I_{X_0})) \\ &\simeq H^0(X_0, \underline{\text{Hom}}(\Omega_{X_0/S_0}^1, I_{X_0})) \\ &= \text{Hom}(\Omega_{X_0/S_0}^1, I_{X_0}), \end{aligned}$$

and the low-degree exact sequence

$$0 \longrightarrow H^1(X_0, \underline{D}_{X_0}^0(I_{X_0})) \longrightarrow \text{Ext}^1 \longrightarrow H^0(X_0, \underline{D}_{X_0}^1(I_{X_0})) \longrightarrow H^2(X_0, \underline{D}_{X_0}^0(I_{X_0})) \longrightarrow \text{Ext}^2.$$

In particular, if $H^2(X_0, \underline{D}_{X_0}^0(I_{X_0})) = 0$ and local flat deformations of X'/S' exist, then compatible local deformations glue to a global flat deformation. More generally, for a prescribed section of the sheaf of isomorphism classes of local flat deformations, the argument of 4.10 places the obstruction to a compatible global deformation in $H^2(X_0, \underline{D}_{X_0}^0(I_{X_0}))$.

The same spectral sequence shows that Ext^2 vanishes whenever

$$(*) \quad H^0(X_0, \underline{D}_{X_0}^2) = 0, \quad H^1(X_0, \underline{D}_{X_0}^1) = 0, \quad H^2(X_0, \underline{D}_{X_0}^0) = 0.$$

The first equality always holds when X_0 is a relative complete intersection over S_0 , since then $\underline{D}_{X_0}^2(M) = 0$ for every $\mathcal{O}_{X_0}$ -module M . The second holds when the support of $\underline{D}_{X_0}^1$ is discrete, as happens when X_0 is smooth over S_0 away from a discrete subset. If all three equalities in $(*)$ hold, then $\underline{F}$ is nonempty by (c), which generalizes 4.12(b).

There is also a geometric interpretation, requiring only the globalized functors $\underline{D}_{X_0}^i$ rather than the global theory of the relative cotangent complex. The three successive obstructions to extending X'/S' flatly over S lie in the three groups displayed in $(*)$. The local theory defines the first obstruction

$$c_1 \in H^0(X_0, \underline{D}_{X_0}^2)$$

as the obstruction to the existence of local solutions. If $c_1 = 0$, the sheaf $\underline{F}$ of isomorphism classes of local solutions is canonically a torsor under $\underline{D}_{X_0}^1$. Its obstruction to having a section is

$$c_2 \in H^1(X_0, \underline{D}_{X_0}^1);$$

such a section is a coherent system of isomorphism classes of local solutions. If $c_2 = 0$, choose a section of $\underline{F}$. The obstruction to a global solution realizing the chosen local classes is then

$$c_3 \in H^2(X_0, \underline{D}_{X_0}^0).$$

Changing the section of $\underline{F}$ by a section of $\underline{D}_{X_0}^1$ gives a map

$$d : H^0(X_0, \underline{D}_{X_0}^1) \longrightarrow H^2(X_0, \underline{D}_{X_0}^0).$$

A solution exists if and only if $c_1 = 0$, $c_2 = 0$, and the class of c_3 in $\text{Coker } d$ is zero (each c_i being defined only after the preceding obstructions vanish).

The three obstruction groups

$$H^0(X_0, \underline{D}_{X_0}^2), \quad H^1(X_0, \underline{D}_{X_0}^1), \quad H^2(X_0, \underline{D}_{X_0}^0)/\text{Im } d$$

are the initial terms $E_2^{0,2}$, $E_2^{1,1}$, and $E_3^{2,0}$ of the preceding spectral sequence; up to sign, d is its differential

$$d_2 : E_2^{0,1} \longrightarrow E_2^{2,0}.$$